

Random Forest Feature Importance — Soil Moisture Response

Random Forest Feature Importance Analysis
Soil Moisture Response Drivers

Event groups: BSC_XX11_Q12, IRRIGATION, IRRIGATION_STRONG

BSC_XX11_Q12 selection:

- BSC bits 3-4 = '11' (cumulative curve HIGH in Q3 & Q4
→ most rain already fell before the 3rd quartile begins)
- Huff quartile 1 or 2 (peak intensity in 1st half of event)
→ Front-loaded impulse: VWC peak is clearly attributable to the initial rainfall burst, not to a sustained late input

Depths: -10 cm / -30 cm / -45 cm

Target: `delta_pct_dyn-{depth}` – VWC rise as % of sensor dyn. range

Model: RandomForestRegressor (150 trees, 3-fold CV)

Importance: permutation importance (10 repeats, signed)

SHAP: signed beeswarm (shap available)

Leakage prevention:

Excluded: `peak_vwc`, `t_peak_min`, `delta_vwc` at all depths

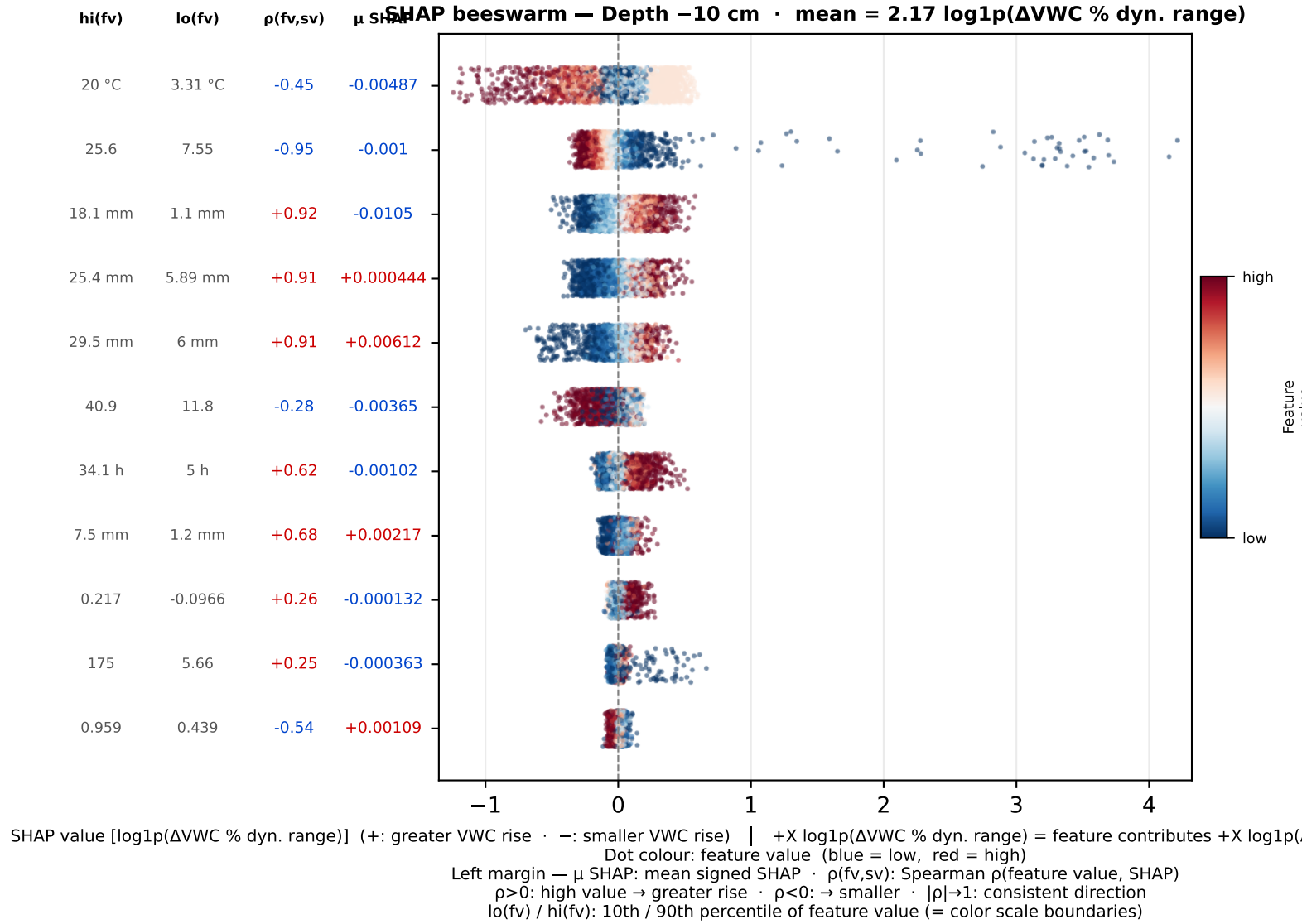
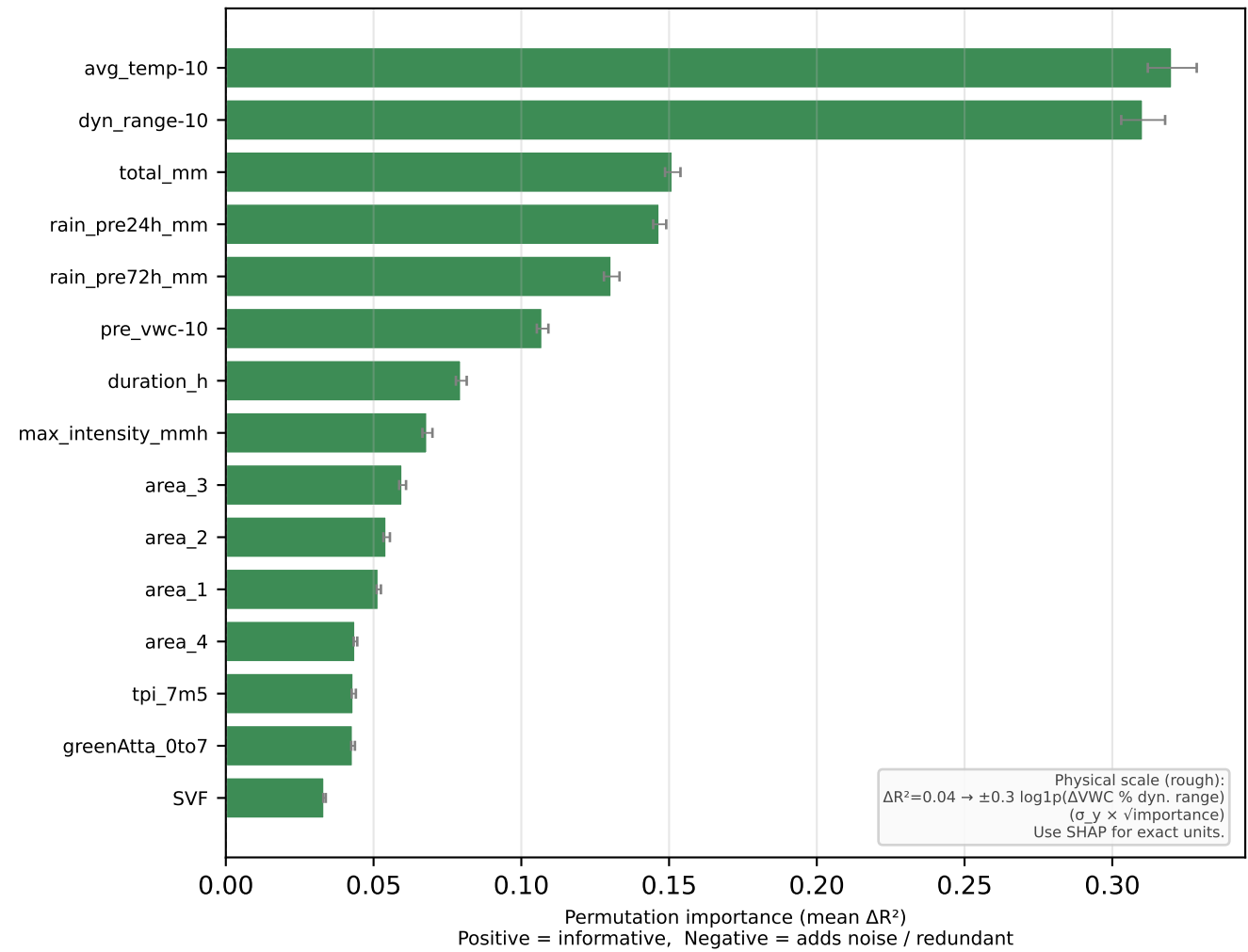
Retained: `pre_vwc`, `dyn_range`, `dry_h` for the target depth only

Excluded: `vw_c_p05/p95` (redundant with `dyn_range = p95 - p05`)

Random Forest Feature Importance — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: Δ VWC as % of dynamic range at each depth (10,911 events)

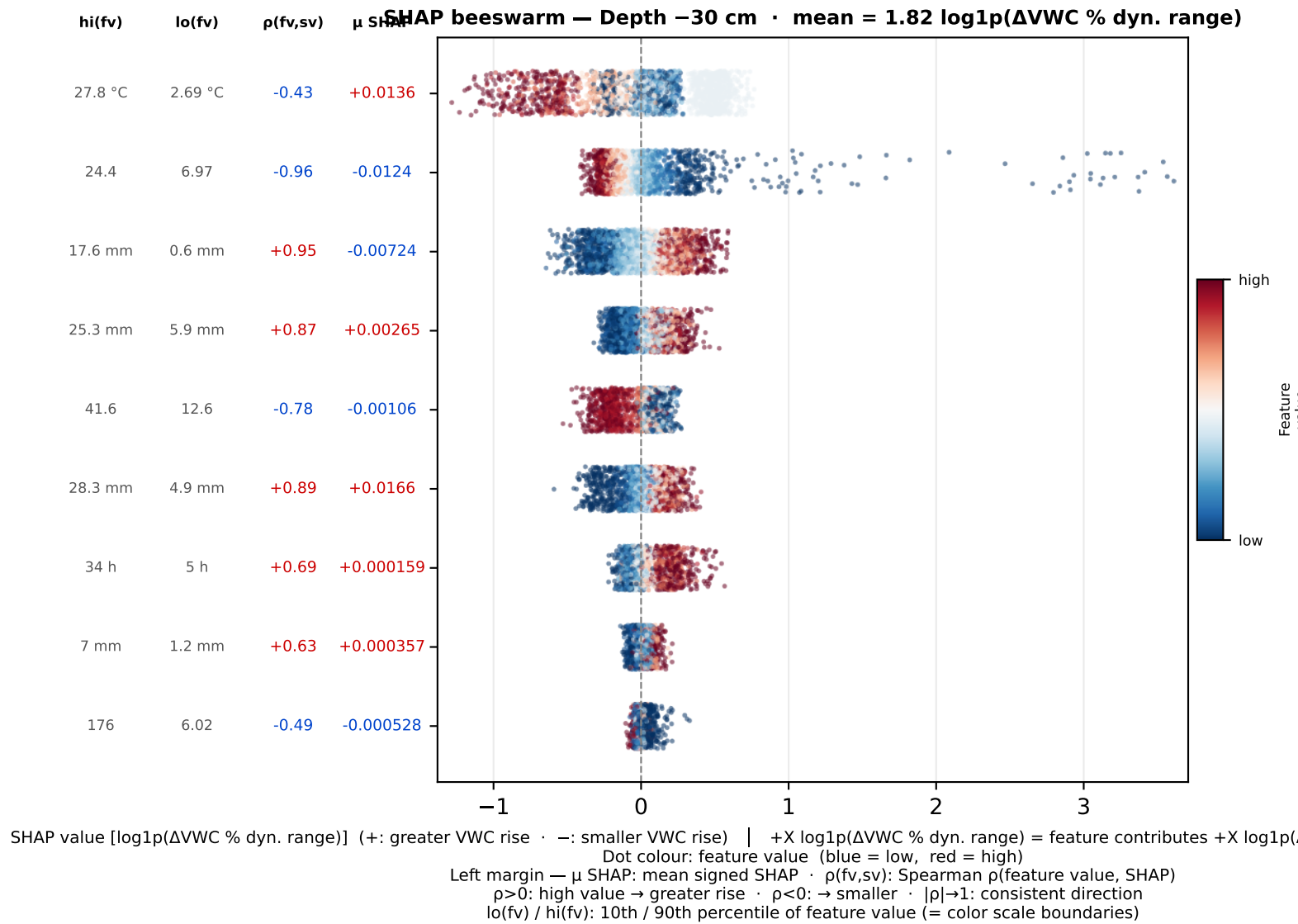
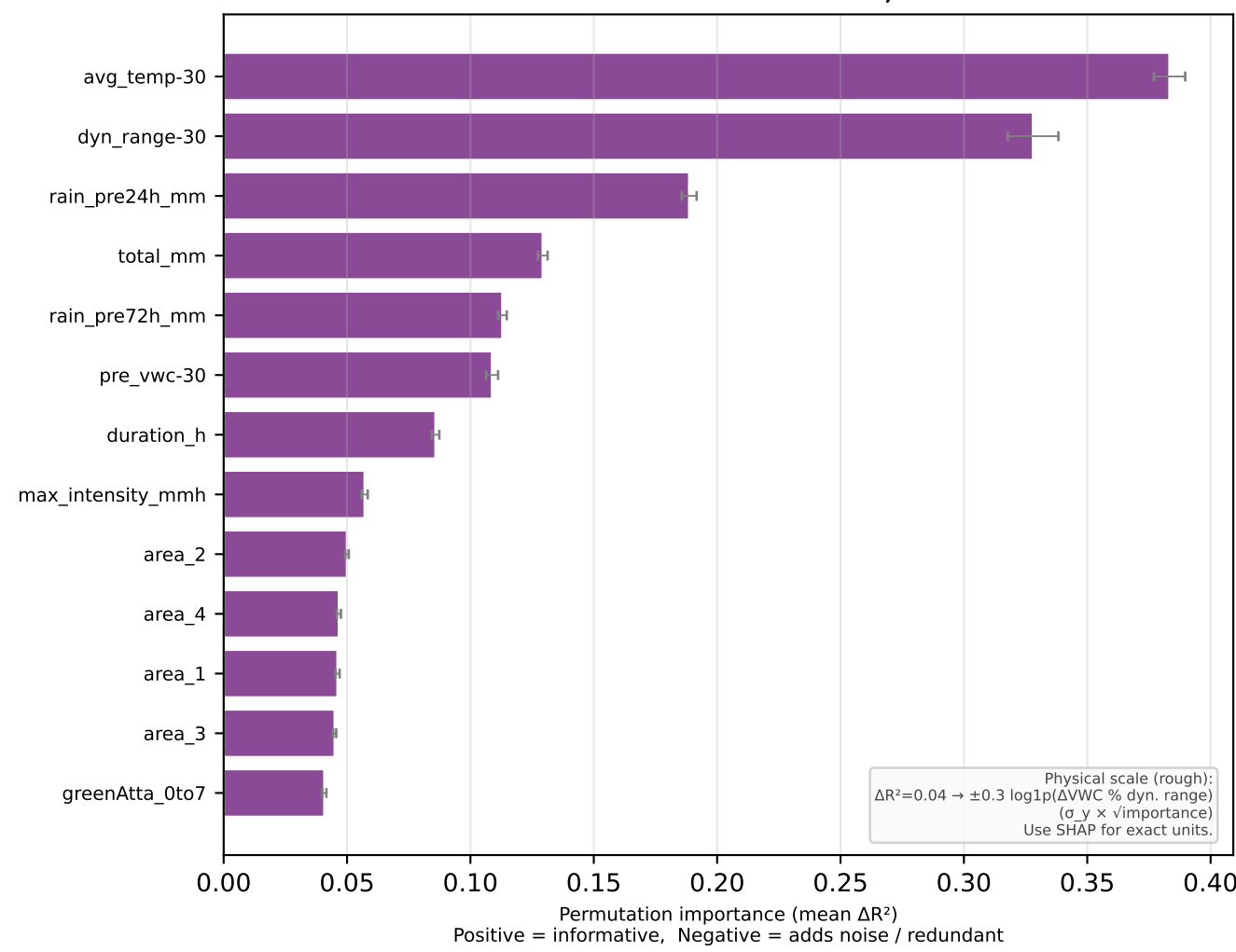
Depth -10 cm — permutation importance

CV $R^2 = 0.513 \pm 0.011$ n = 4,499



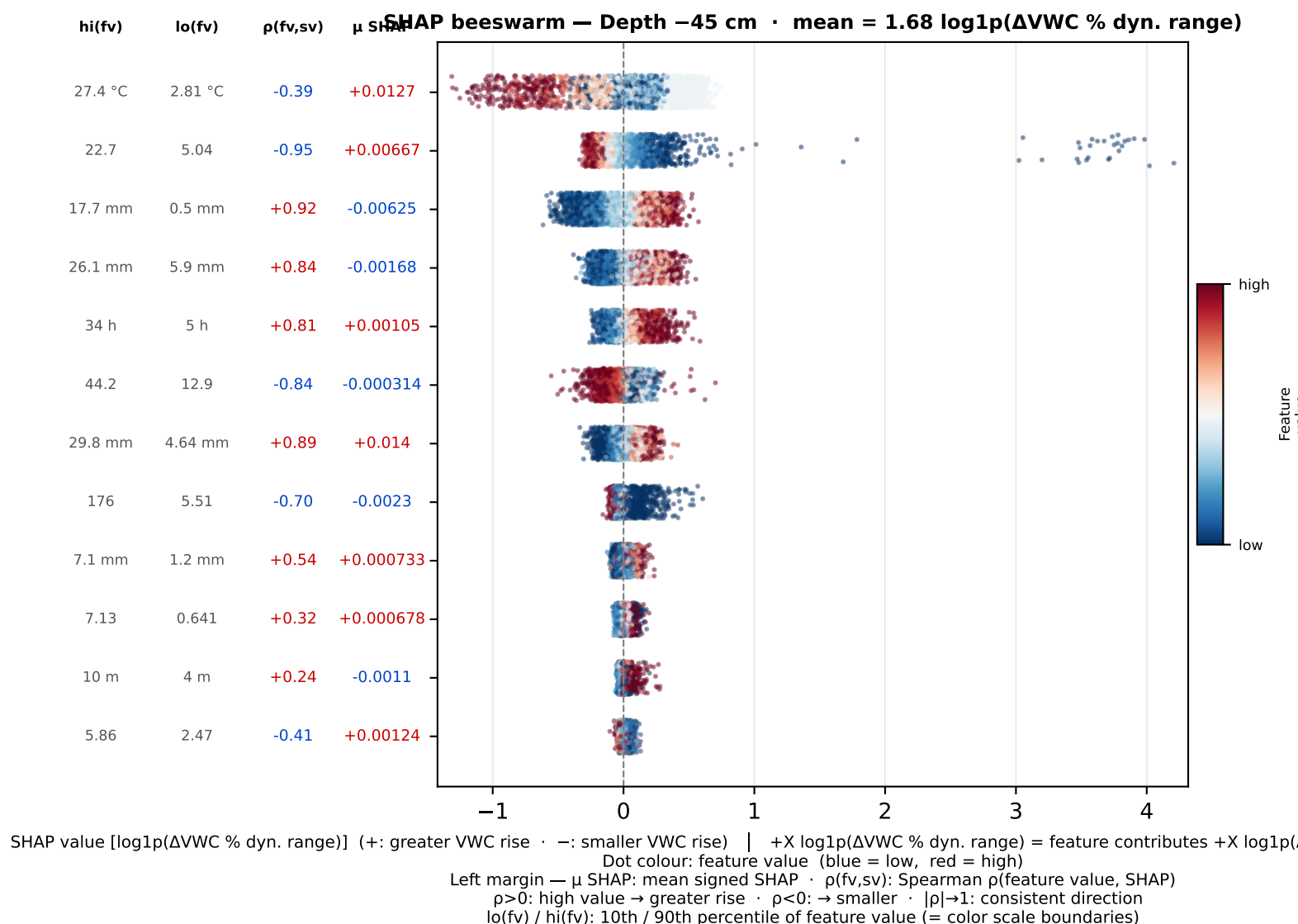
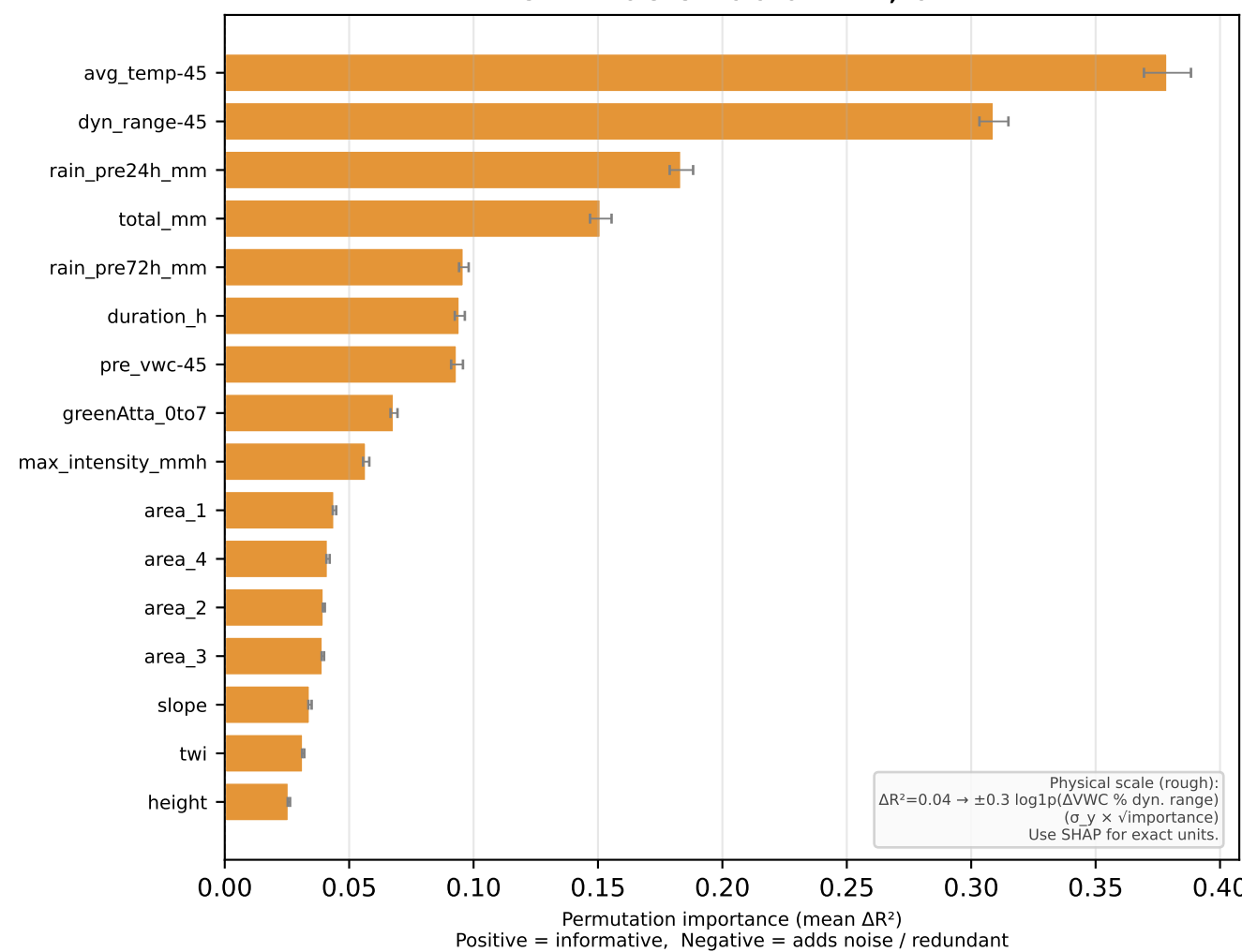
Depth -30 cm — permutation importance

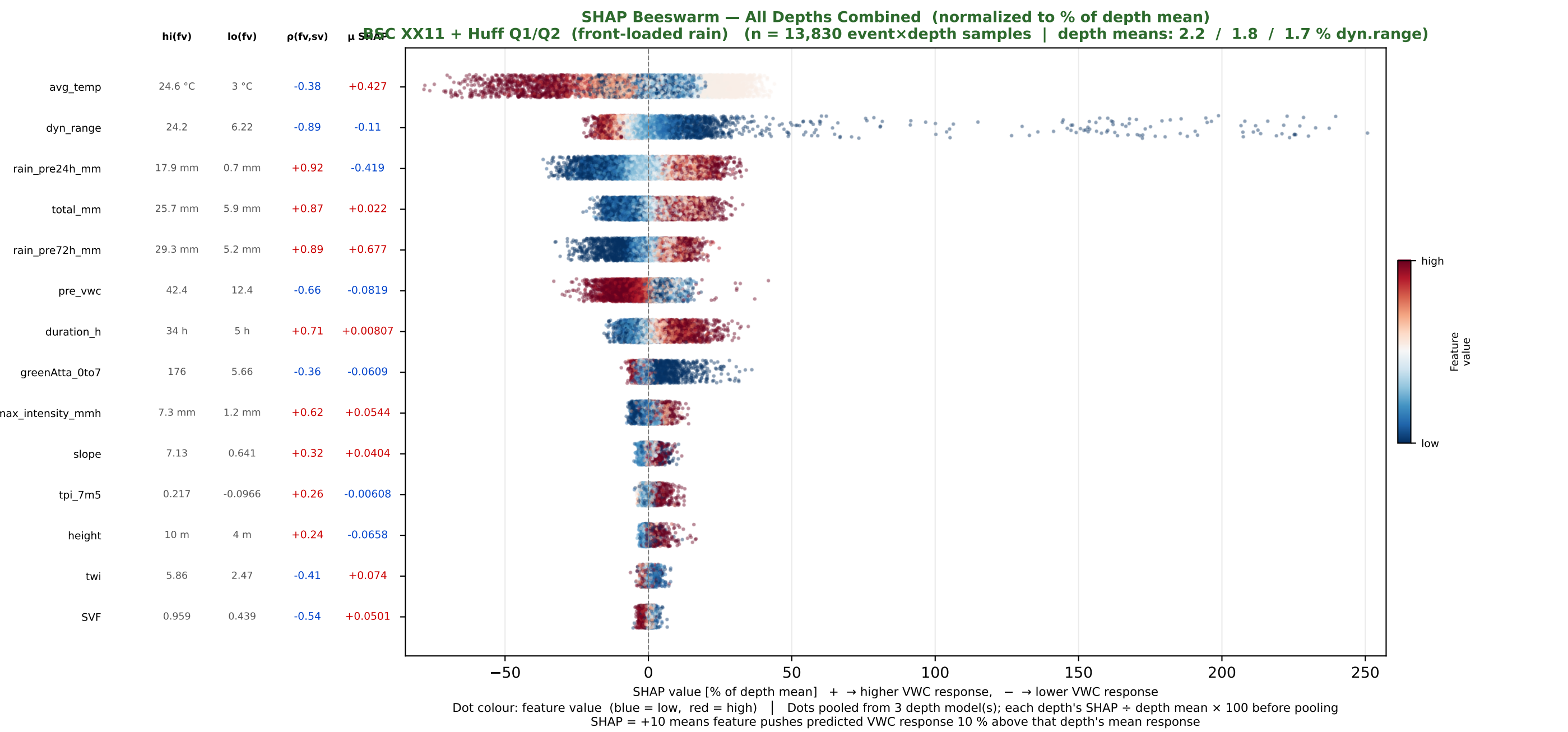
CV $R^2 = 0.554 \pm 0.014$ n = 4,867



Depth -45 cm — permutation importance

CV $R^2 = 0.515 \pm 0.020$ n = 4,464





HOW TO READ THIS CHART (Combined depth view – SHAP normalized to % of depth mean)

SHAP value (x-axis) – unit: % of depth-specific mean VWC response
Each depth's SHAP values were divided by that depth's training mean before pooling.
This makes all depths directly comparable regardless of their absolute response magnitude.

- +10 % → this feature pushes the predicted VWC response 10 % above that depth's mean.
- -10 % → this feature pulls the predicted VWC response 10 % below that depth's mean.

Per-depth means used for normalization (3 depth(s)):

Depth 1: mean = 2.2 % dyn.range	→ SHAP +10 % of mean = +0.2 % dyn.range	≈ +0.039 % abs. VWC	(at dyn.range = 18 %)
Depth 2: mean = 1.8 % dyn.range	→ SHAP +10 % of mean = +0.2 % dyn.range	≈ +0.033 % abs. VWC	(at dyn.range = 18 %)
Depth 3: mean = 1.7 % dyn.range	→ SHAP +10 % of mean = +0.2 % dyn.range	≈ +0.030 % abs. VWC	(at dyn.range = 18 %)

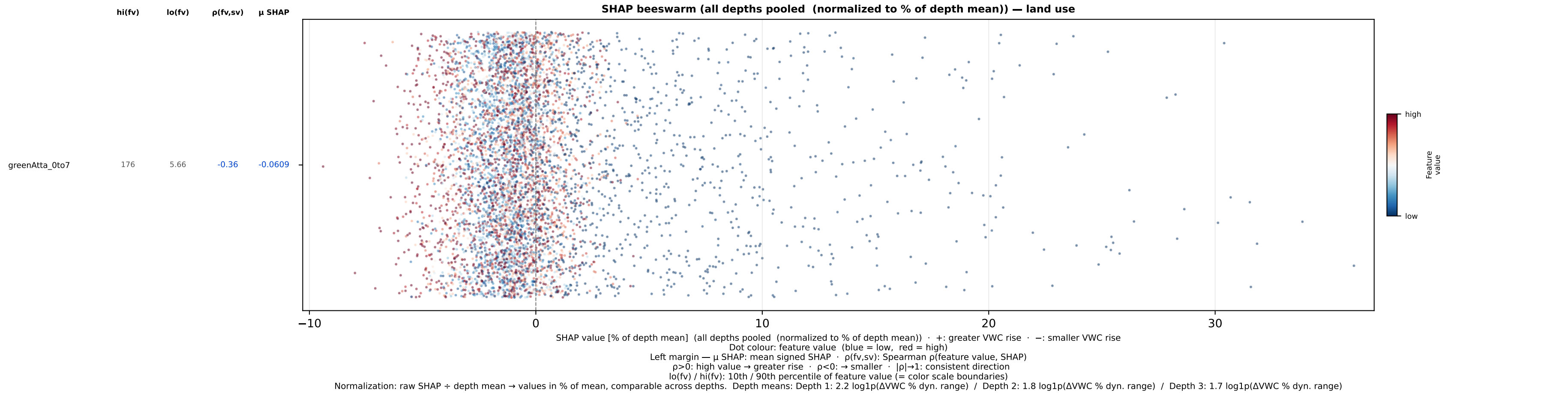
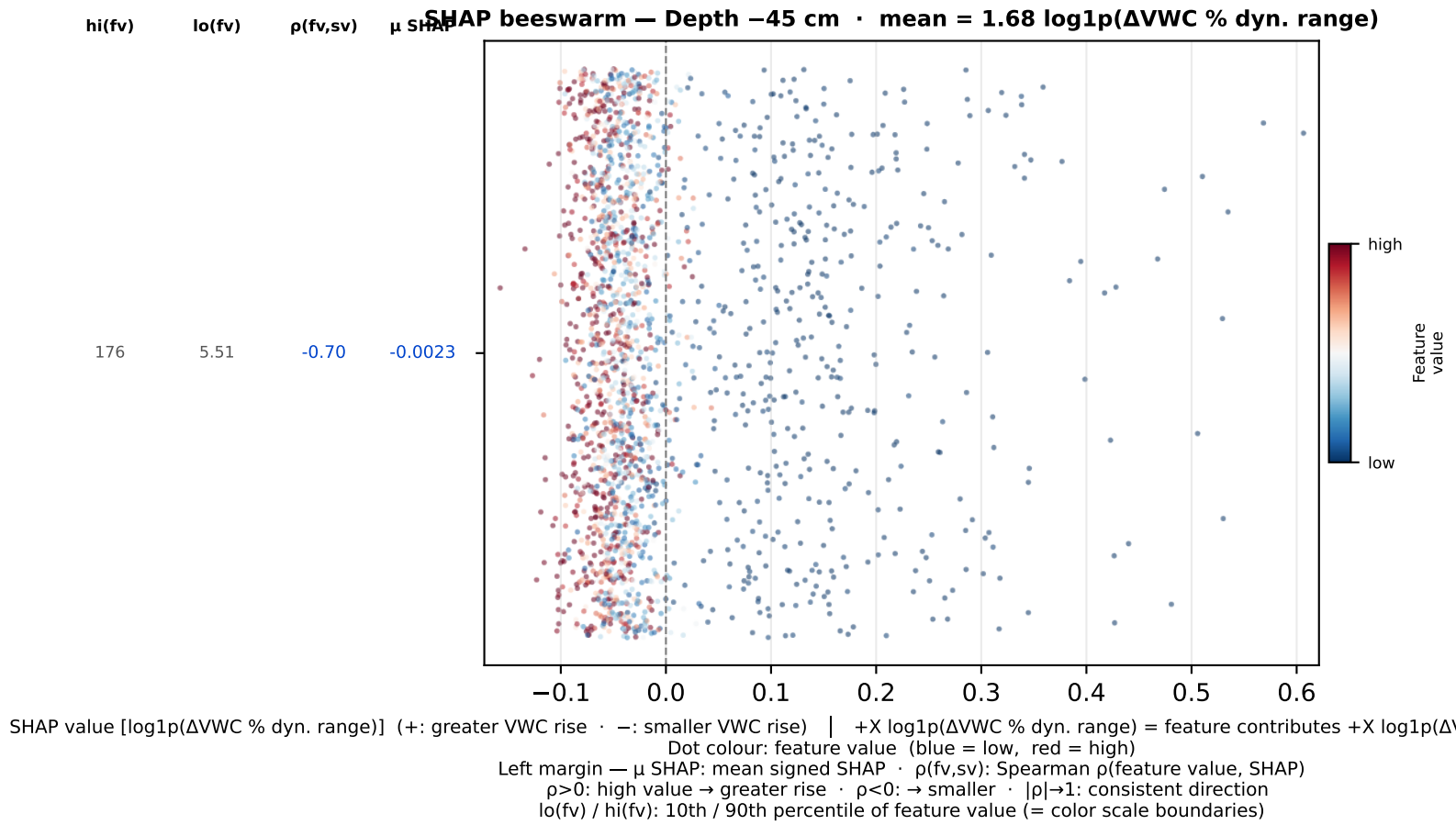
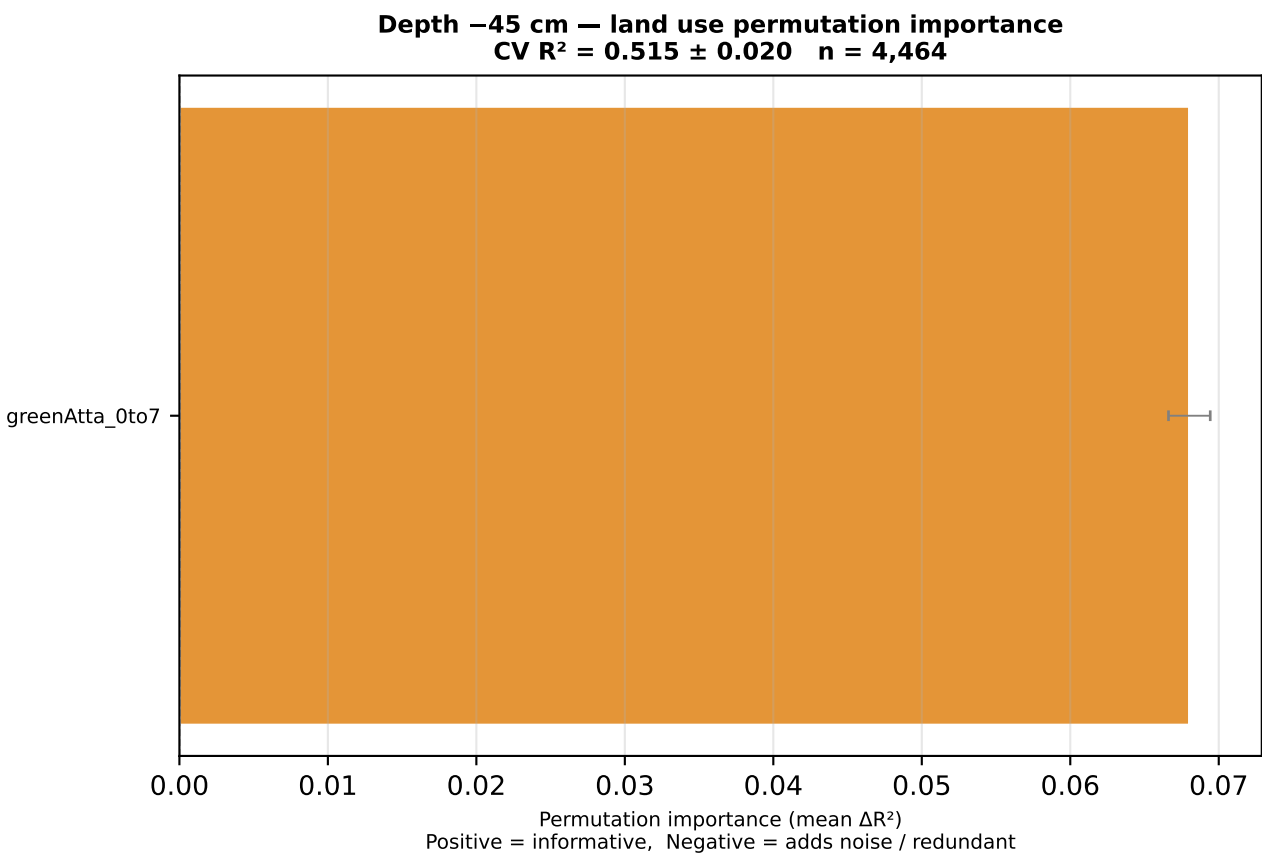
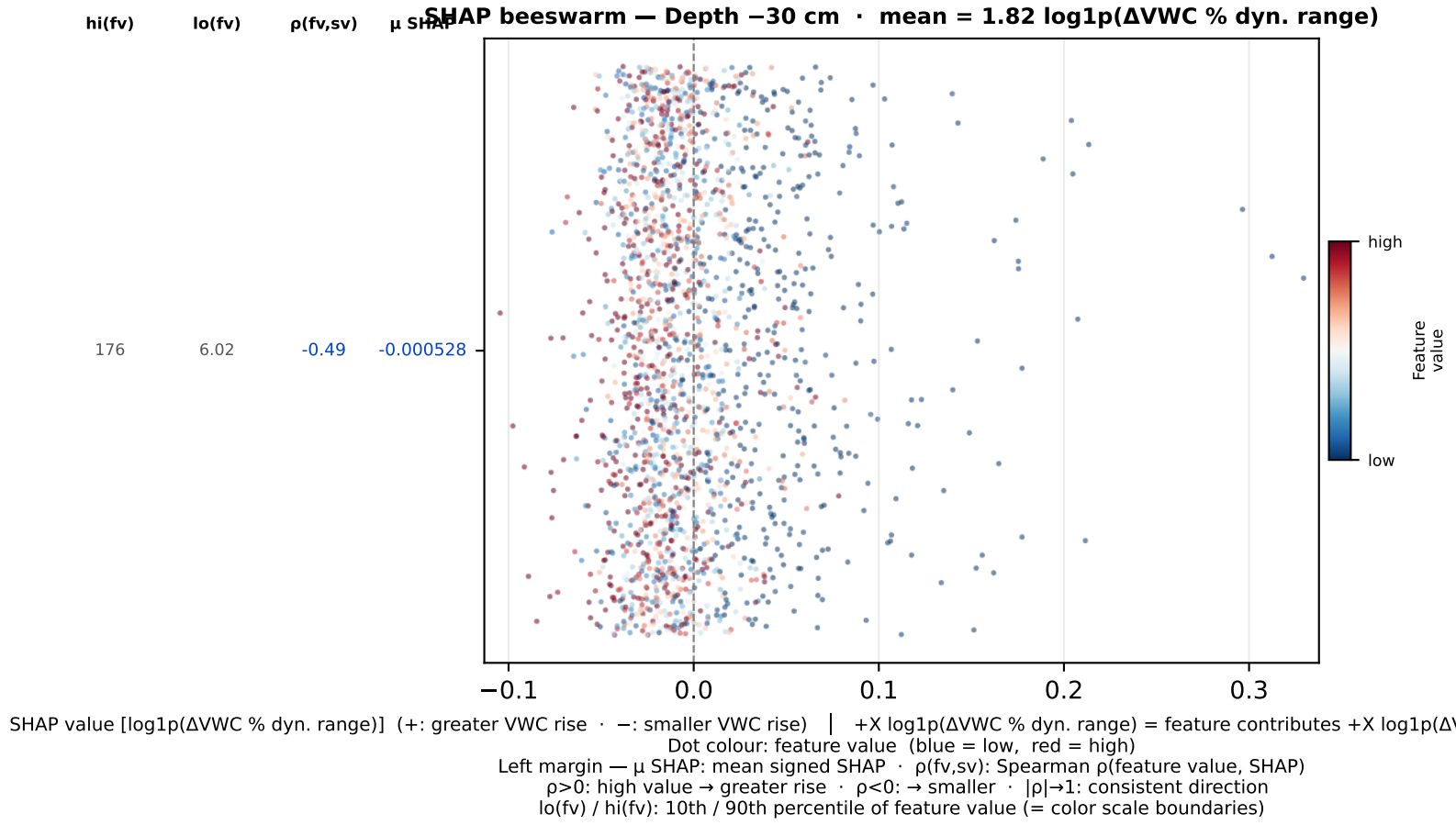
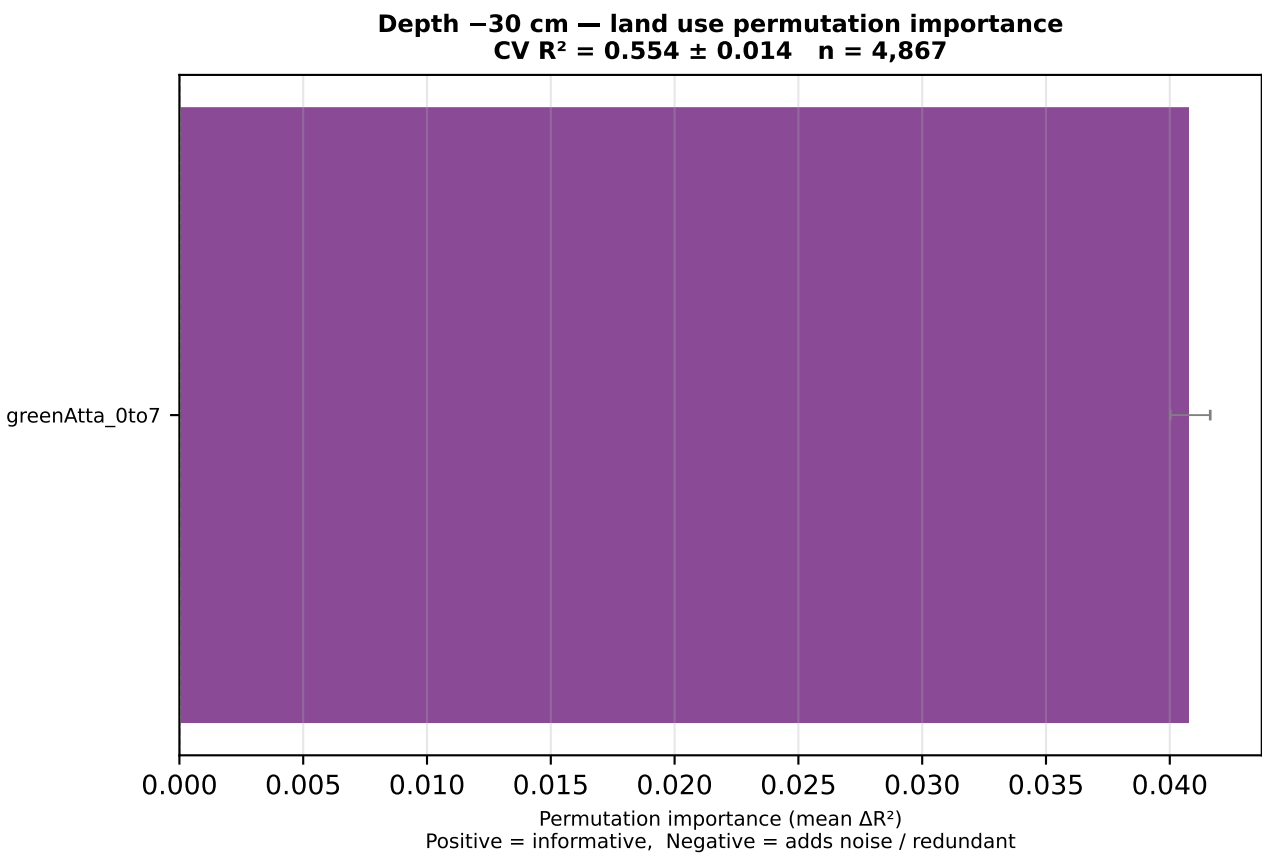
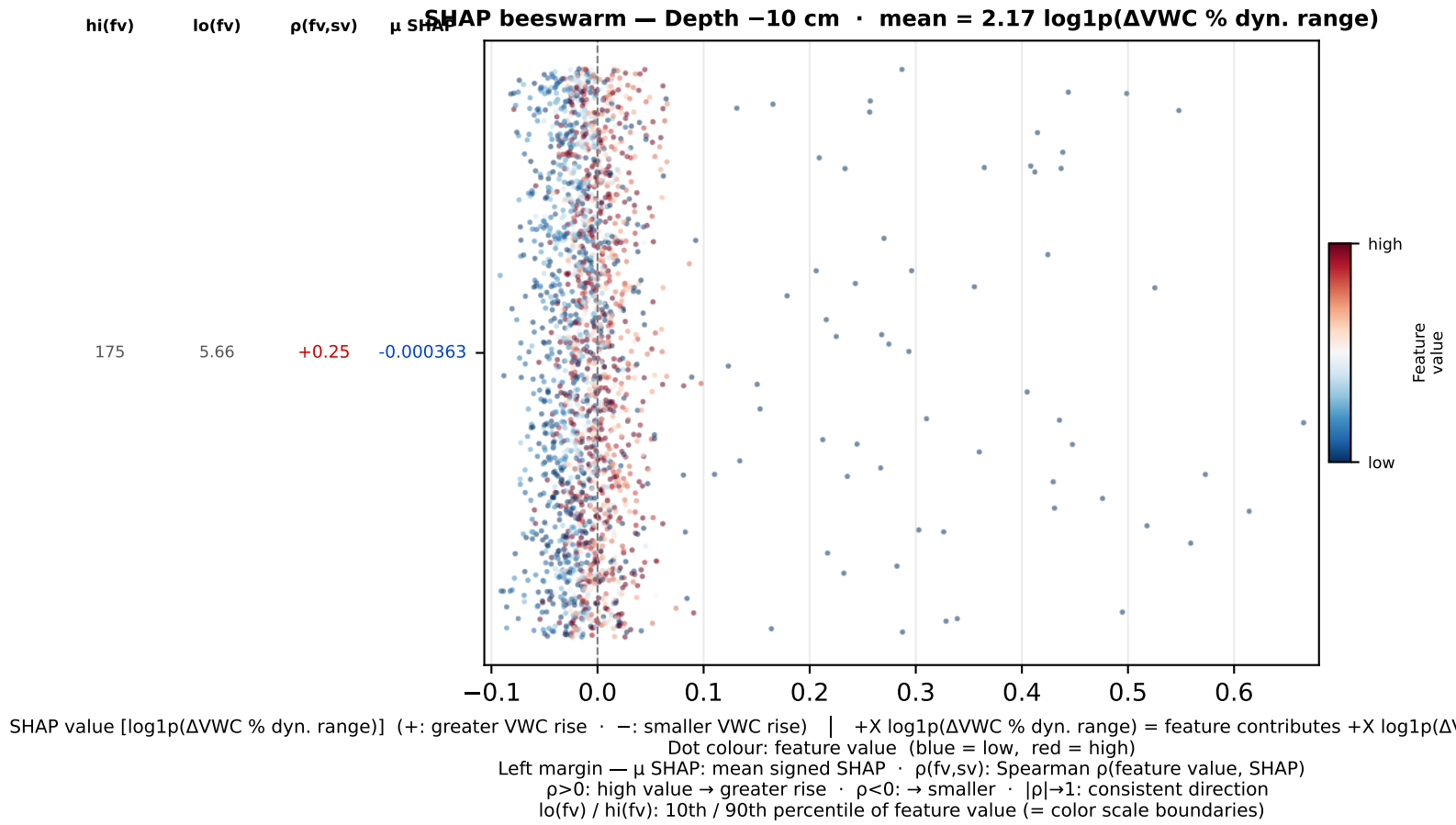
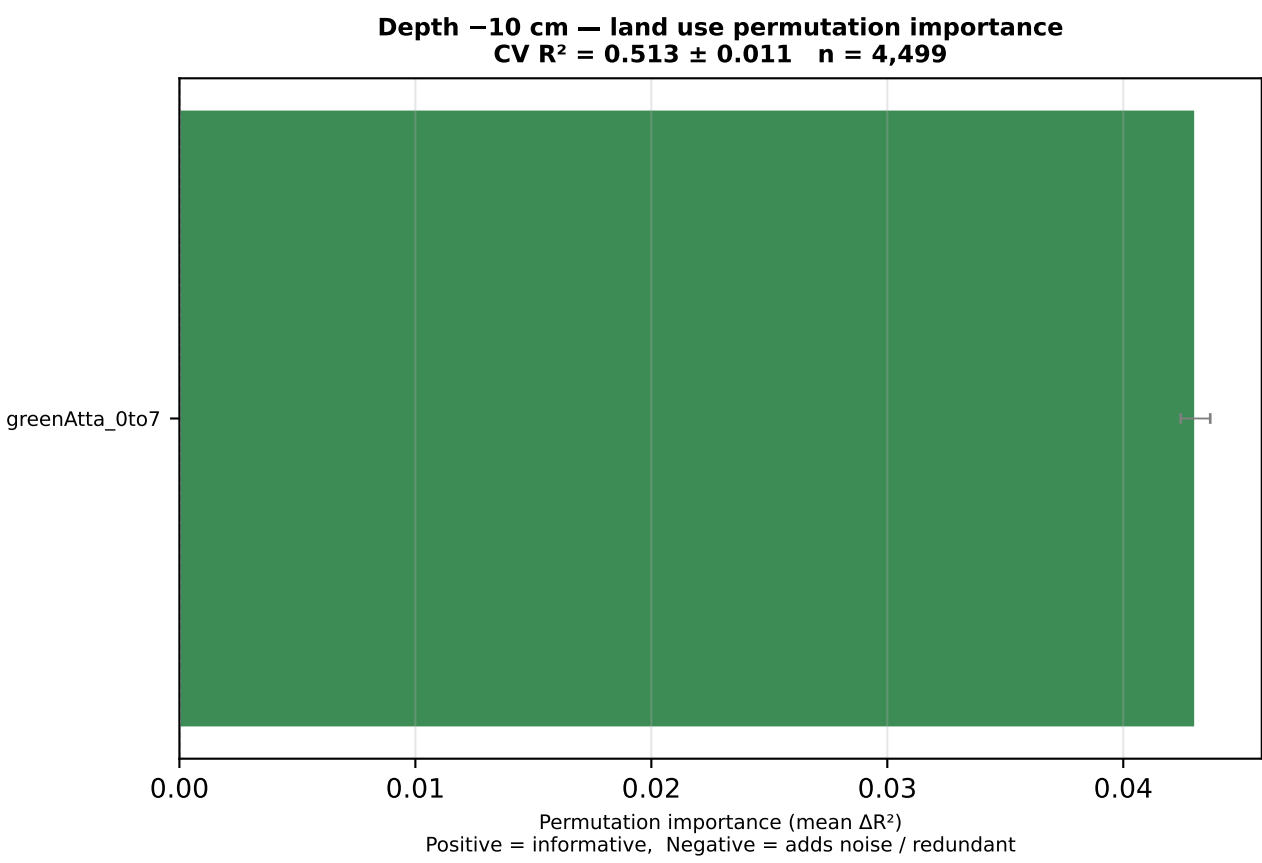
Example – top feature "avg_temp" (normalized SHAP IQR: -12.5 to +17.5 % of depth mean):
Low feature value → SHAP ≈ -12.5% High feature value → SHAP ≈ +17.5%
Difference: ~30.0 % of each depth's mean → in physical units per depth:

Depth 1 (mean 2.2 % dyn.range):	Δ ≈ 0.7 % dyn.range	≈ 0.117 % abs. VWC
Depth 2 (mean 1.8 % dyn.range):	Δ ≈ 0.5 % dyn.range	≈ 0.098 % abs. VWC
Depth 3 (mean 1.7 % dyn.range):	Δ ≈ 0.5 % dyn.range	≈ 0.091 % abs. VWC

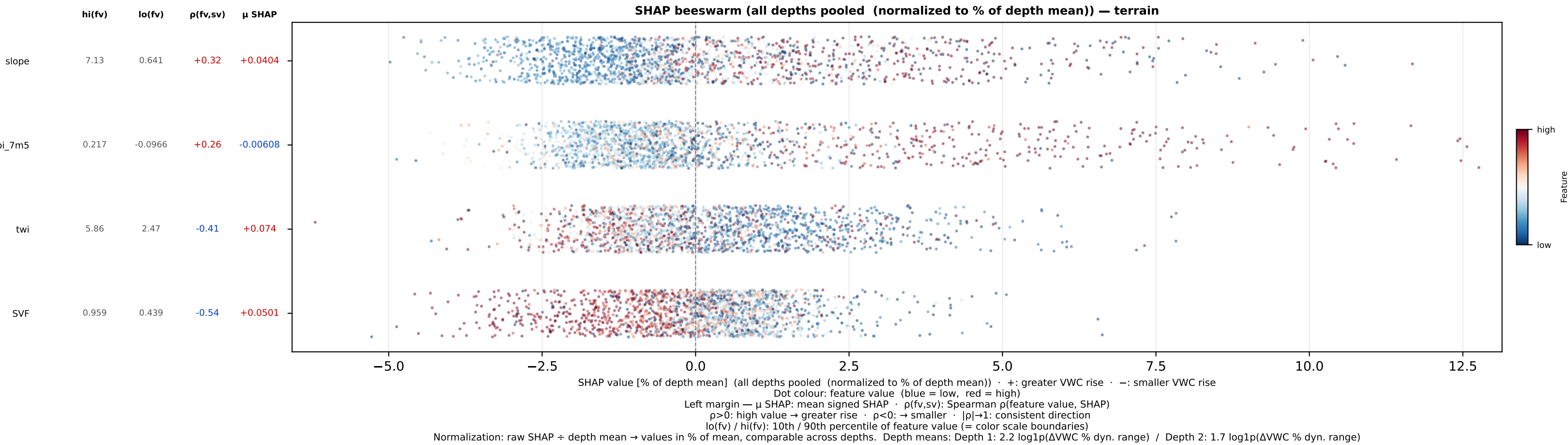
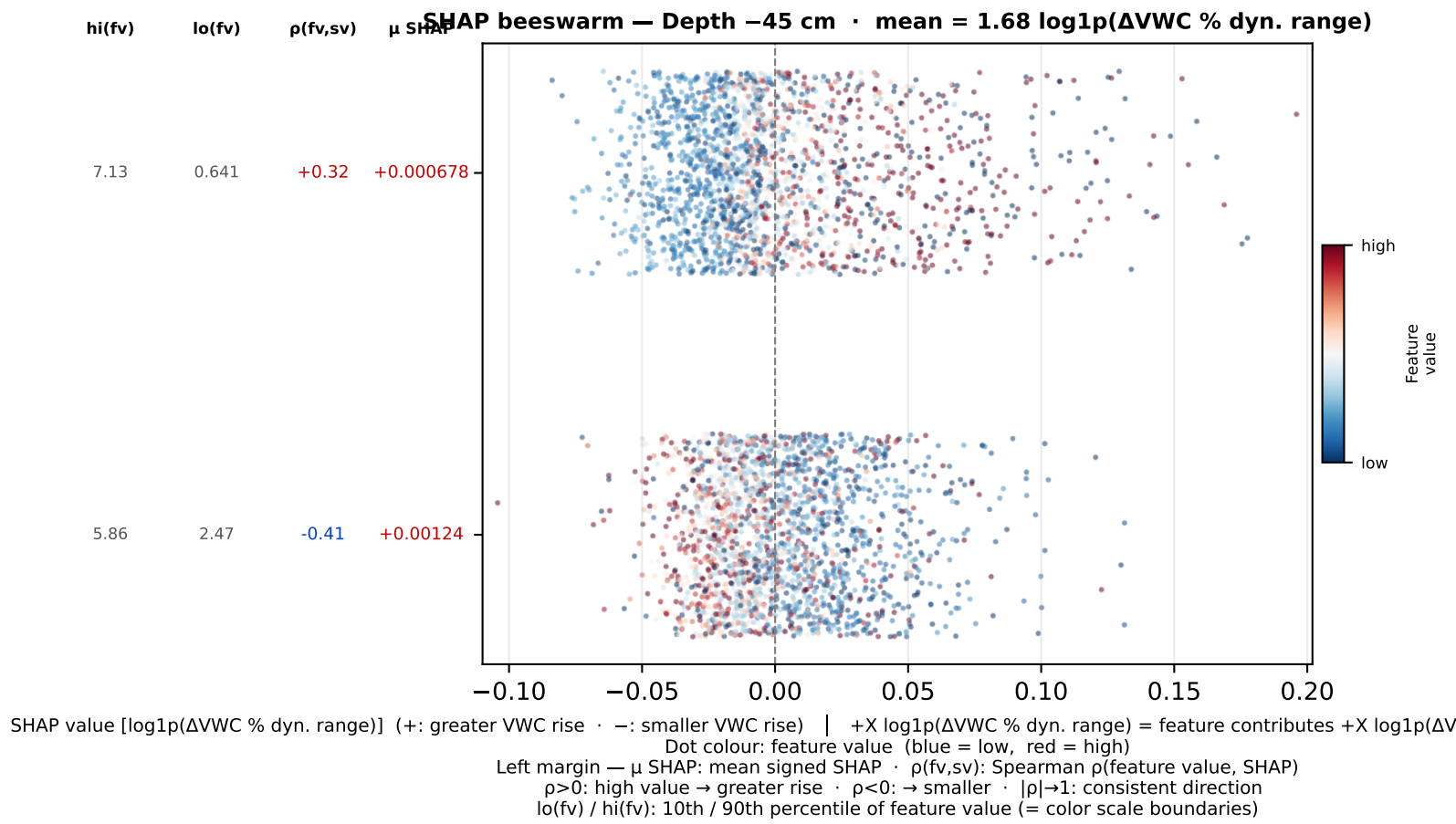
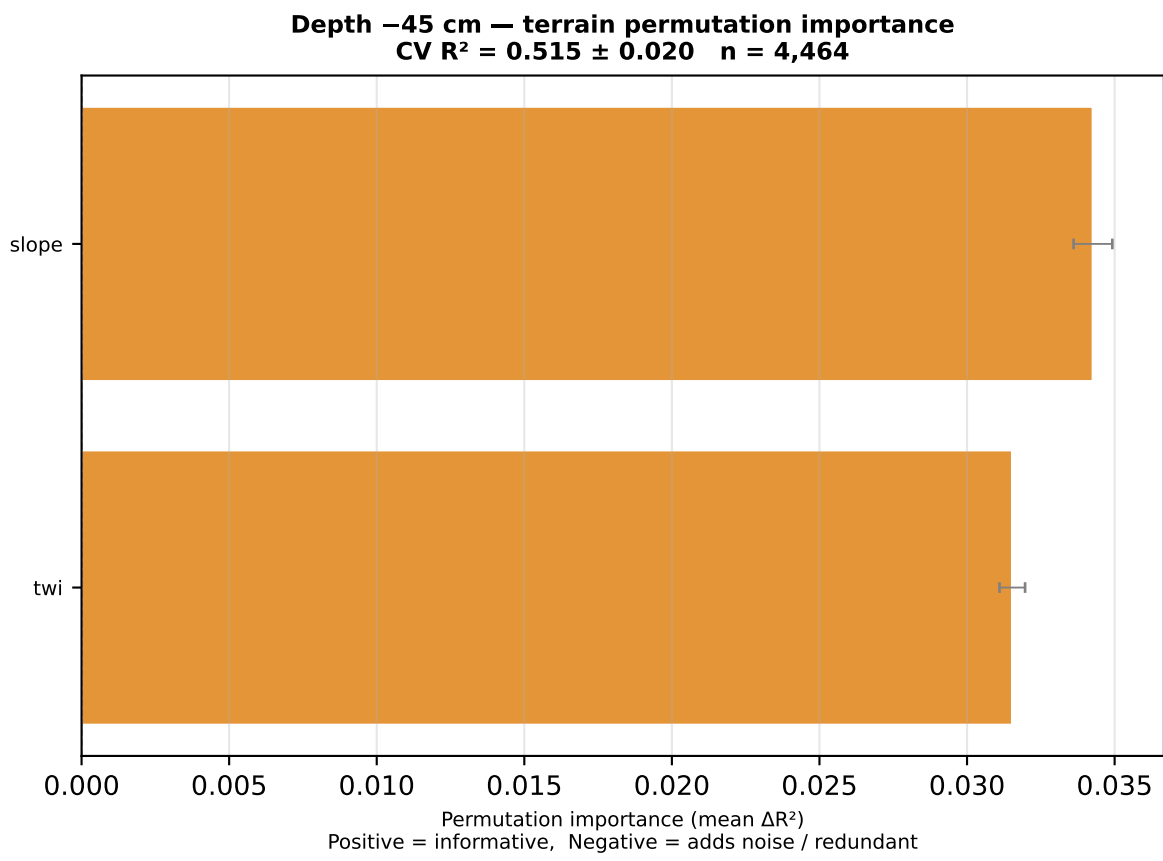
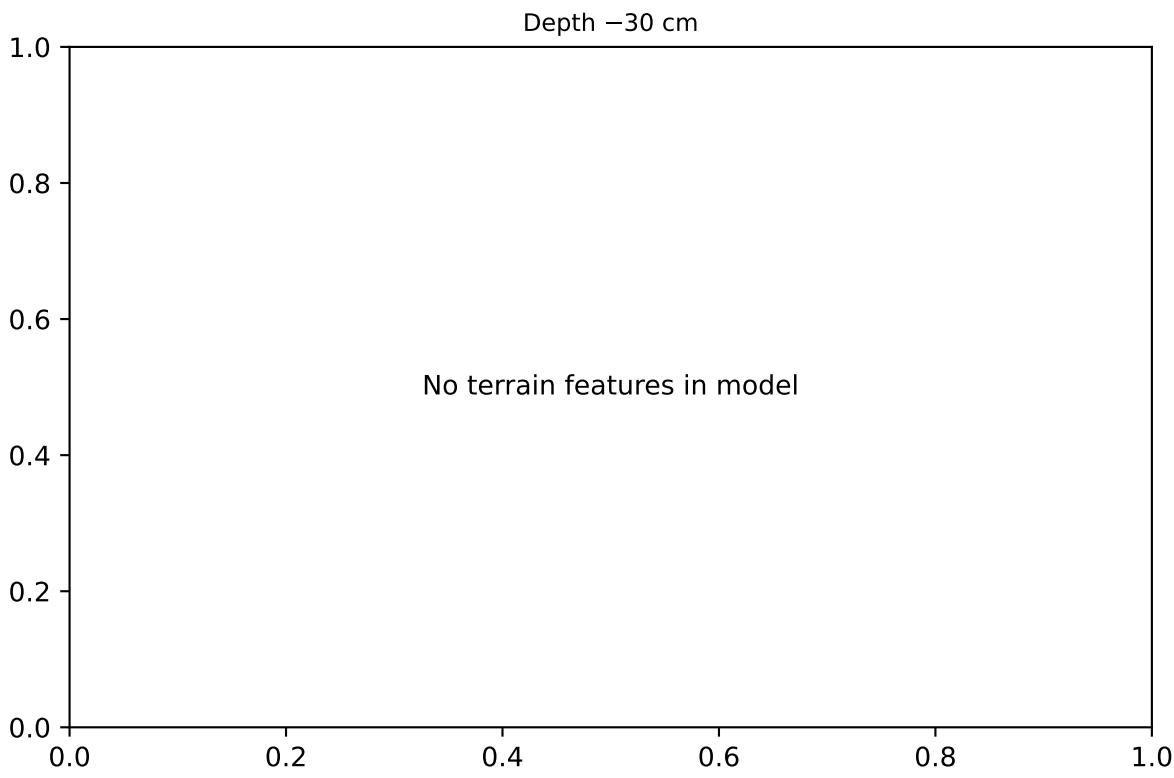
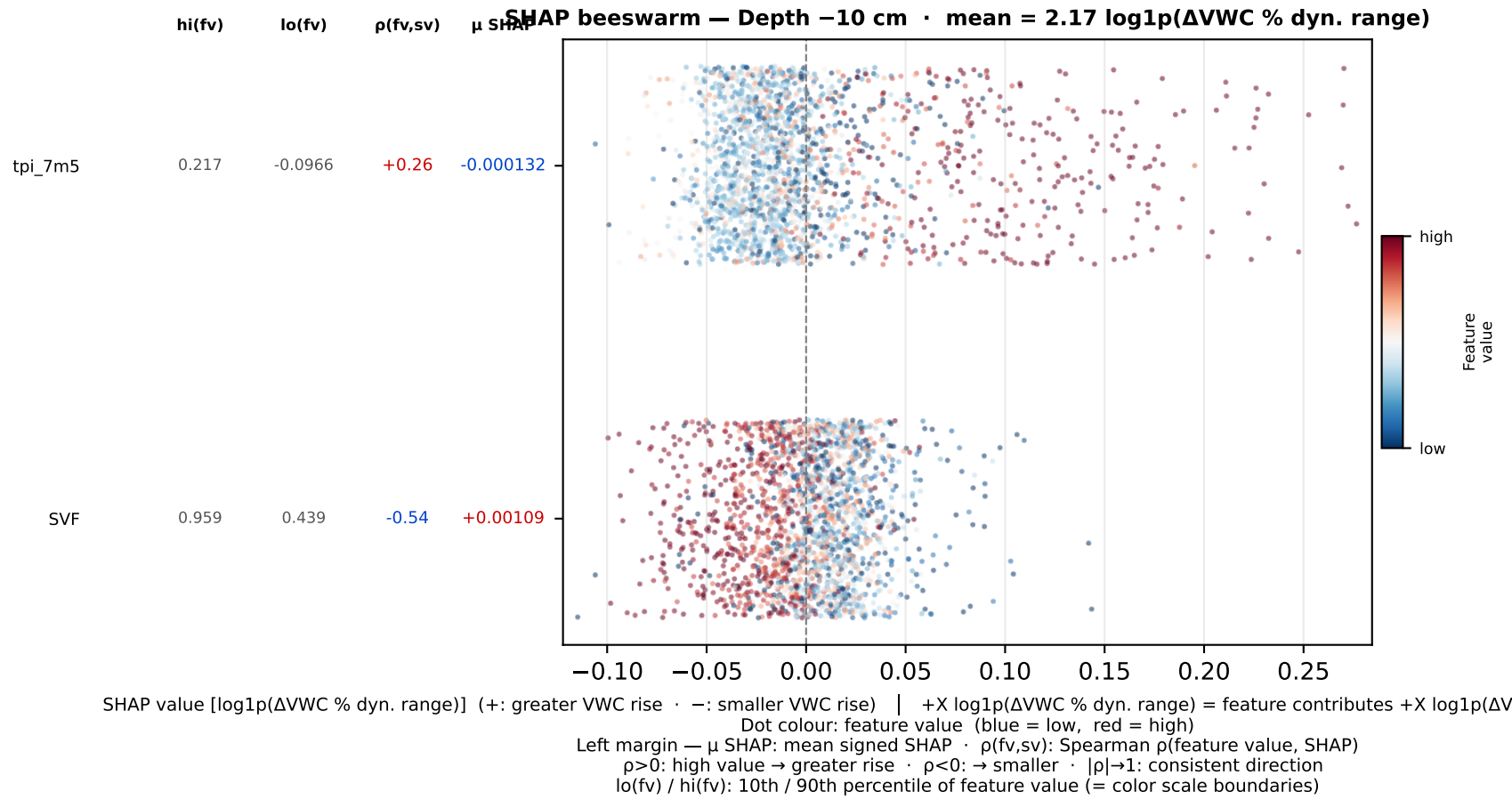
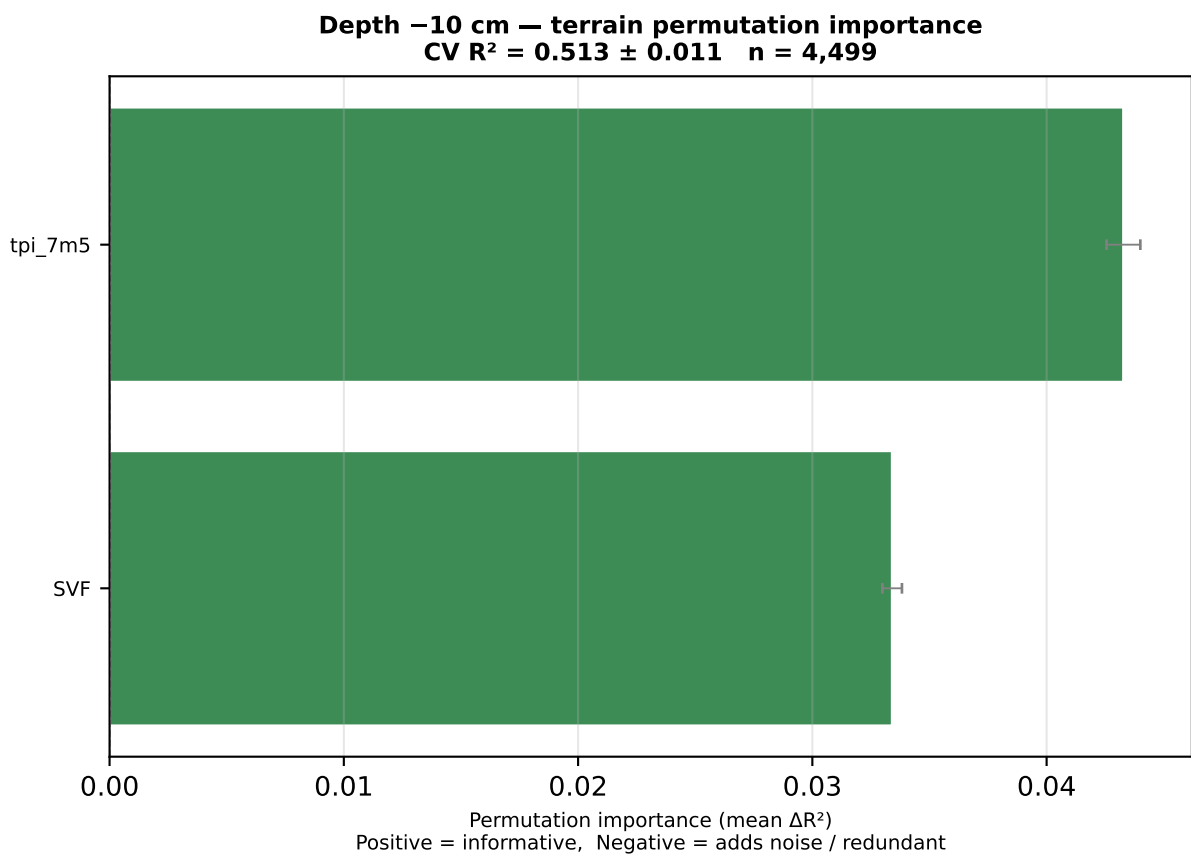
Total samples: 13,830 pooled across 3 depth(s) | μ SHAP and ρ in the left margin are computed on the normalized values.

Permutation importance (bar charts on preceding pages) – unit: mean ΔR² (NOT in % dyn.range!)
Physical scale estimate per depth: contribution ≈ σ_y × √importance
(use SHAP values, not permutation importance, for direct attribution in physical units)

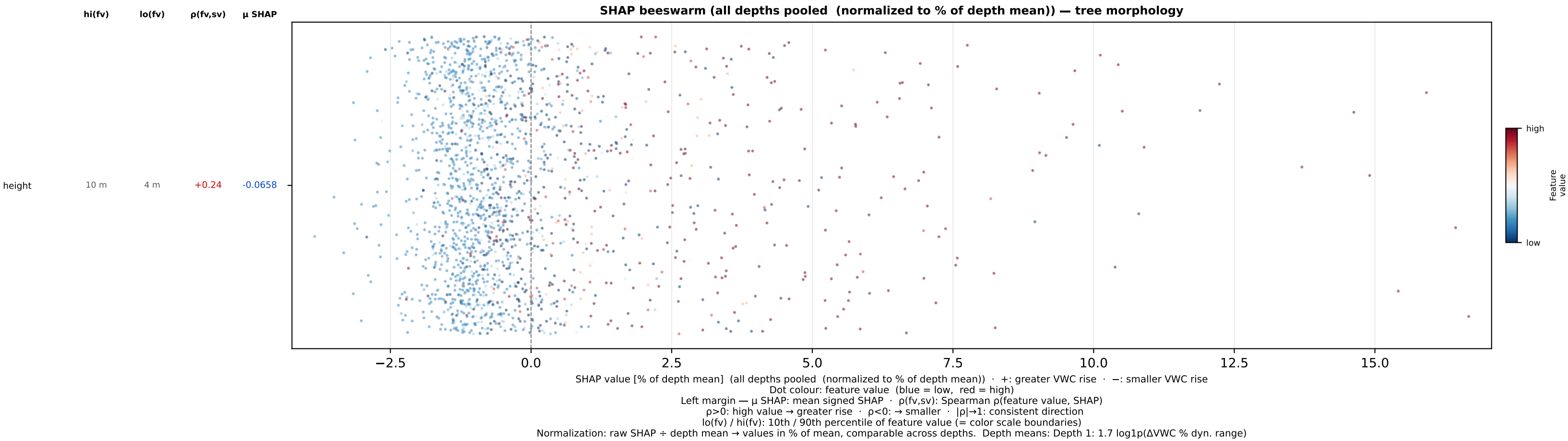
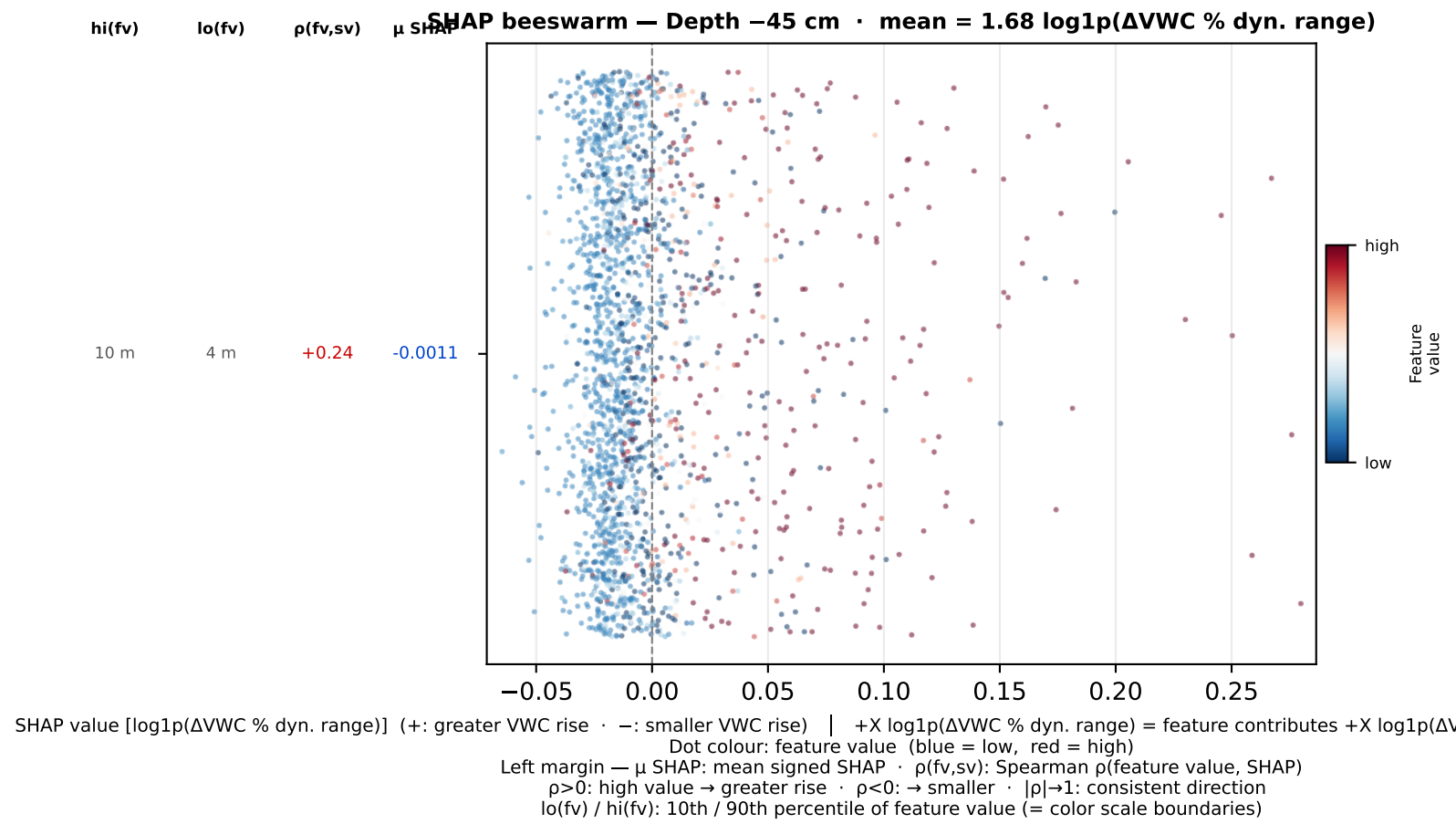
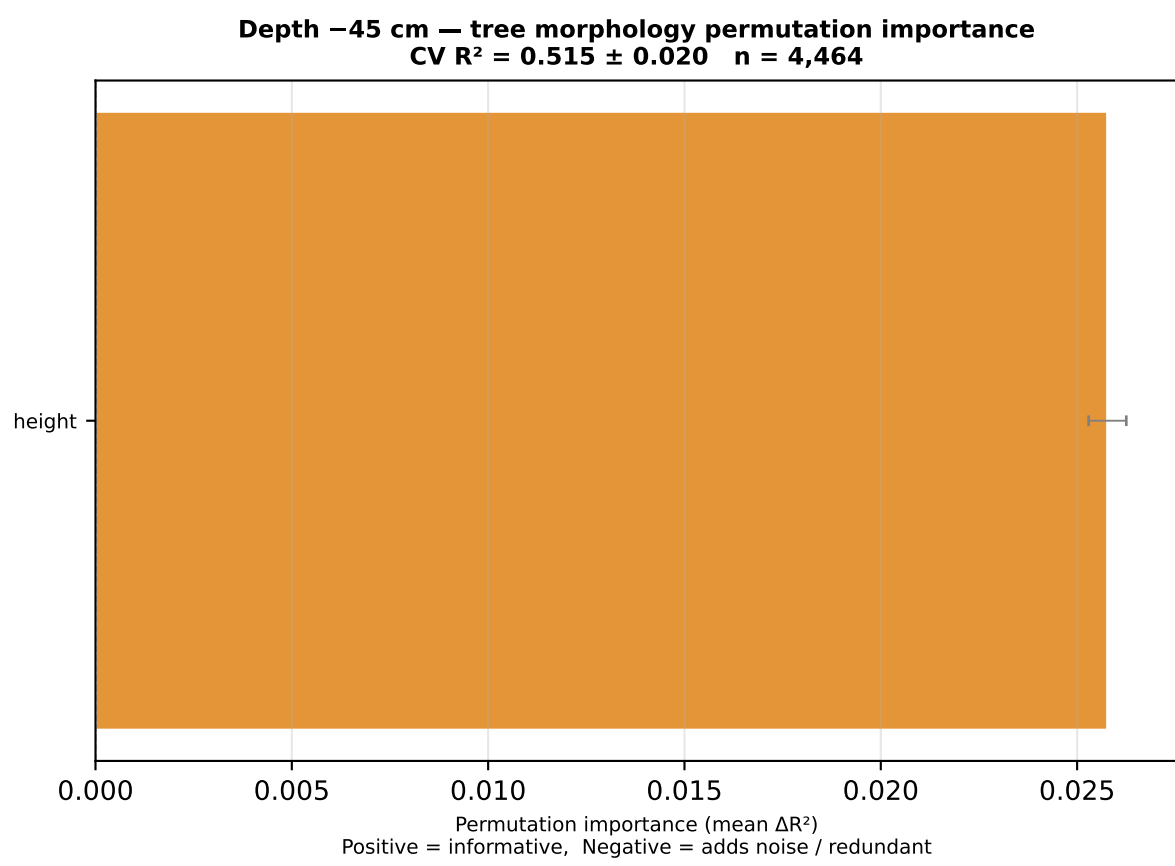
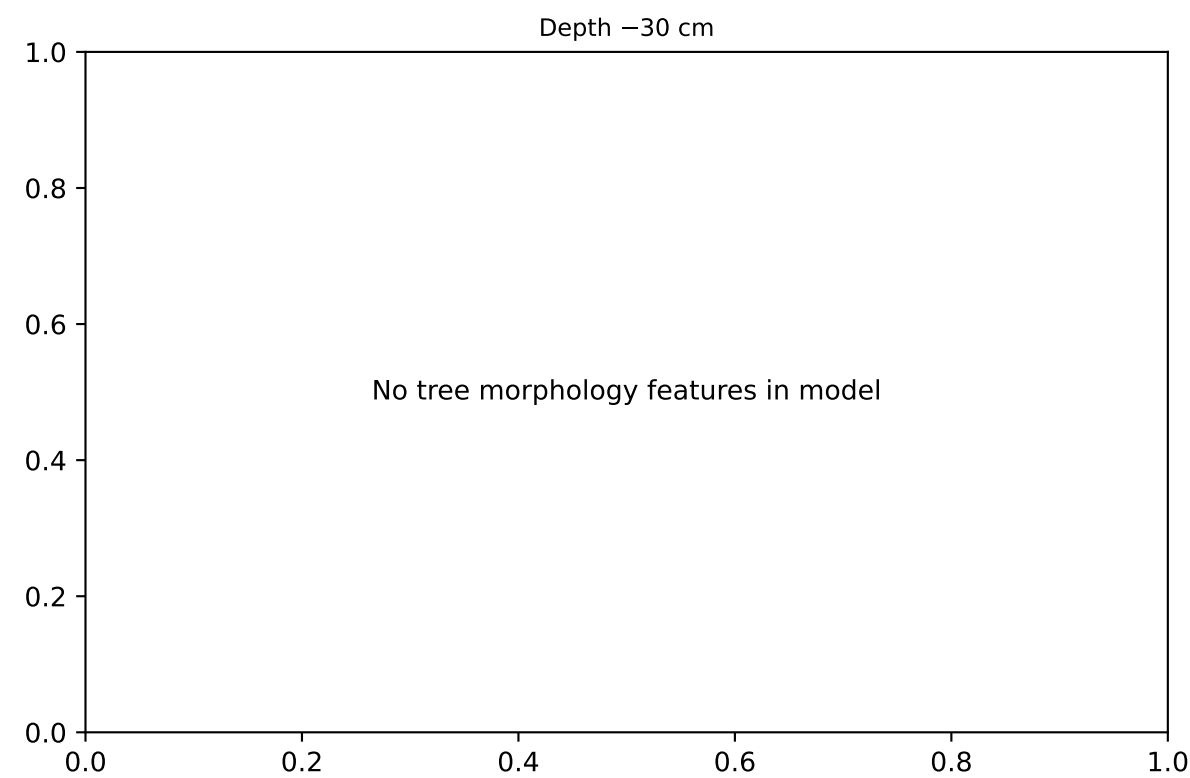
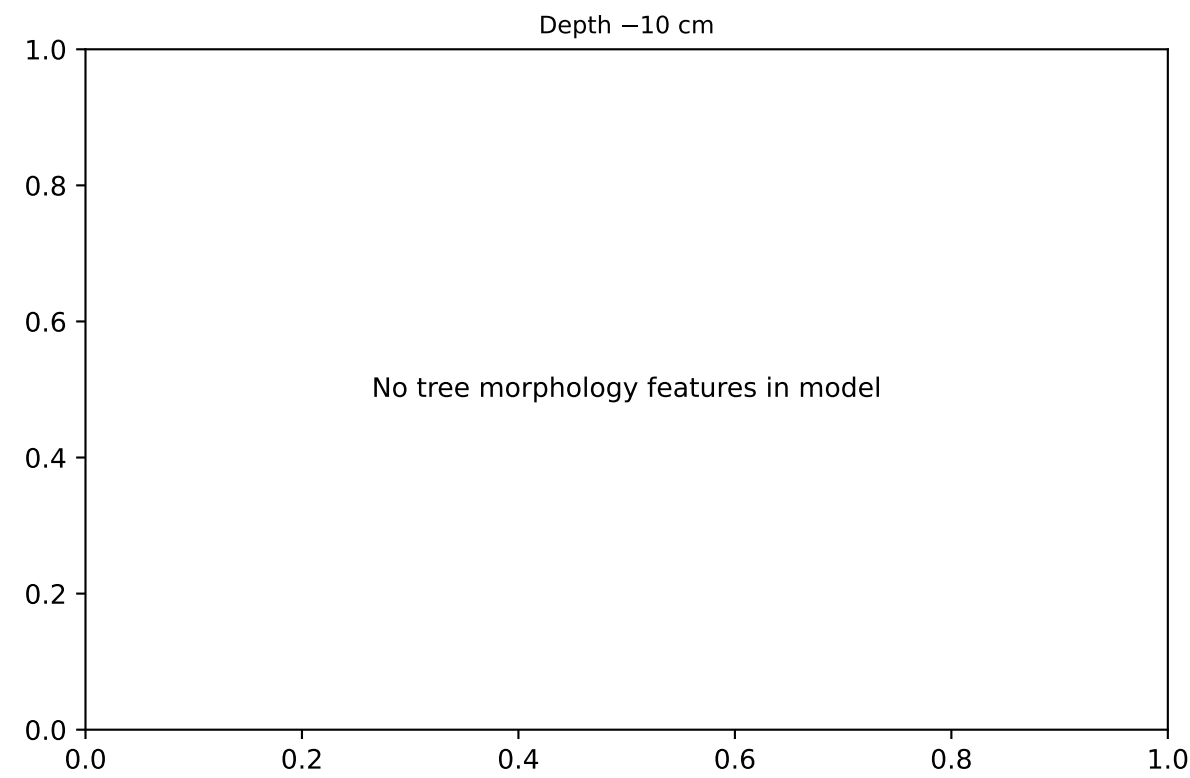
Land Use Feature Analysis — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: soil moisture response [log1p(ΔVWC % dyn. range)] per depth (10,911 events)
Features: green (combined) / greenAttached / greenDetached / buildings / sealedSurface (2.5 / 5 / 7.5 m rings, 0→5 m + 0→7.5 m aggregates, crown-diameter)



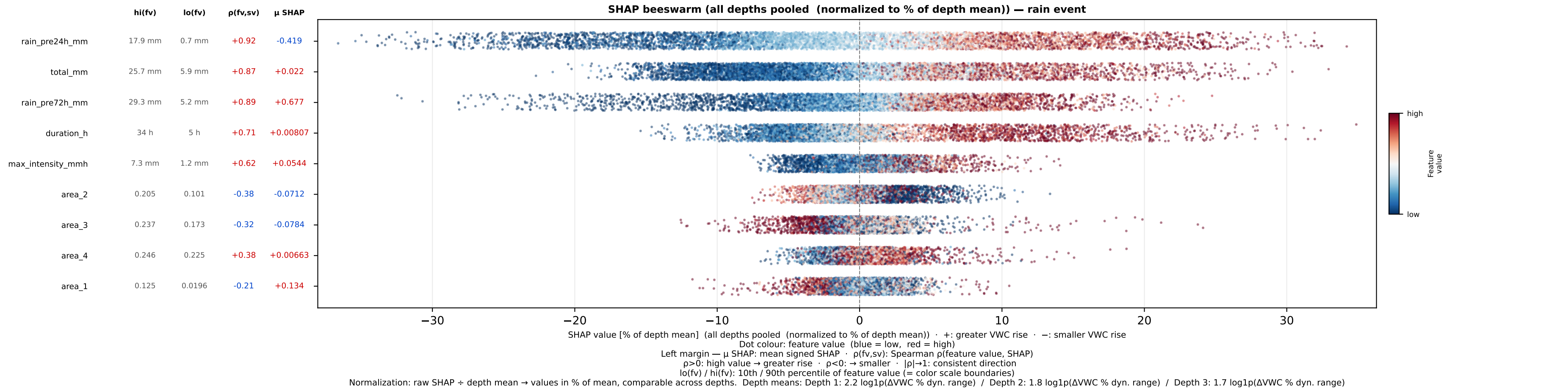
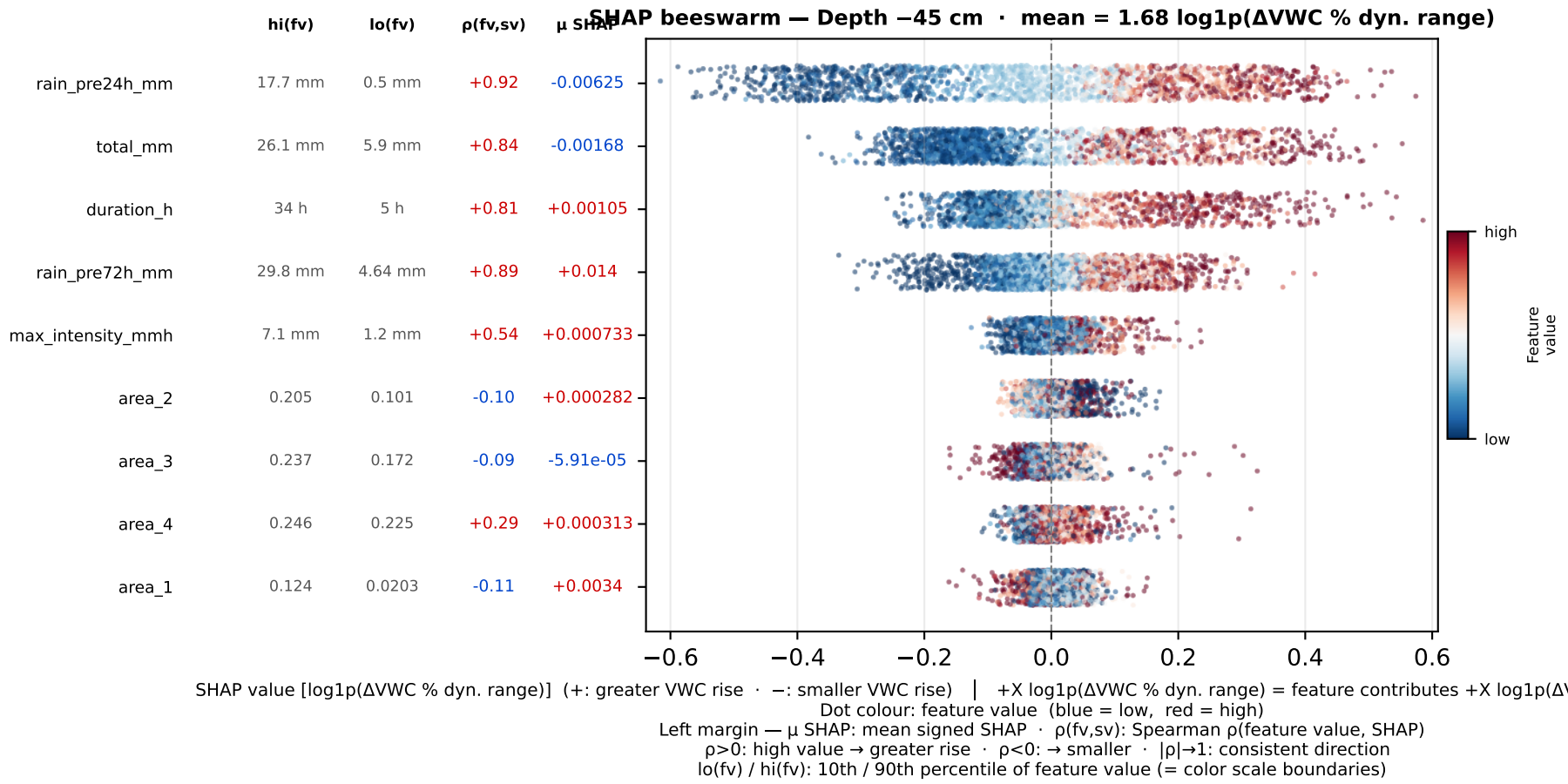
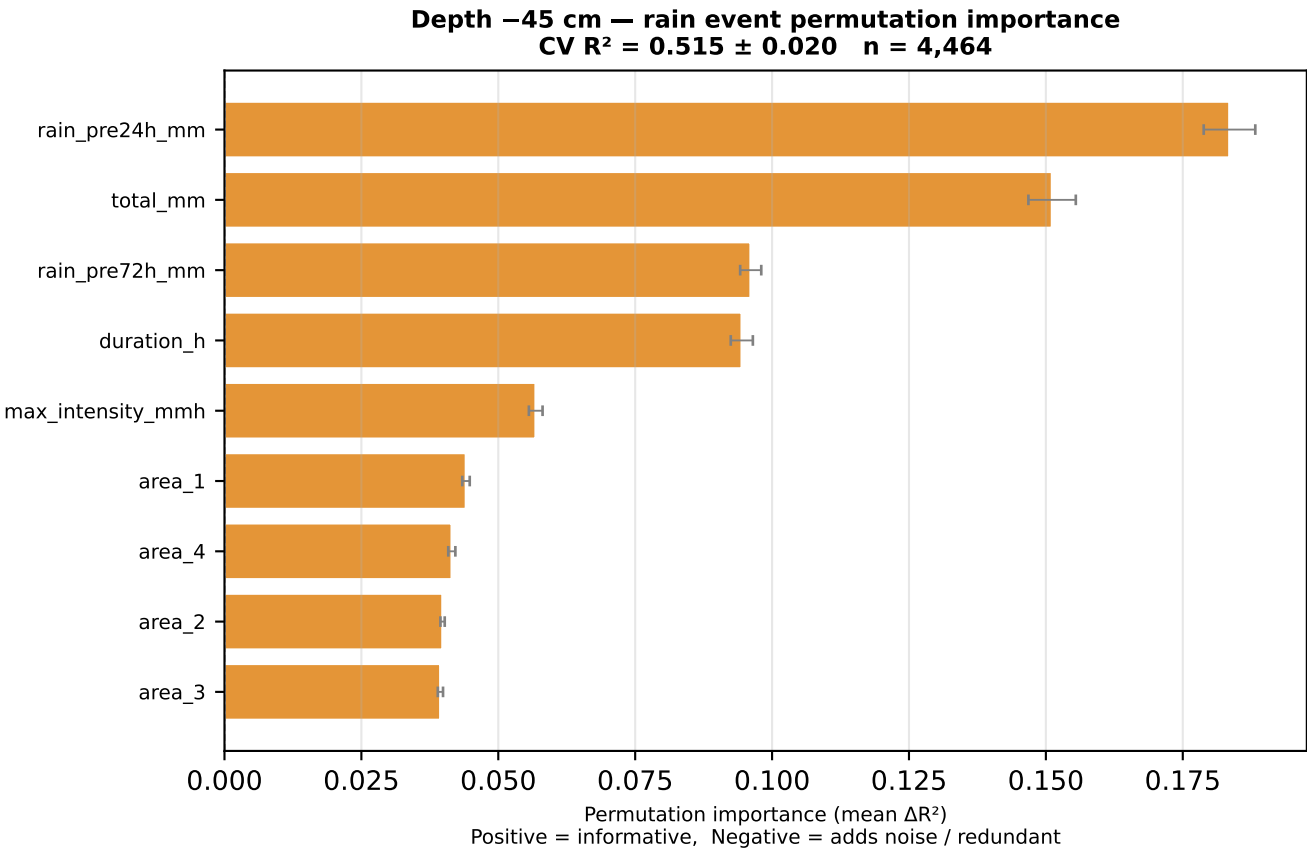
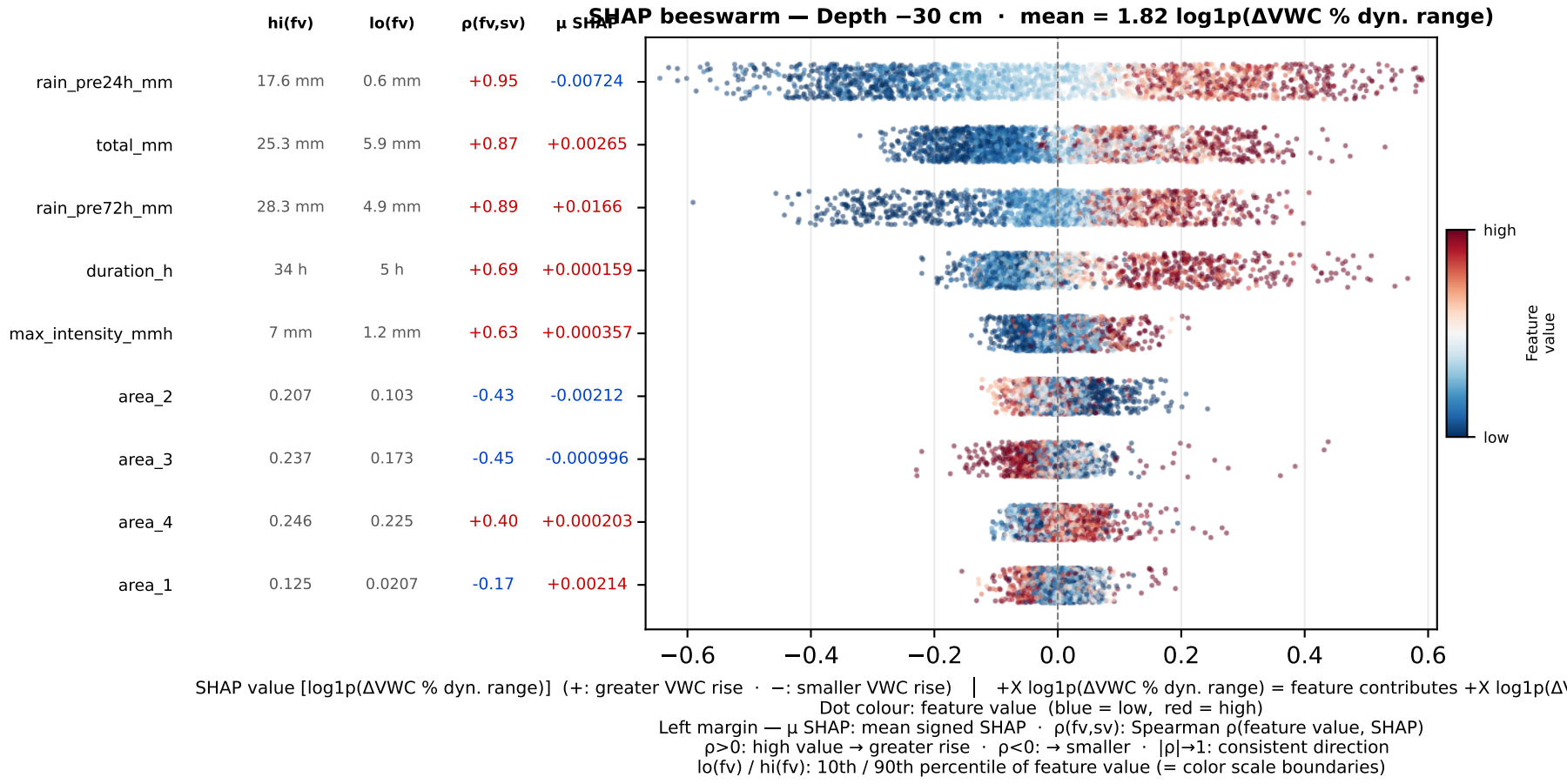
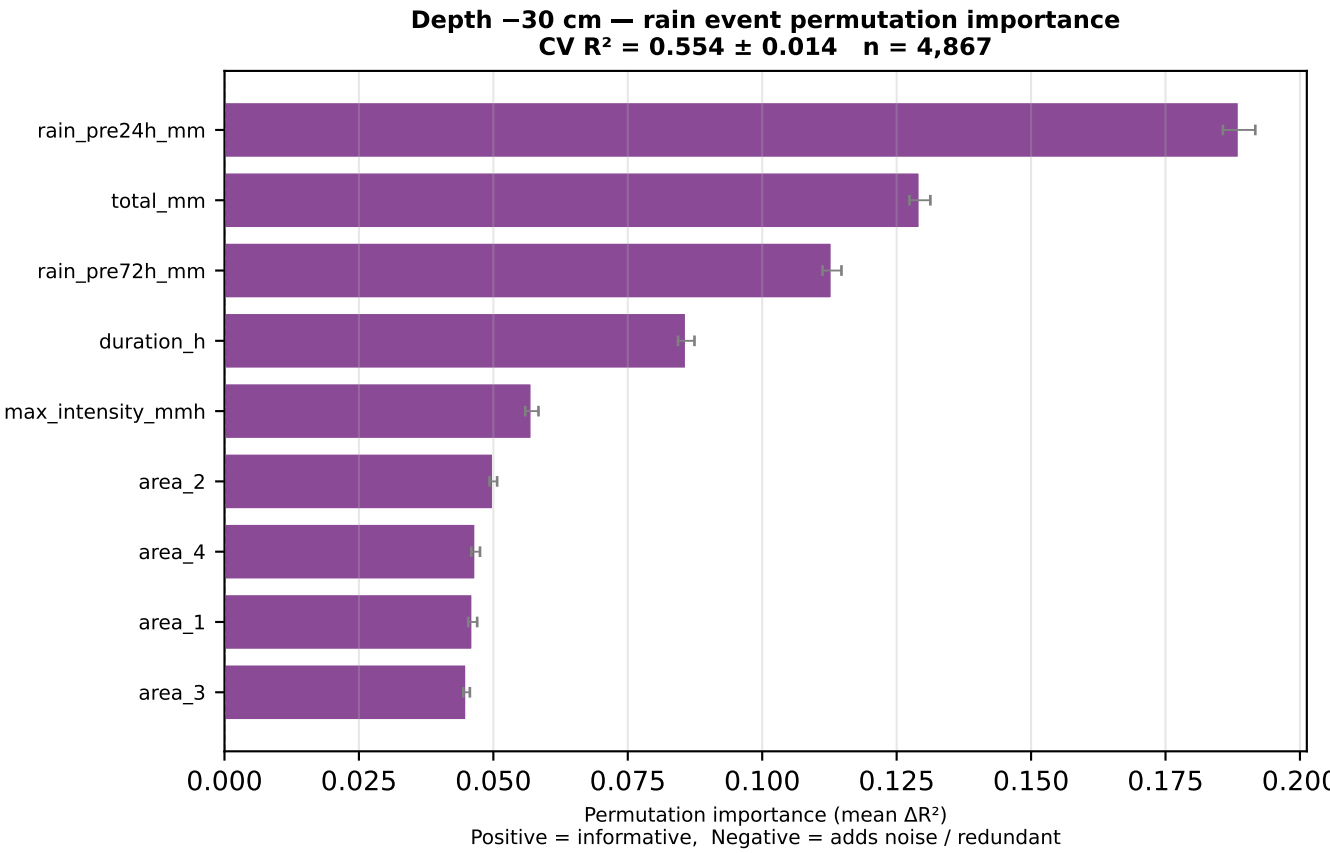
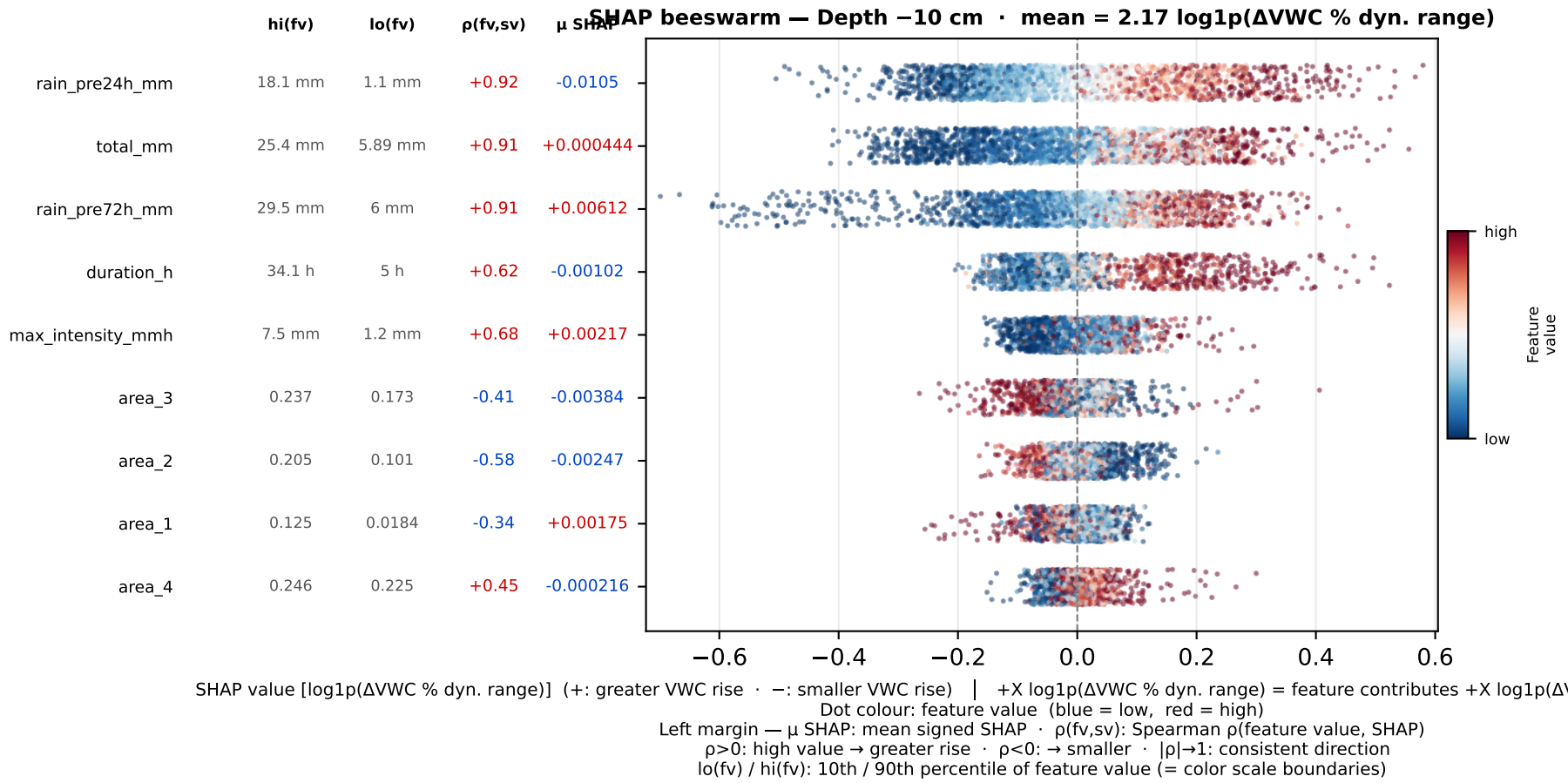
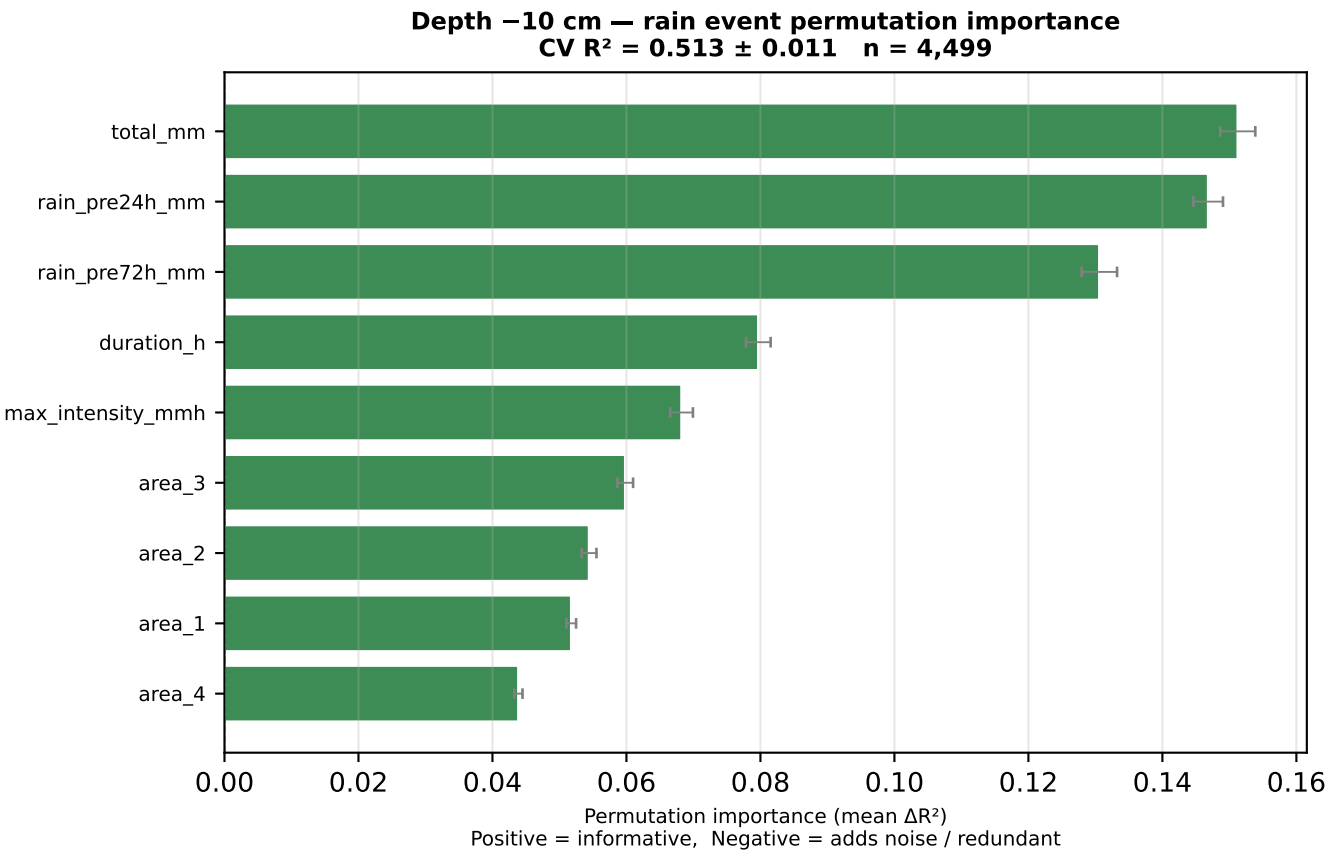
Terrain Feature Analysis — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (10,911 events)
Features: TWI · TPI (2.5/5/7.5 m + crown-diam) · slope · amsl · flow accumulation · SVF · depth to groundwater



Tree Morphology Feature Analysis — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (10,911 events)
Features: crown diameter · height · stem diameter · vitality



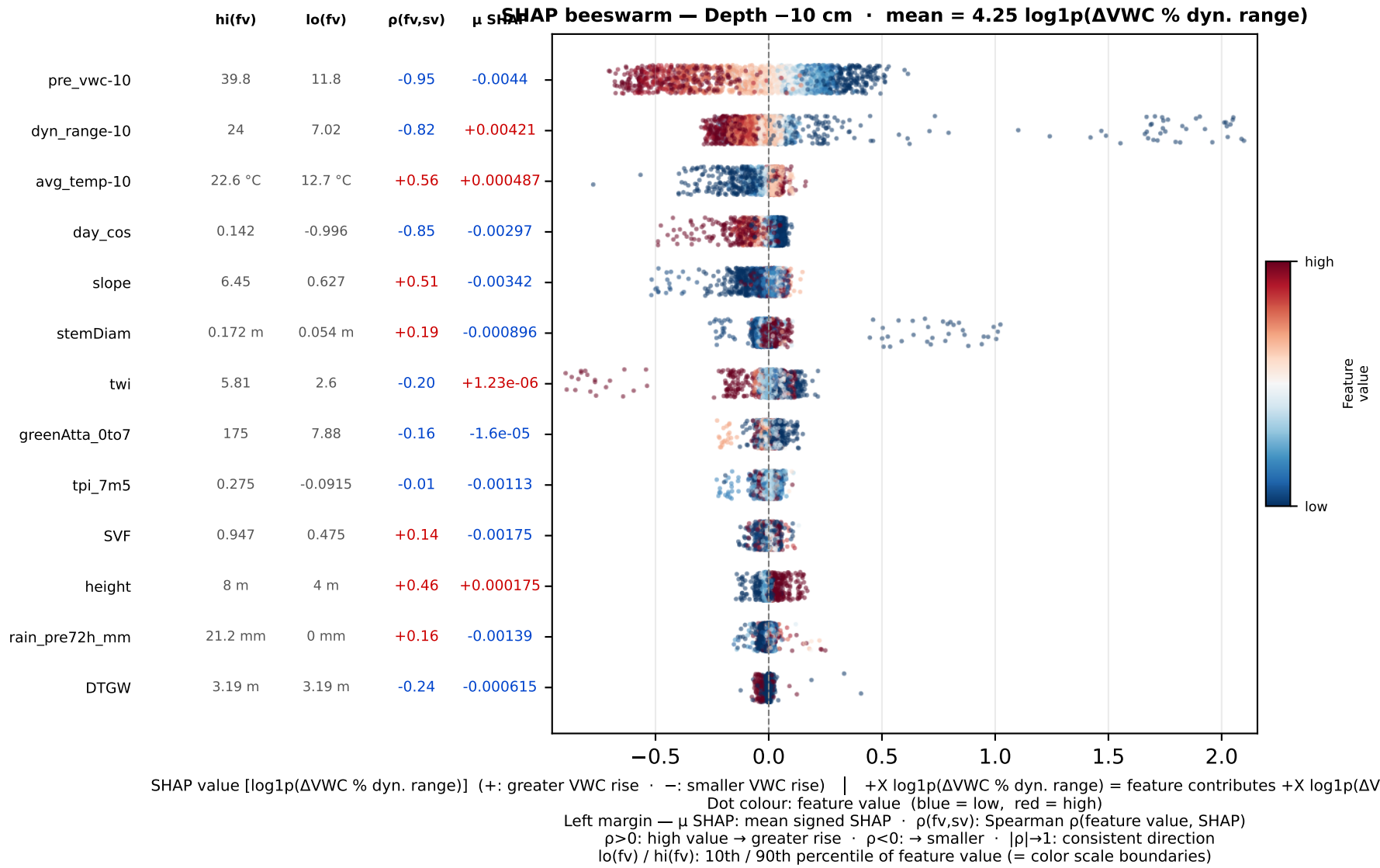
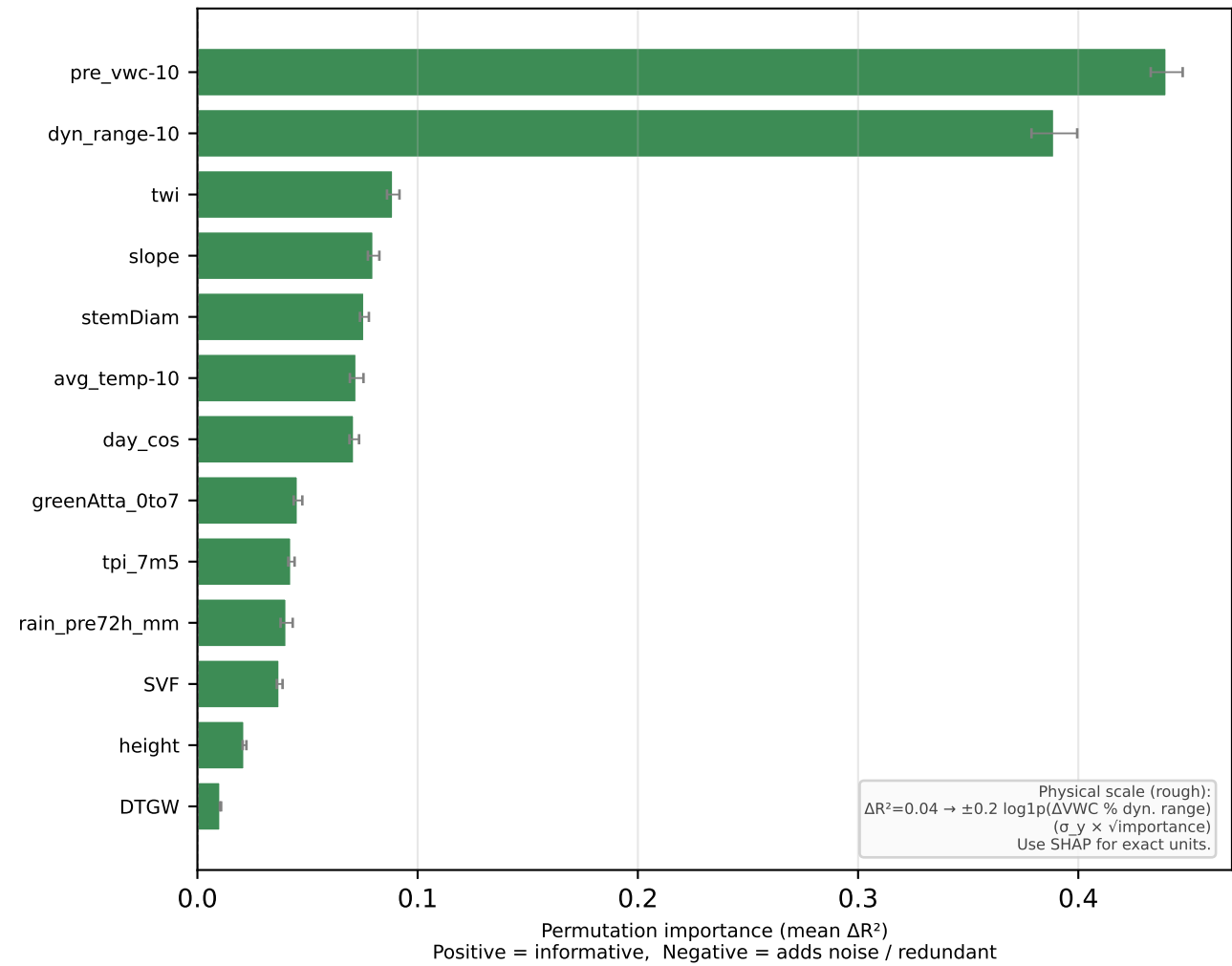
Rain Event Feature Analysis — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (10,911 events)
Features: total rain · pre/post event rain · peak intensity · StAcc · duration · Huff quartile · rain distribution percentages · day-of-year



Random Forest Feature Importance — Irrigation
Target: ΔVWC as % of dynamic range at each depth (5,242 events)

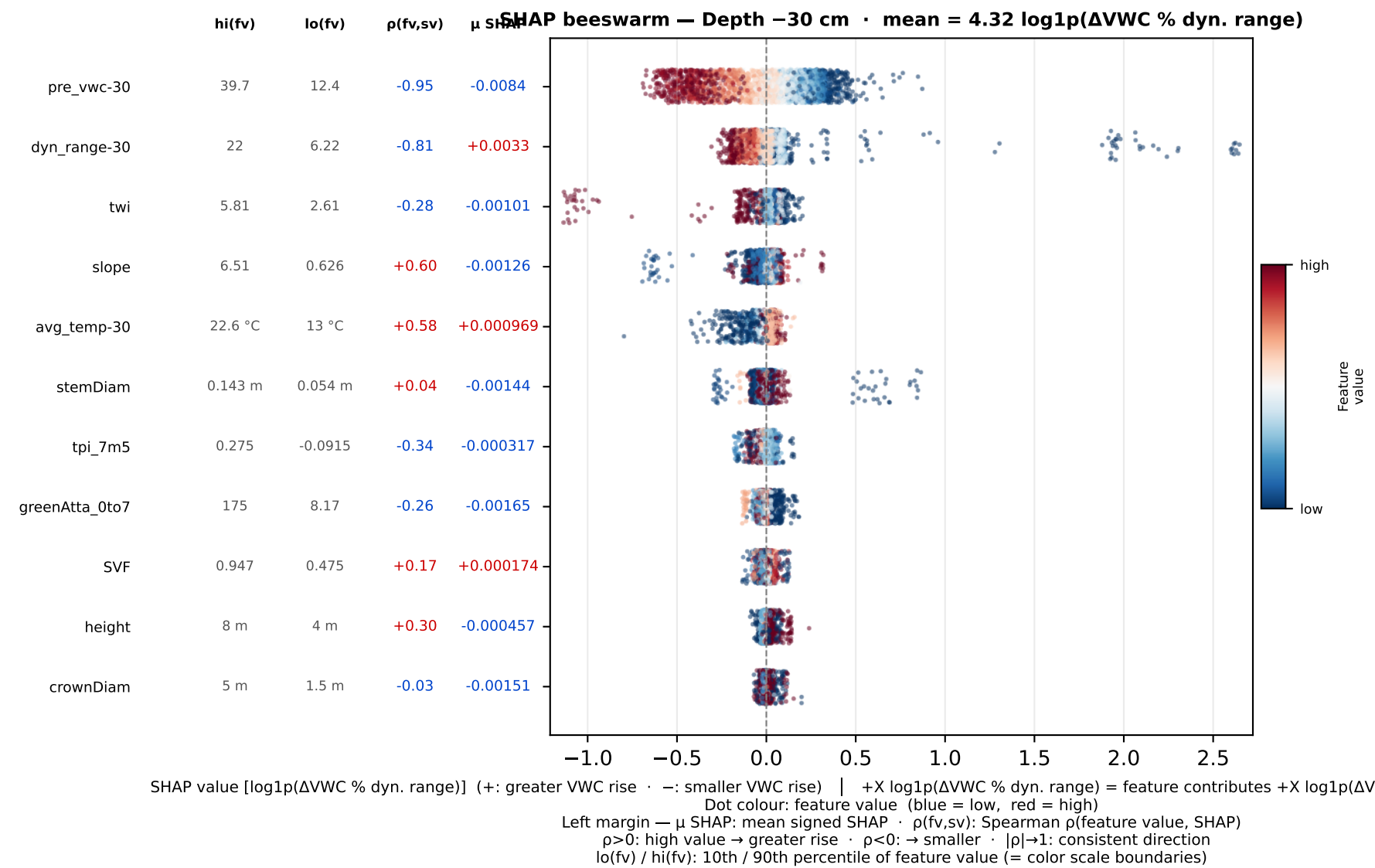
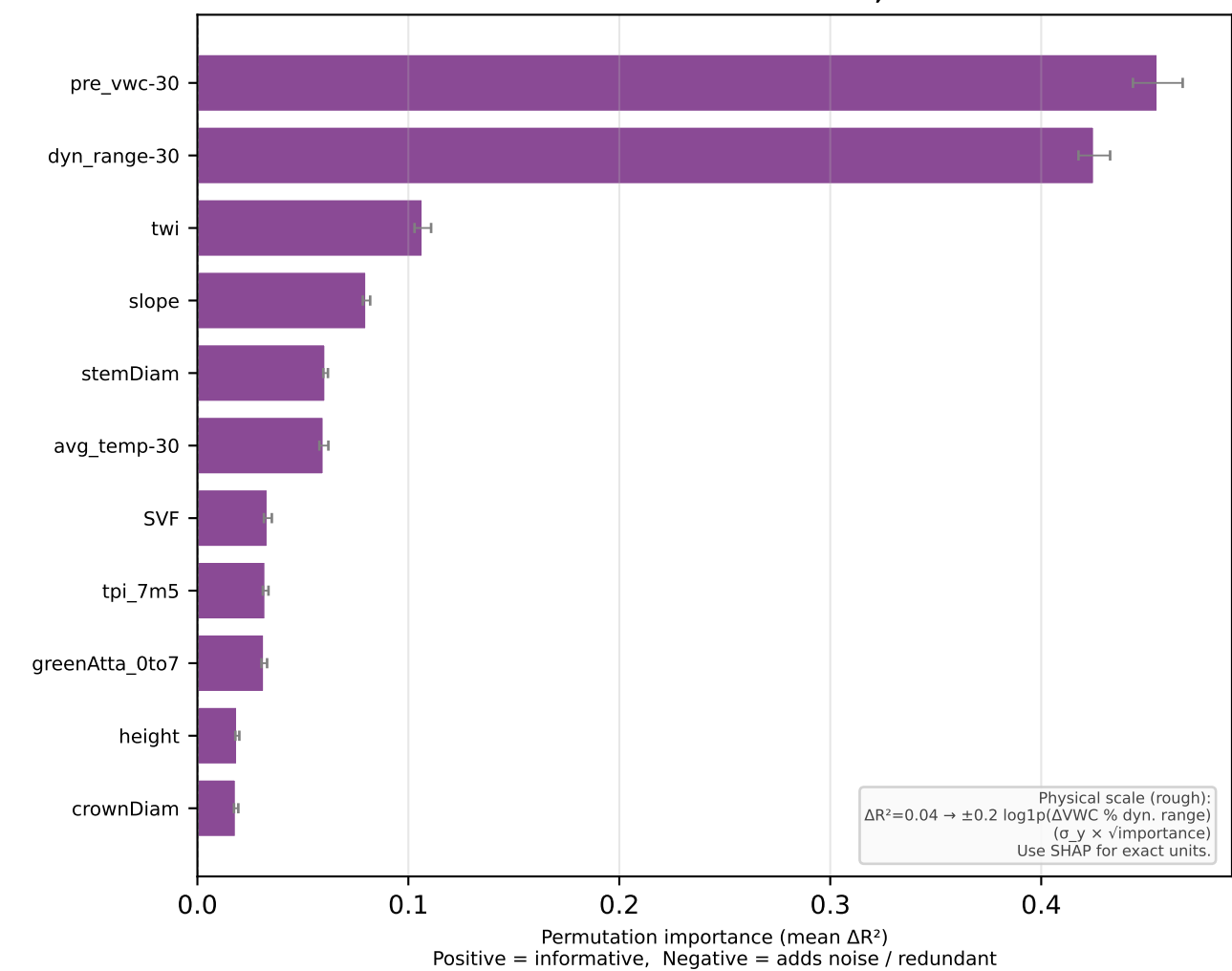
Depth -10 cm — permutation importance

CV R² = 0.627 ± 0.033 n = 3,051



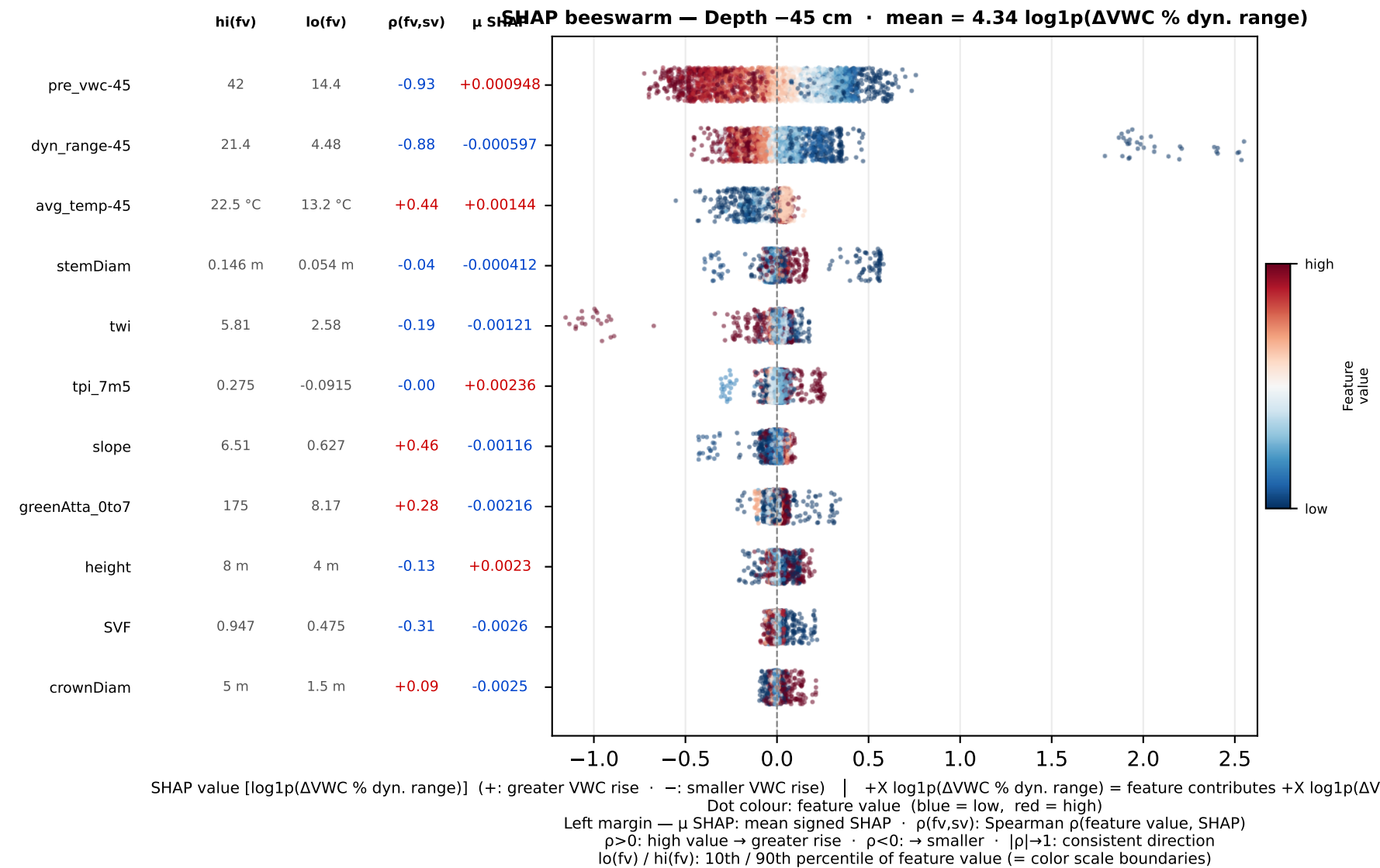
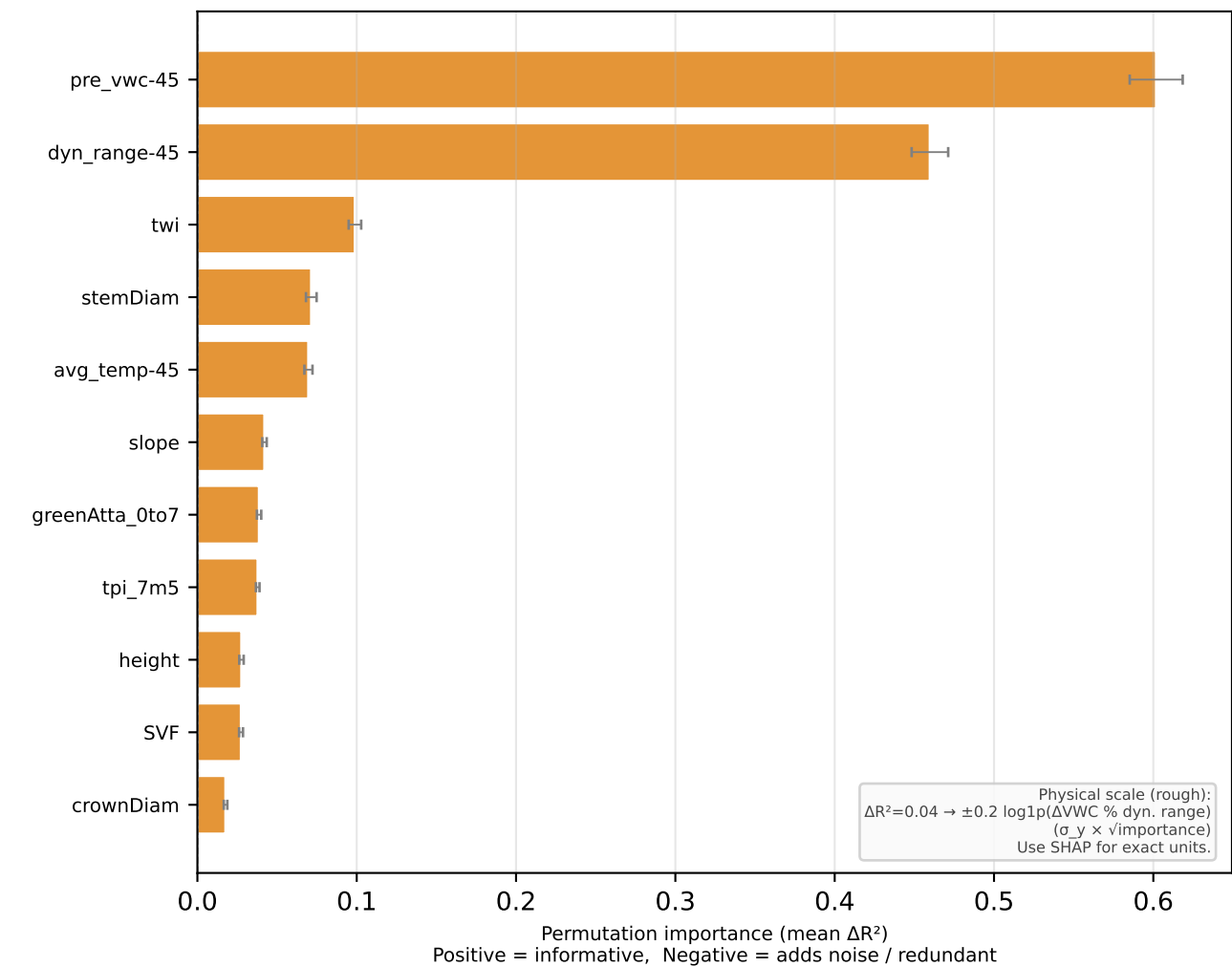
Depth -30 cm — permutation importance

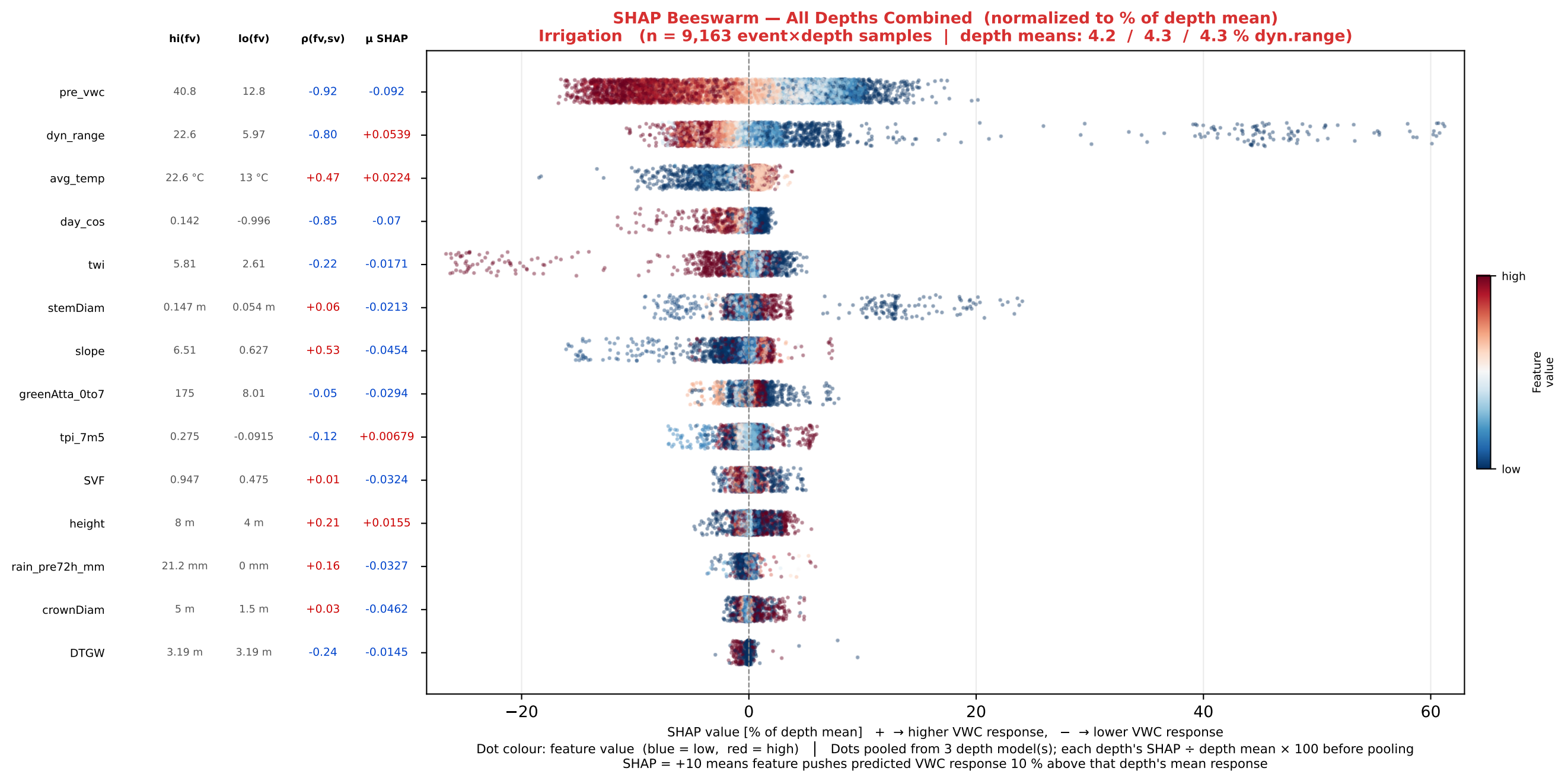
CV R² = 0.686 ± 0.026 n = 3,057



Depth -45 cm — permutation importance

CV R² = 0.690 ± 0.030 n = 3,055





HOW TO READ THIS CHART (Combined depth view – SHAP normalized to % of depth mean)

SHAP value (x-axis) – unit: % of depth-specific mean VWC response

Each depth's SHAP values were divided by that depth's training mean before pooling.

This makes all depths directly comparable regardless of their absolute response magnitude.

- +10 % → this feature pushes the predicted VWC response 10 % above that depth's mean.
- -10 % → this feature pulls the predicted VWC response 10 % below that depth's mean.

Per-depth means used for normalization (3 depth(s)):

Depth 1: mean = 4.2 % dyn.range → SHAP +10 % of mean = +0.4 % dyn.range ≈ +0.076 % abs. VWC (at dyn.range = 18 %)

Depth 2: mean = 4.3 % dyn.range → SHAP +10 % of mean = +0.4 % dyn.range ≈ +0.078 % abs. VWC (at dyn.range = 18 %)

Depth 3: mean = 4.3 % dyn.range → SHAP +10 % of mean = +0.4 % dyn.range ≈ +0.078 % abs. VWC (at dyn.range = 18 %)

Example – top feature "pre_vwc" (normalized SHAP IQR: -5.5 to +5.2 % of depth mean):

Low feature value → SHAP ≈ -5.5% High feature value → SHAP ≈ +5.2%

Difference: ~10.7 % of each depth's mean → in physical units per depth:

Depth 1 (mean 4.2 % dyn.range): Δ ≈ 0.5 % dyn.range ≈ 0.082 % abs. VWC

Depth 2 (mean 4.3 % dyn.range): Δ ≈ 0.5 % dyn.range ≈ 0.083 % abs. VWC

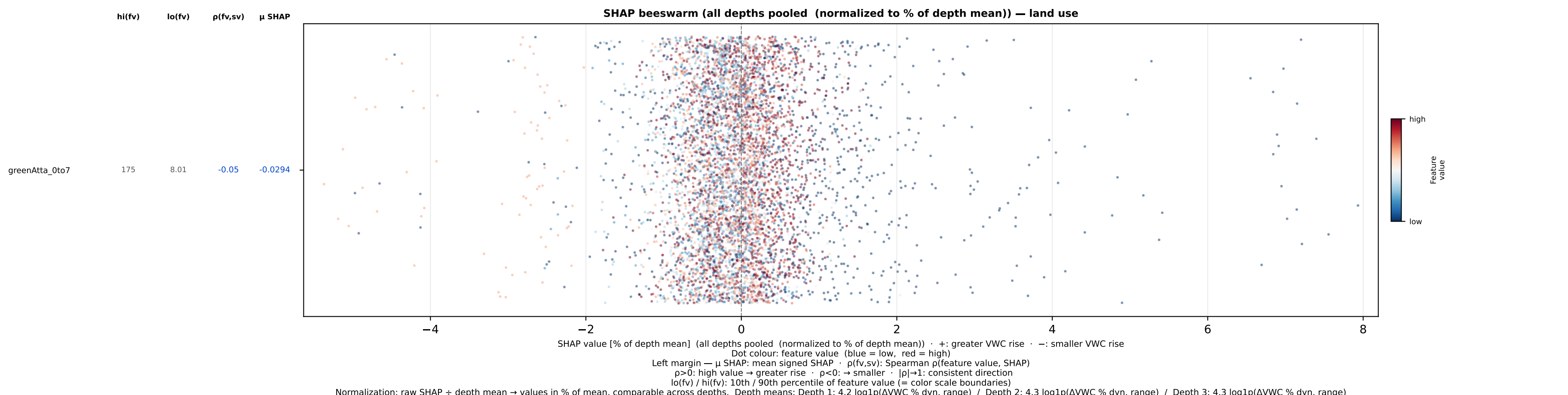
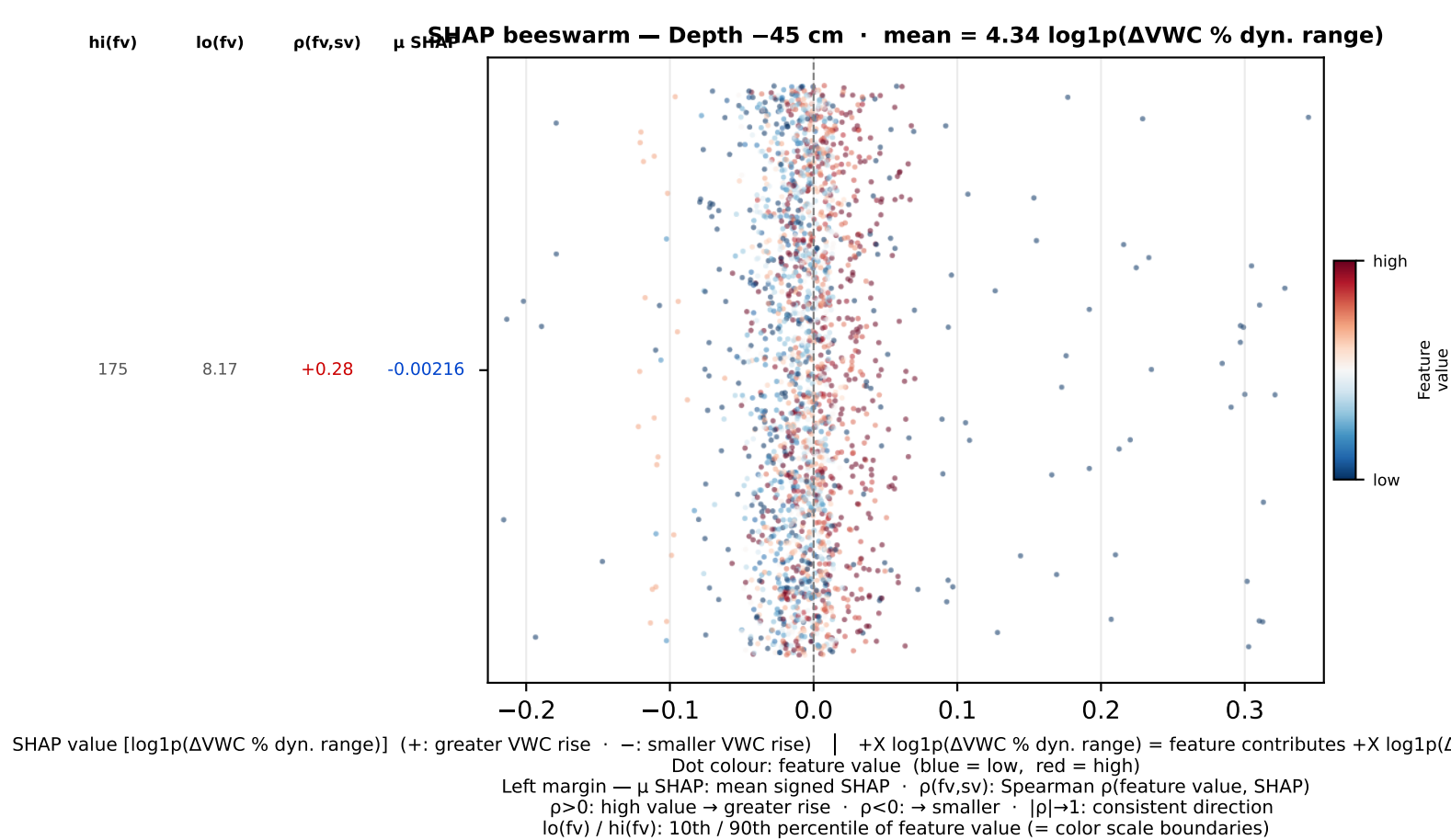
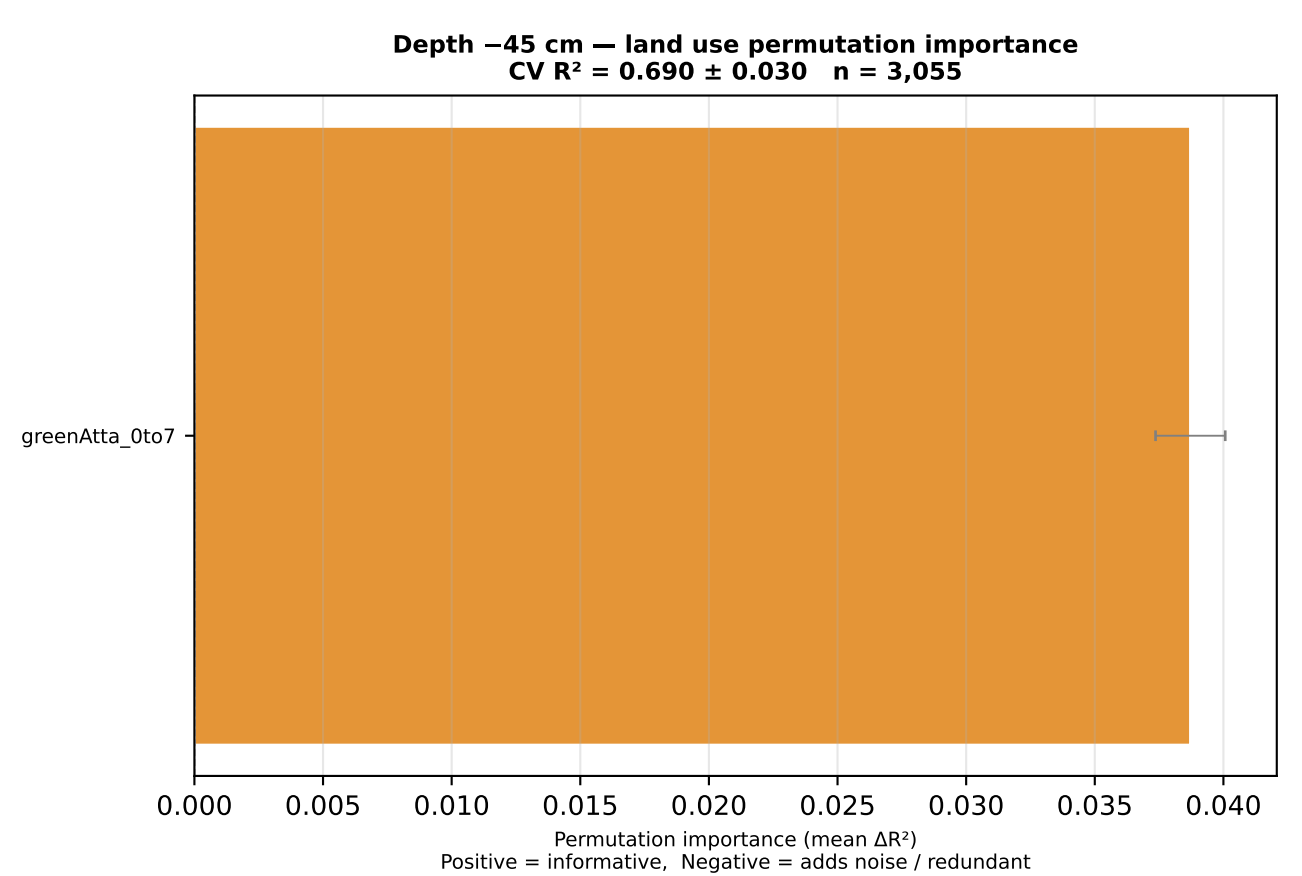
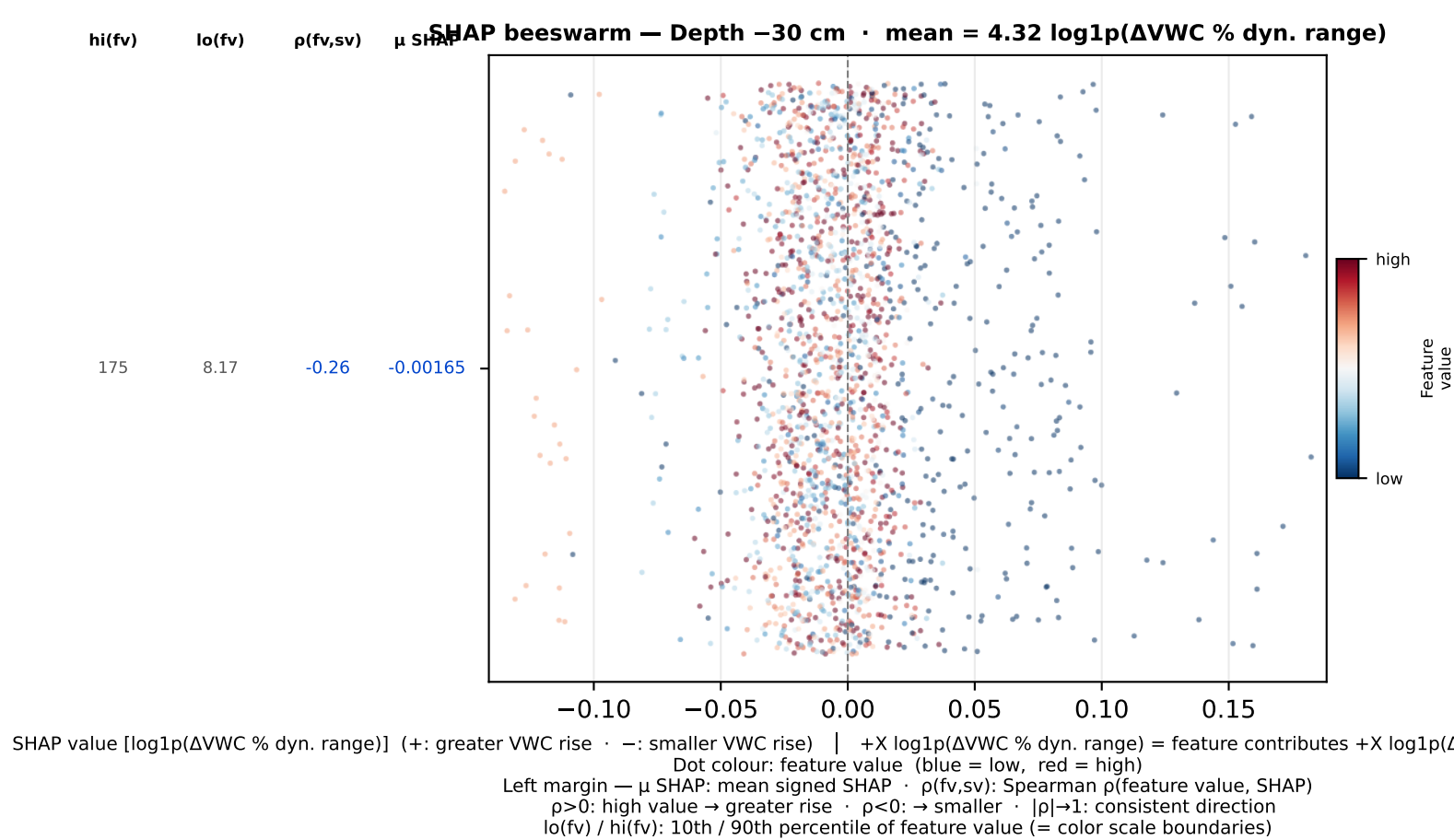
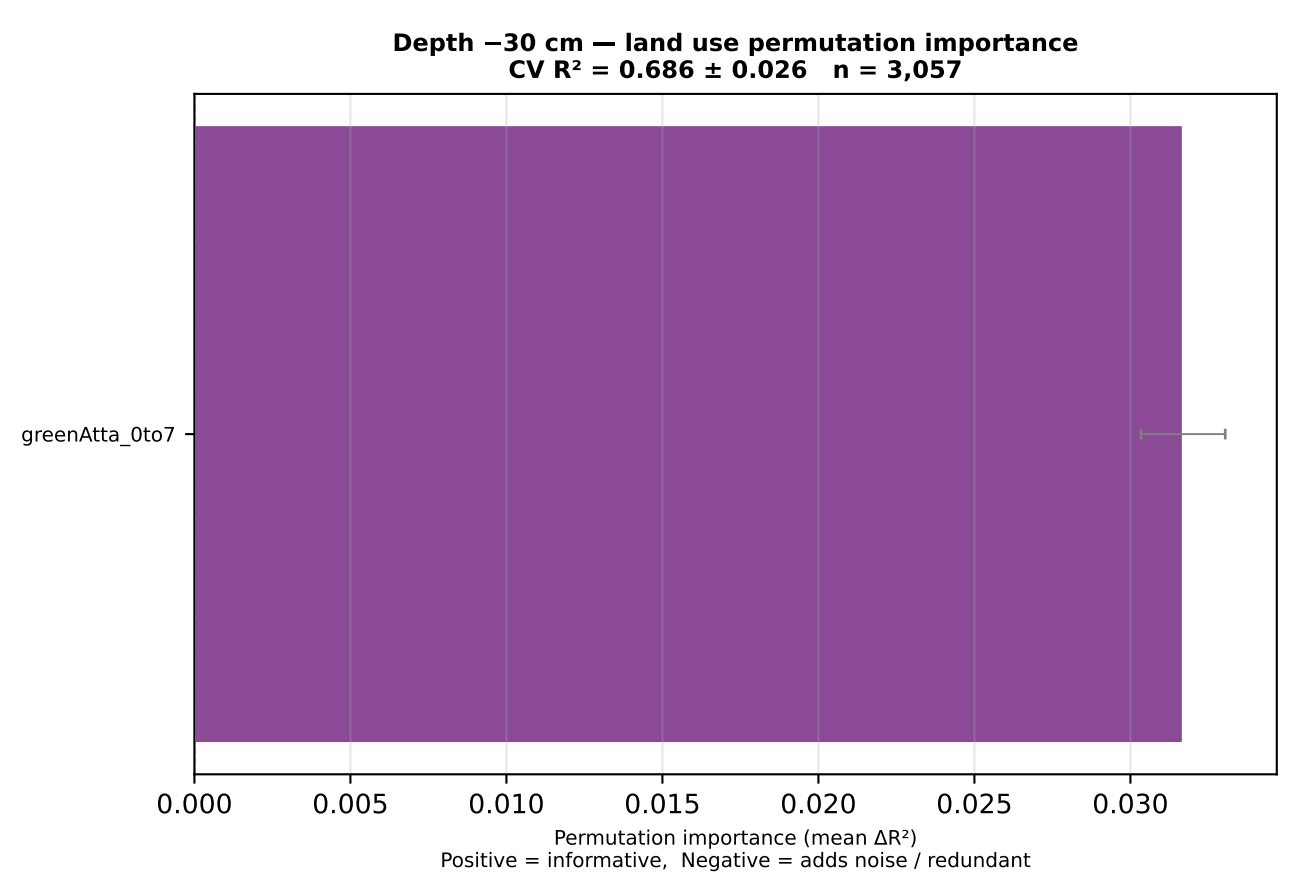
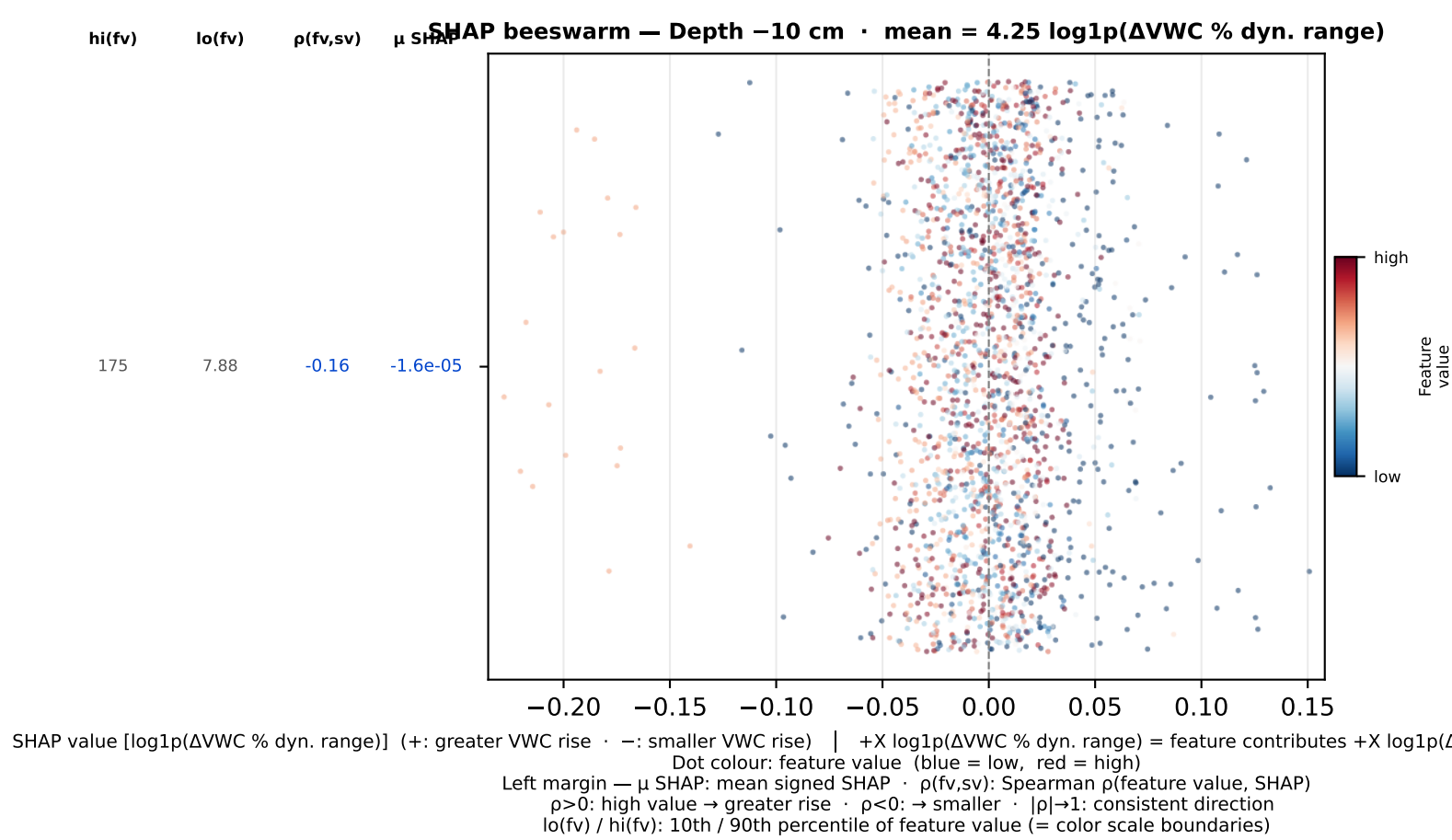
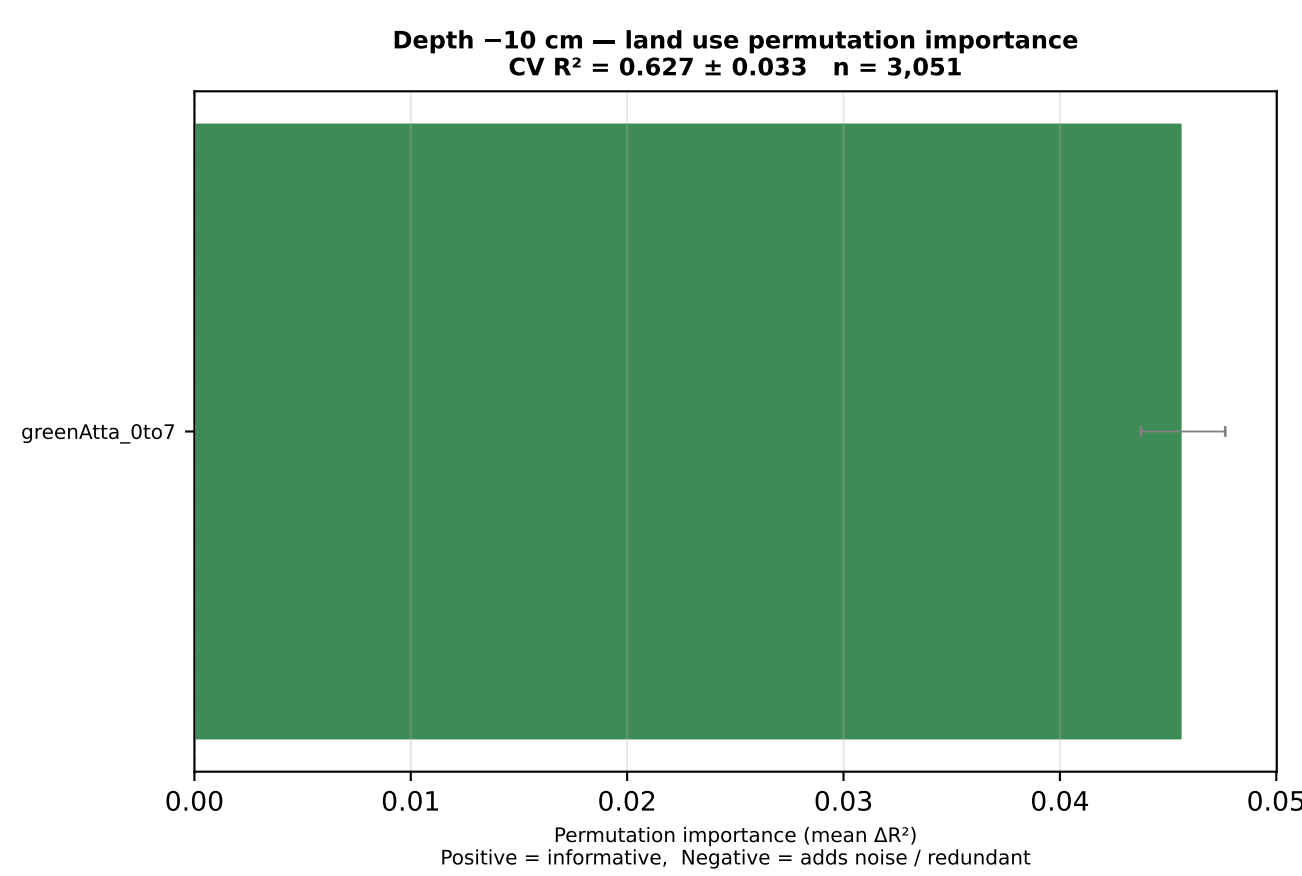
Depth 3 (mean 4.3 % dyn.range): Δ ≈ 0.5 % dyn.range ≈ 0.084 % abs. VWC

Total samples: 9,163 pooled across 3 depth(s) | μ SHAP and p in the left margin are computed on the normalized values.

Permutation importance (bar charts on preceding pages) – unit: mean ΔR² (NOT in % dyn.range!)

Physical scale estimate per depth: contribution ≈ σ_y × √importance

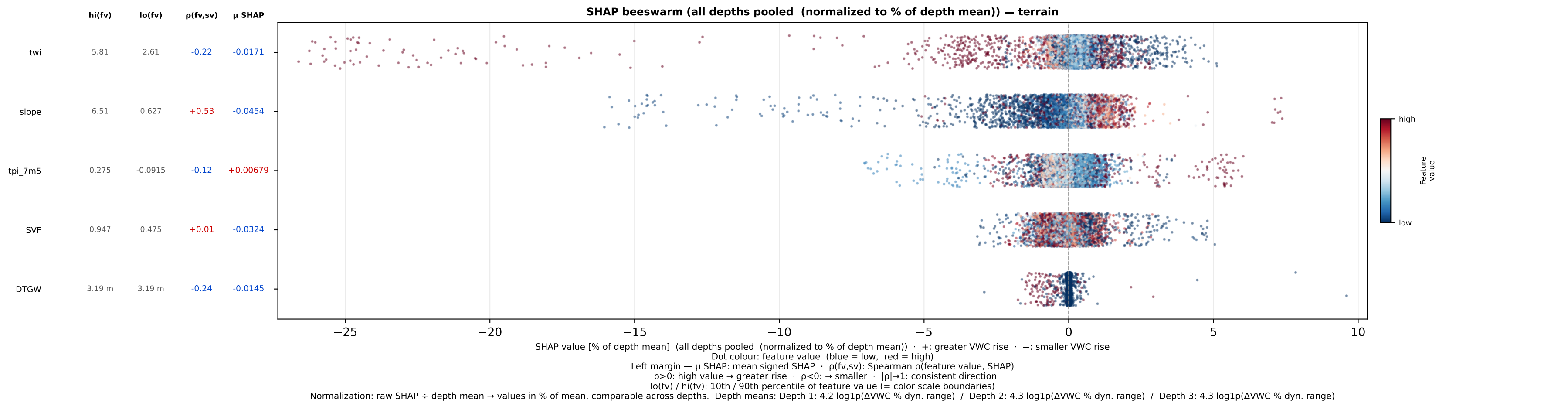
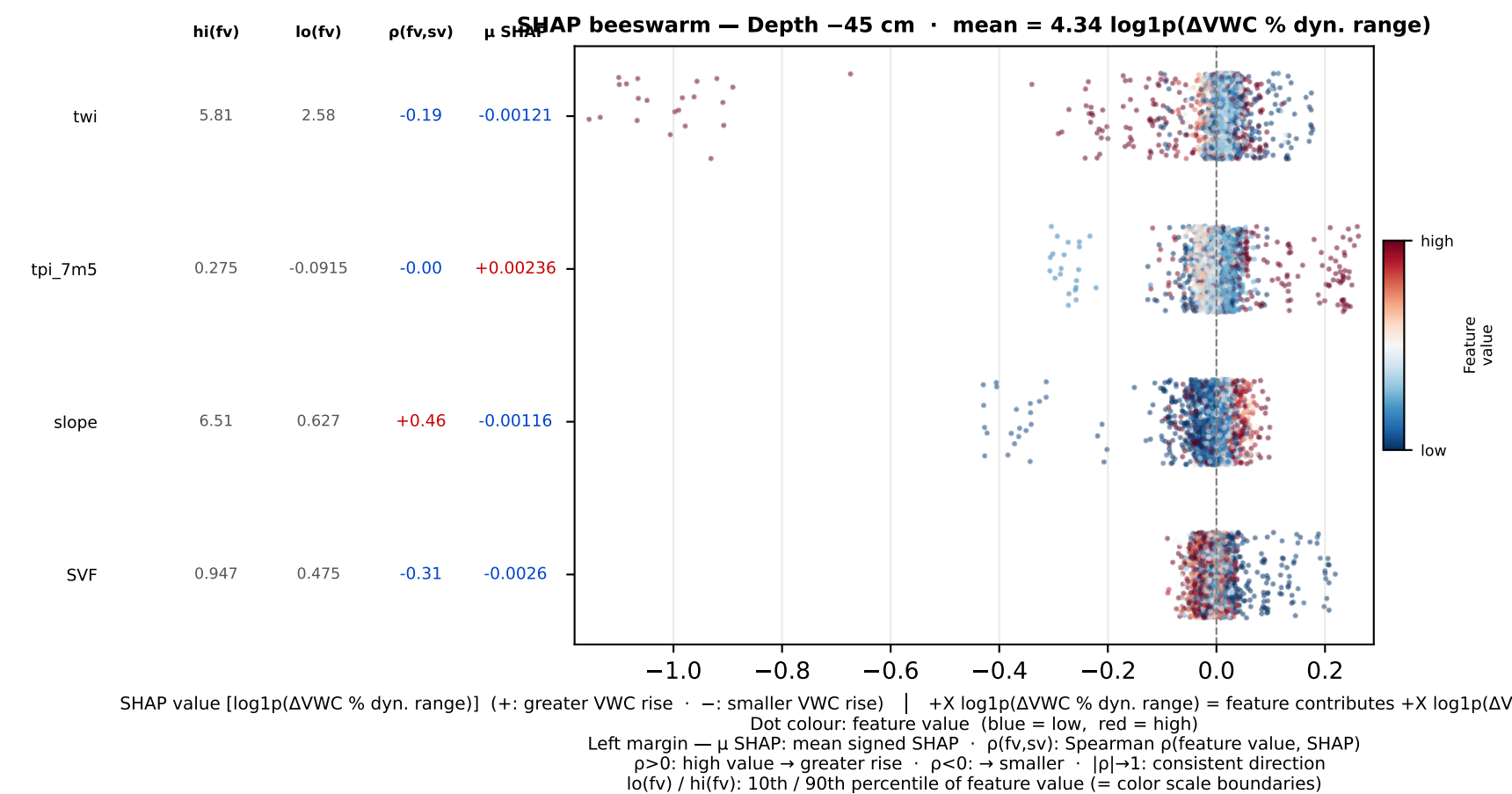
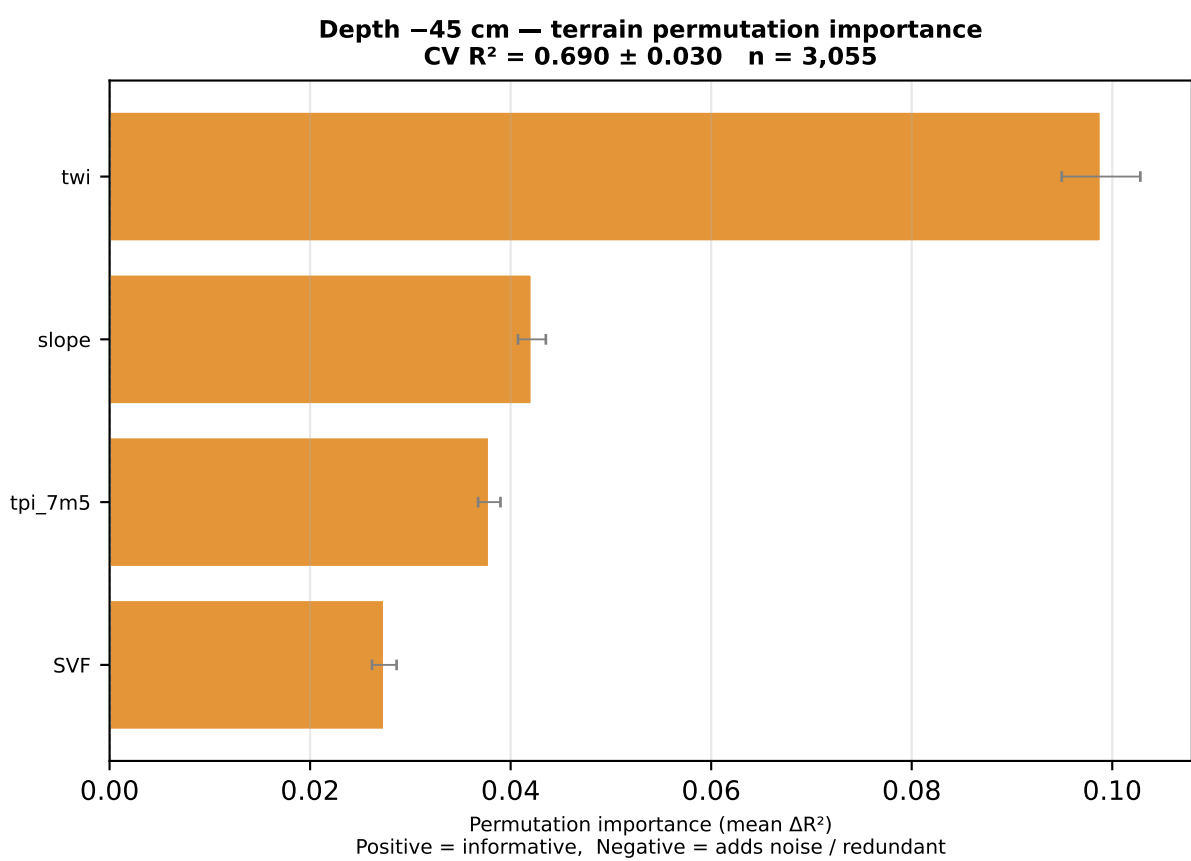
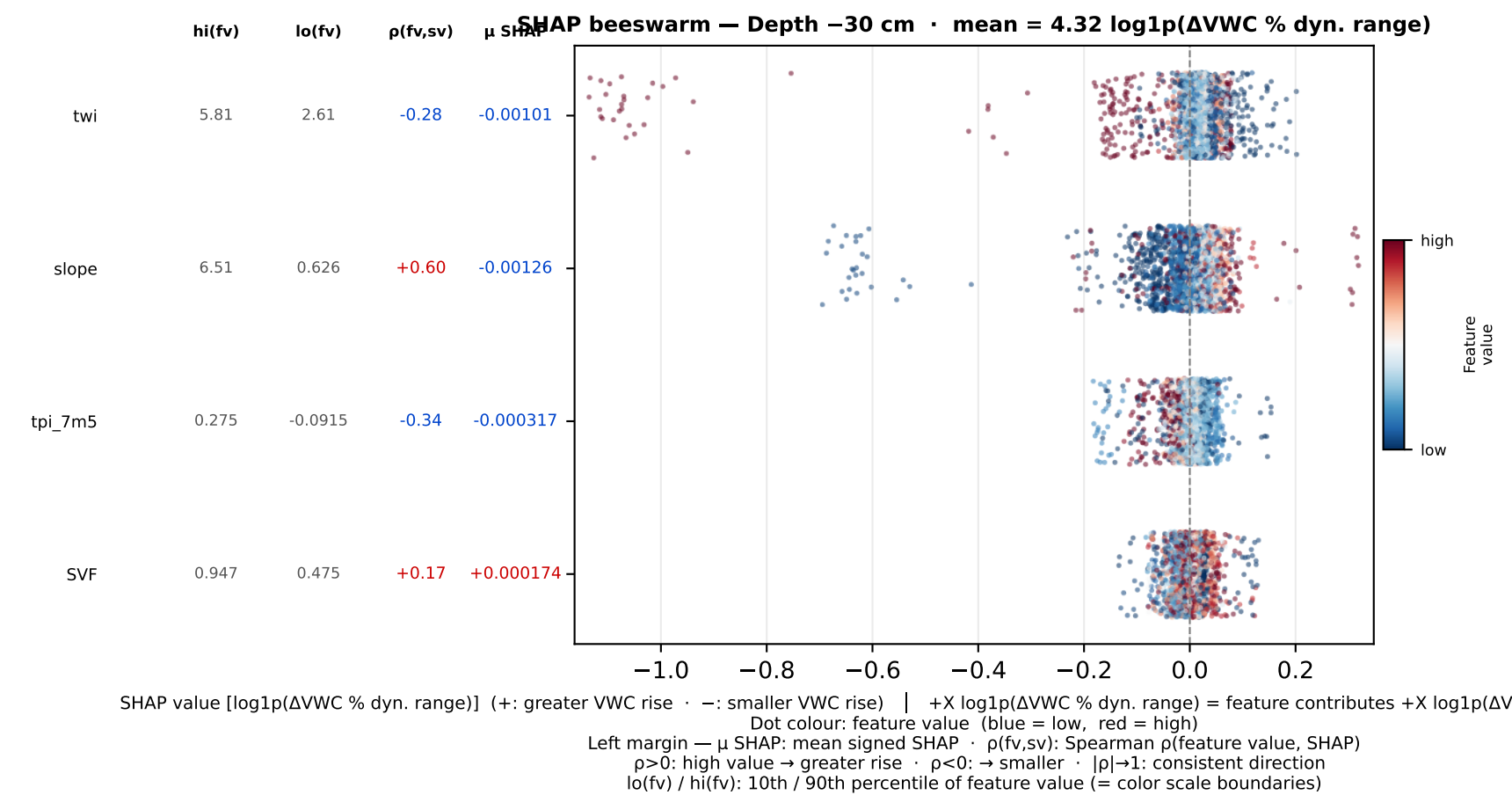
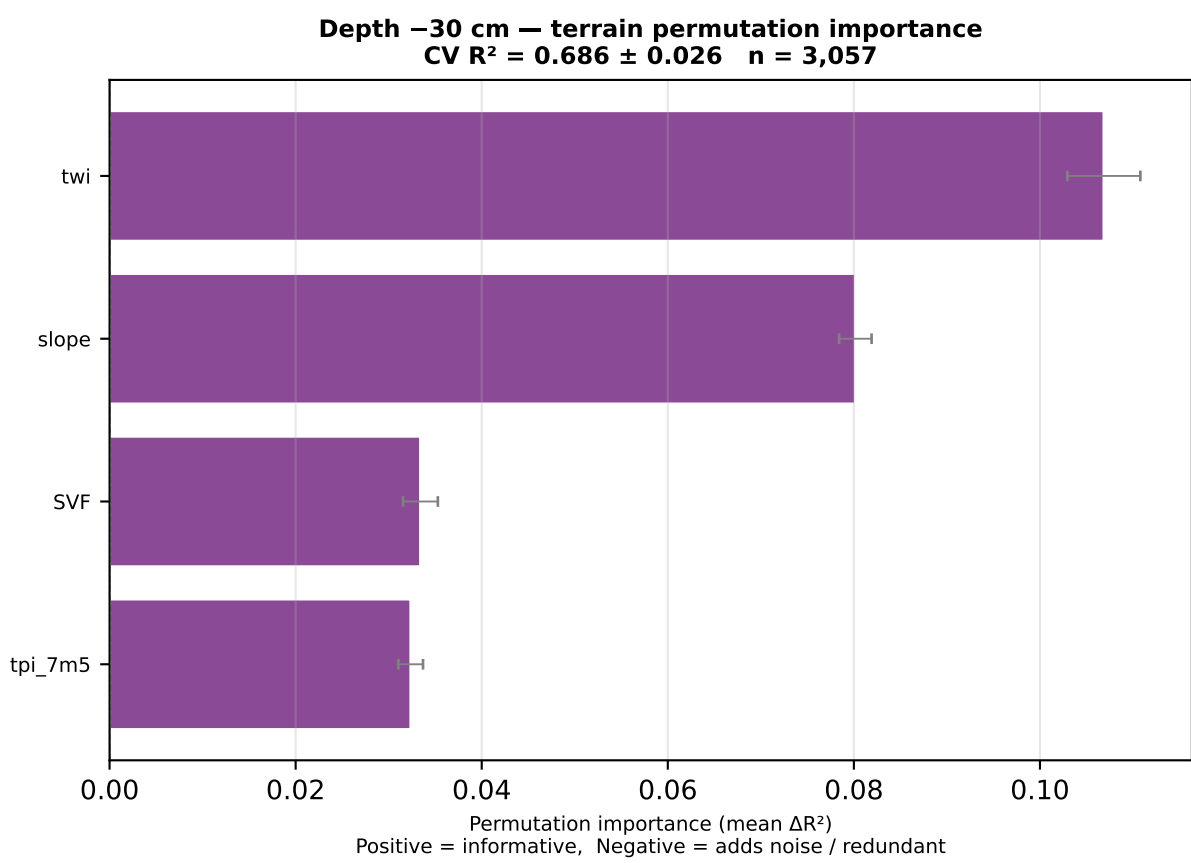
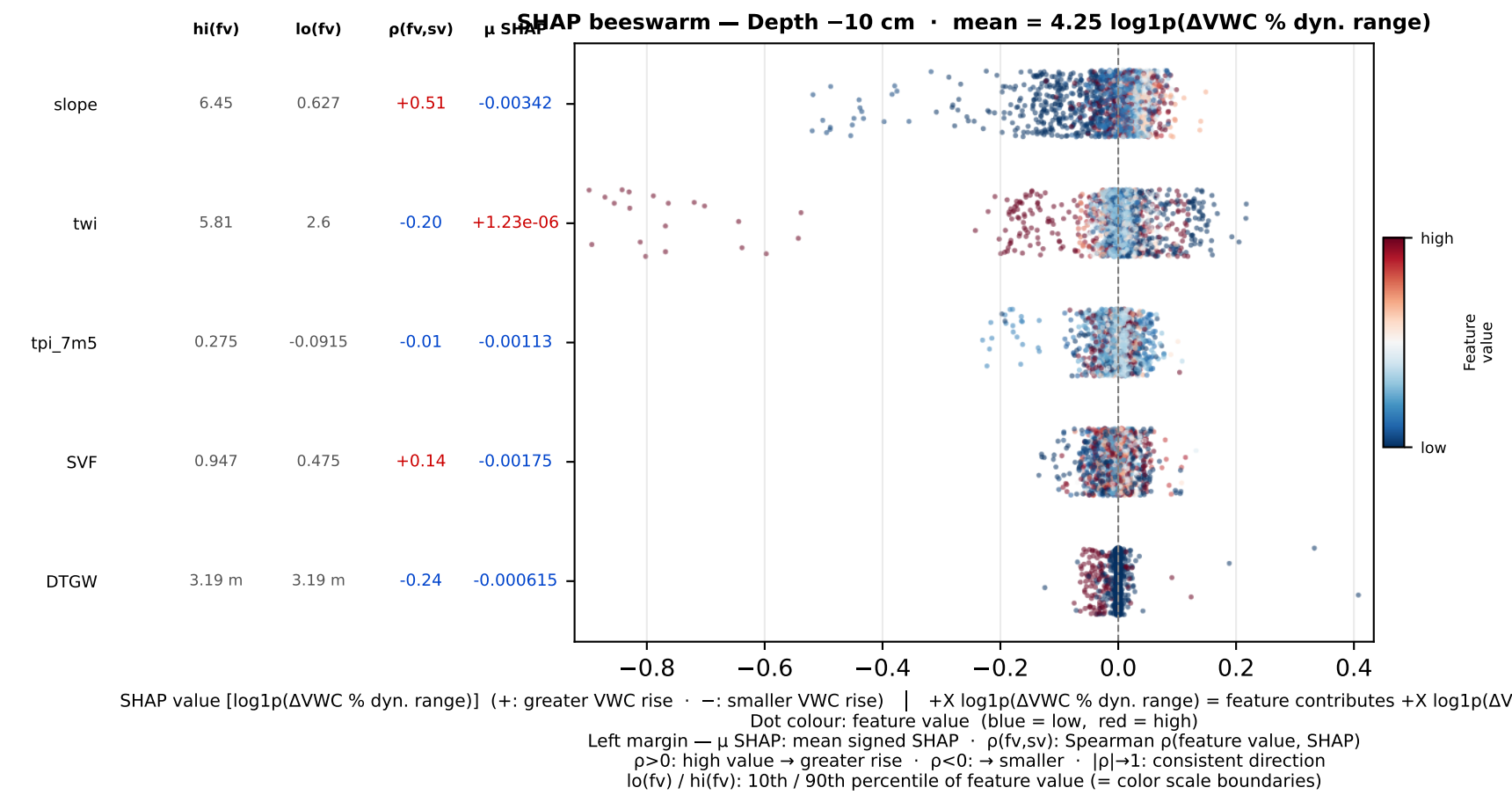
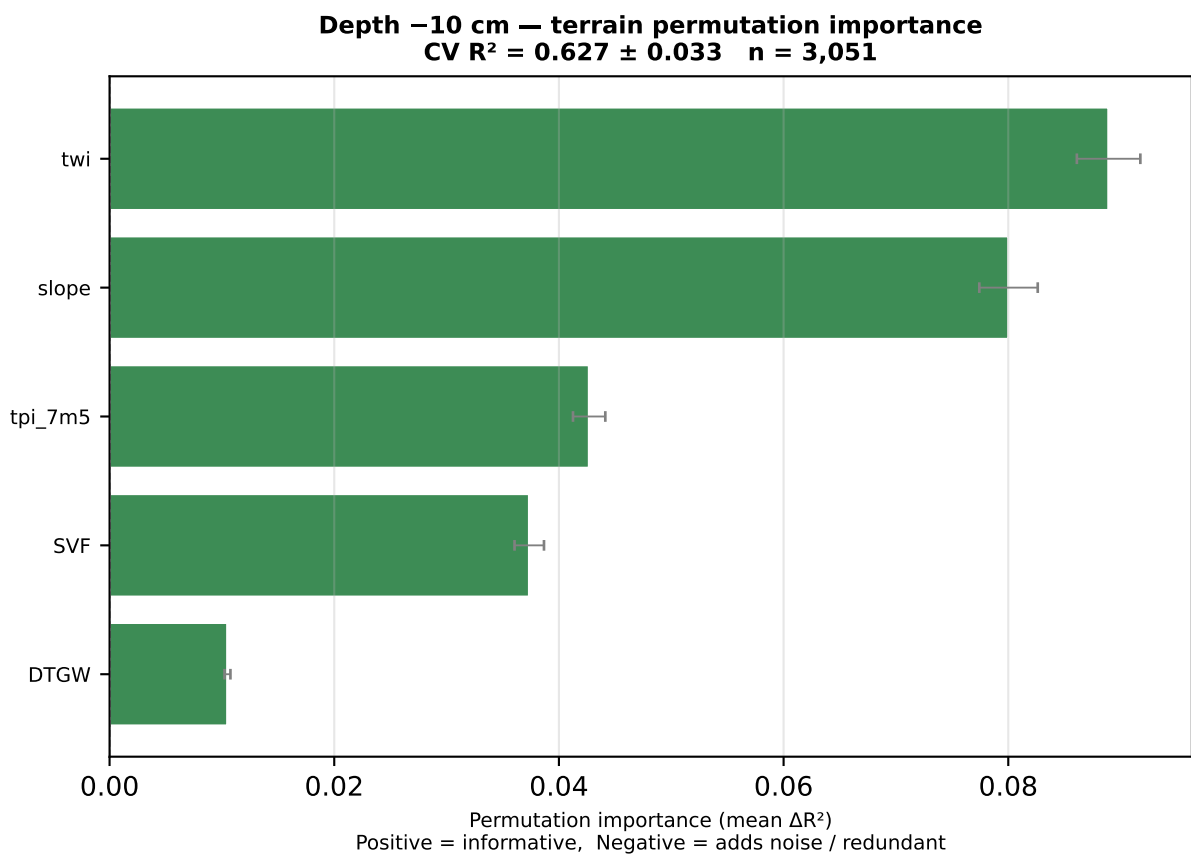
(use SHAP values, not permutation importance, for direct attribution in physical units)



Terrain Feature Analysis — Irrigation

Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (5,242 events)

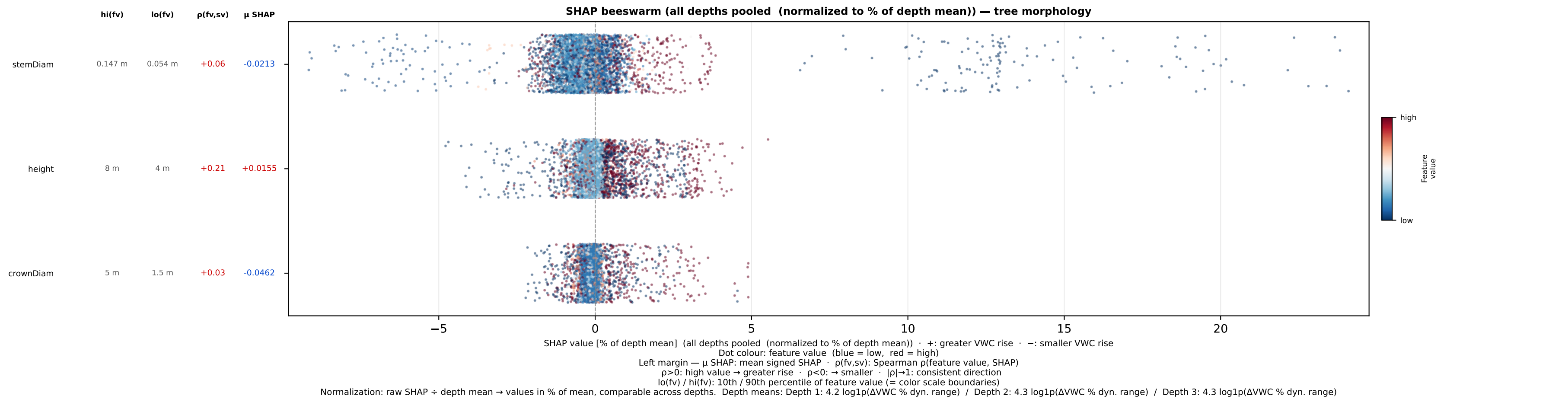
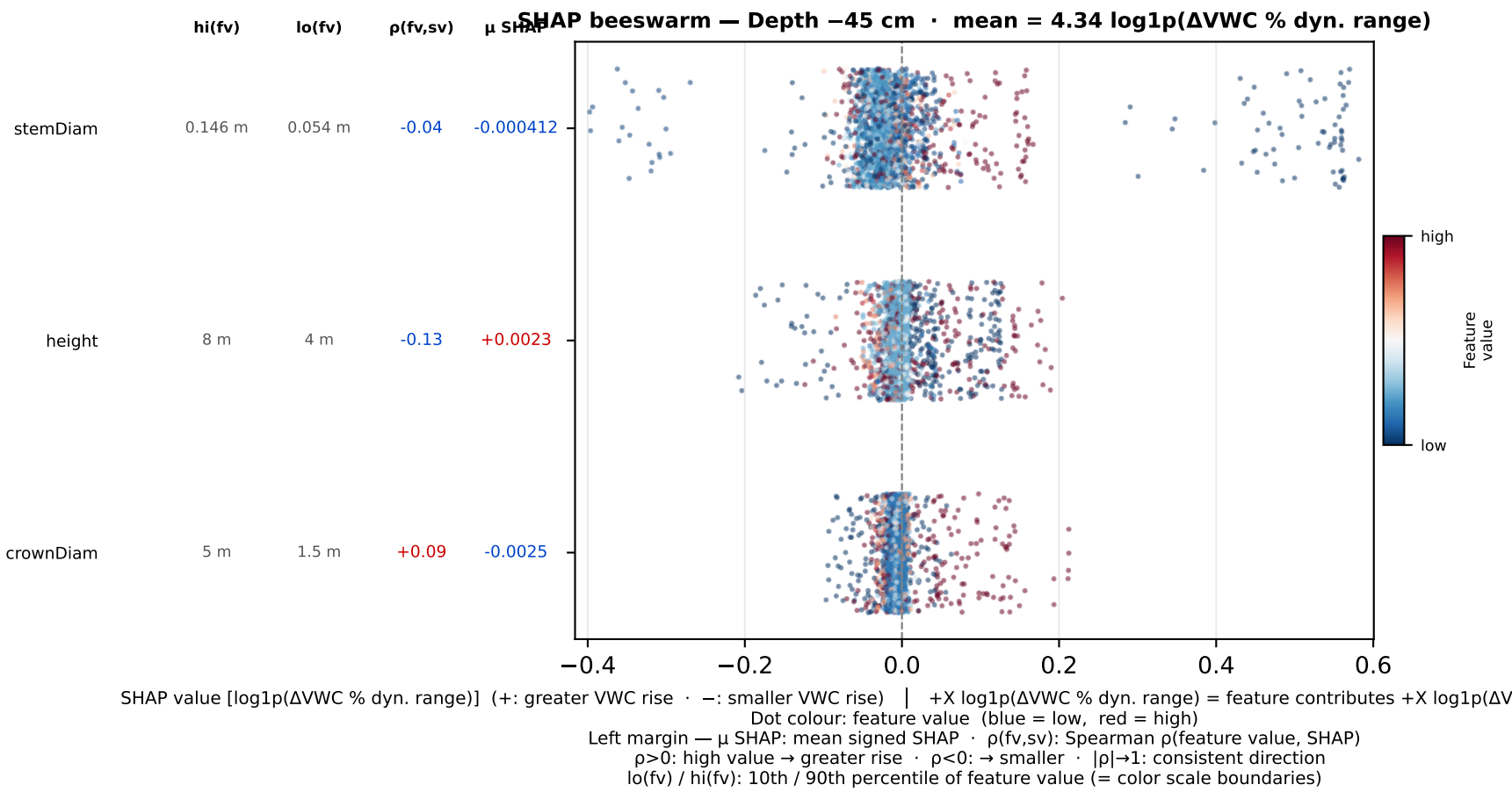
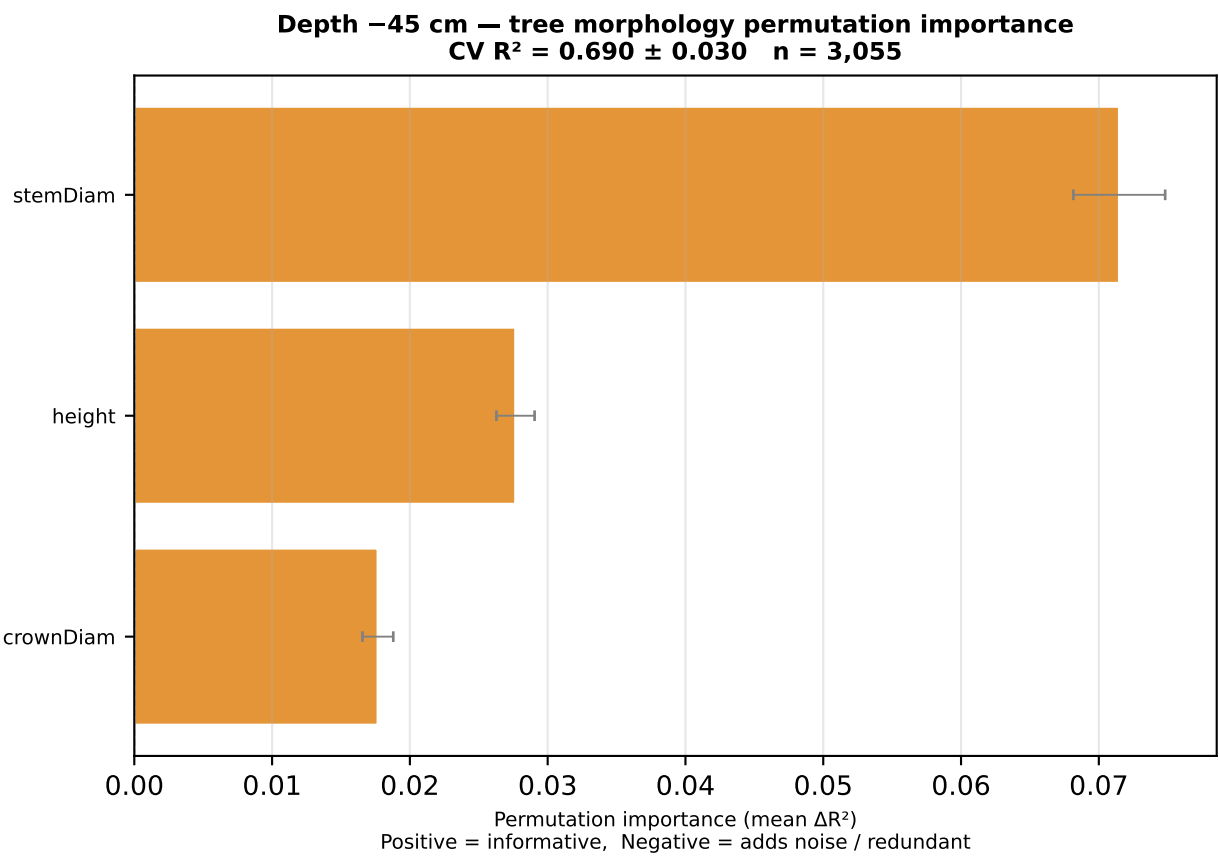
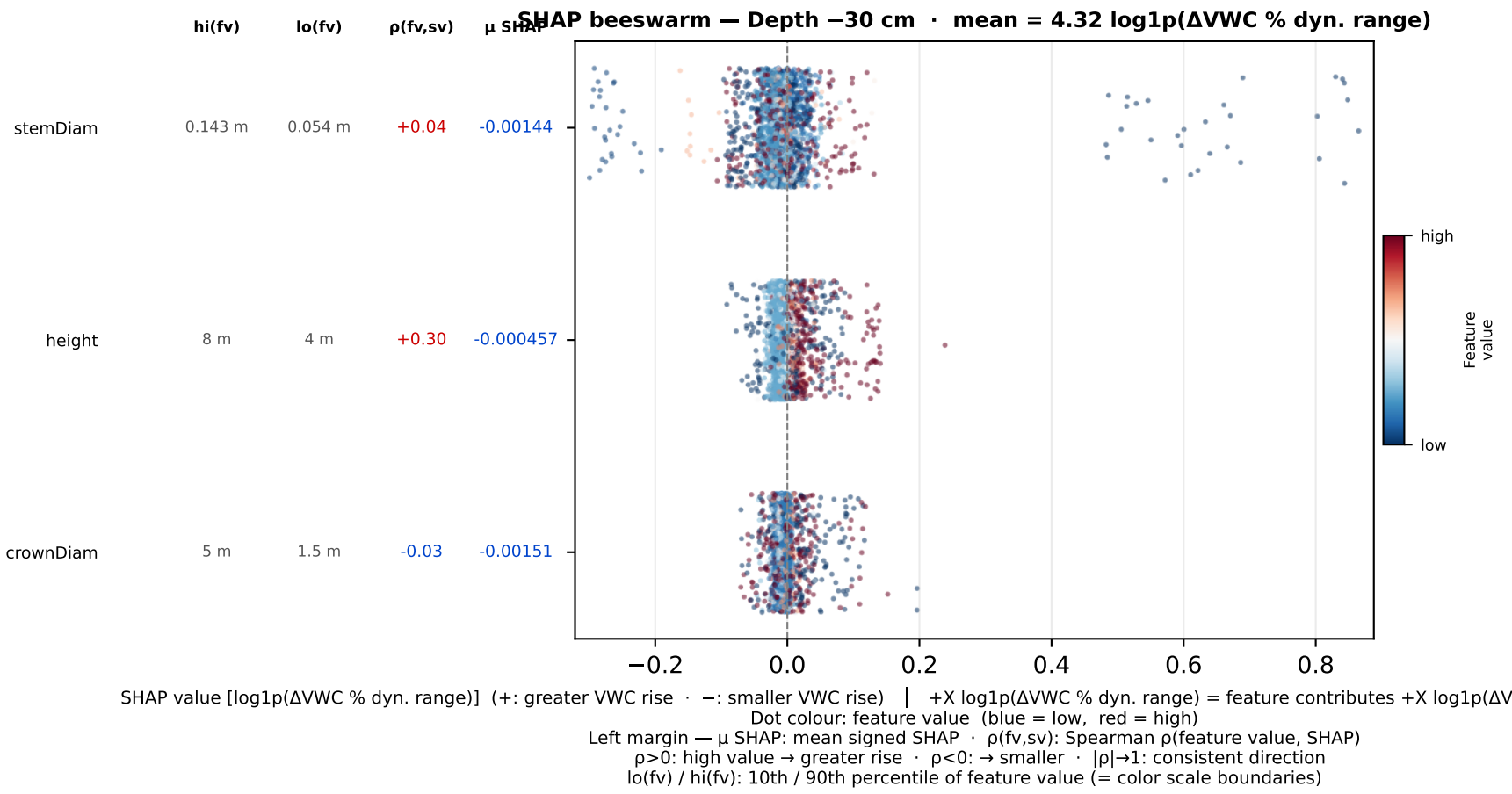
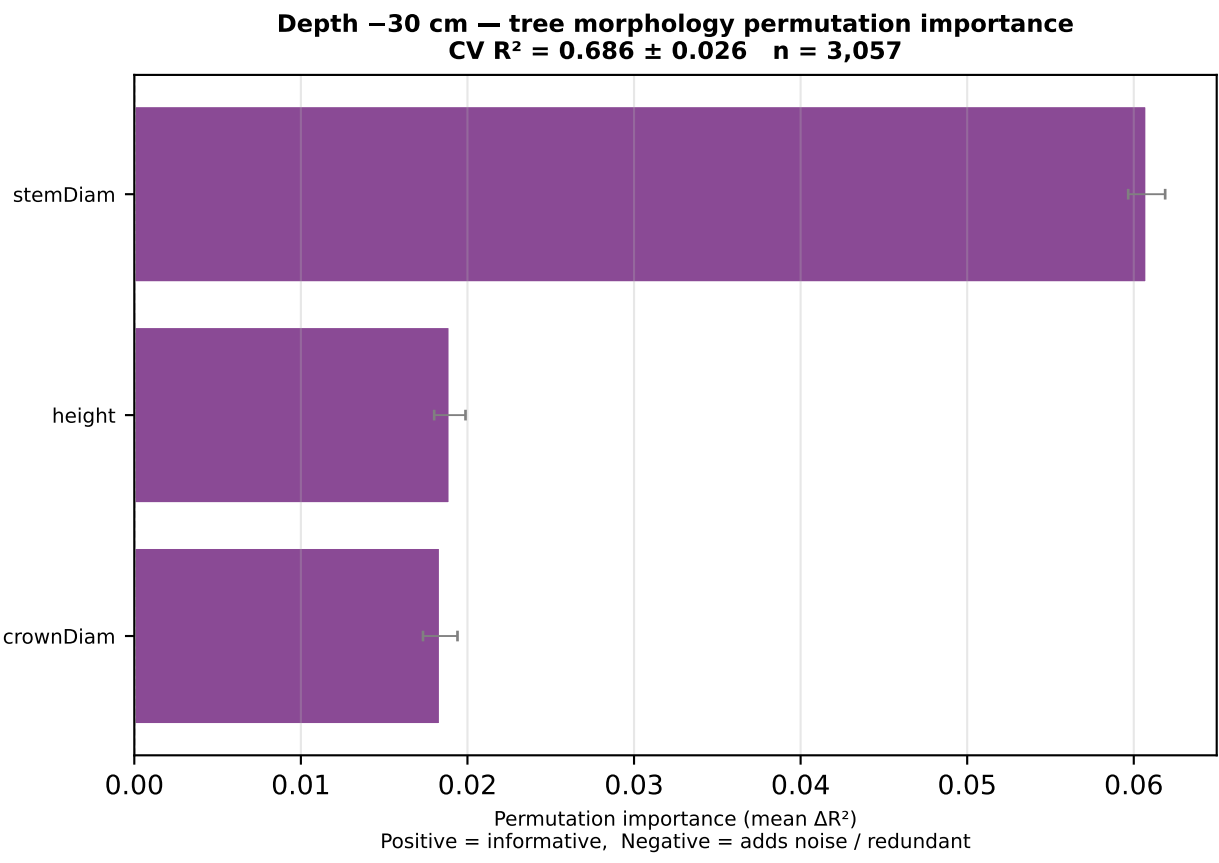
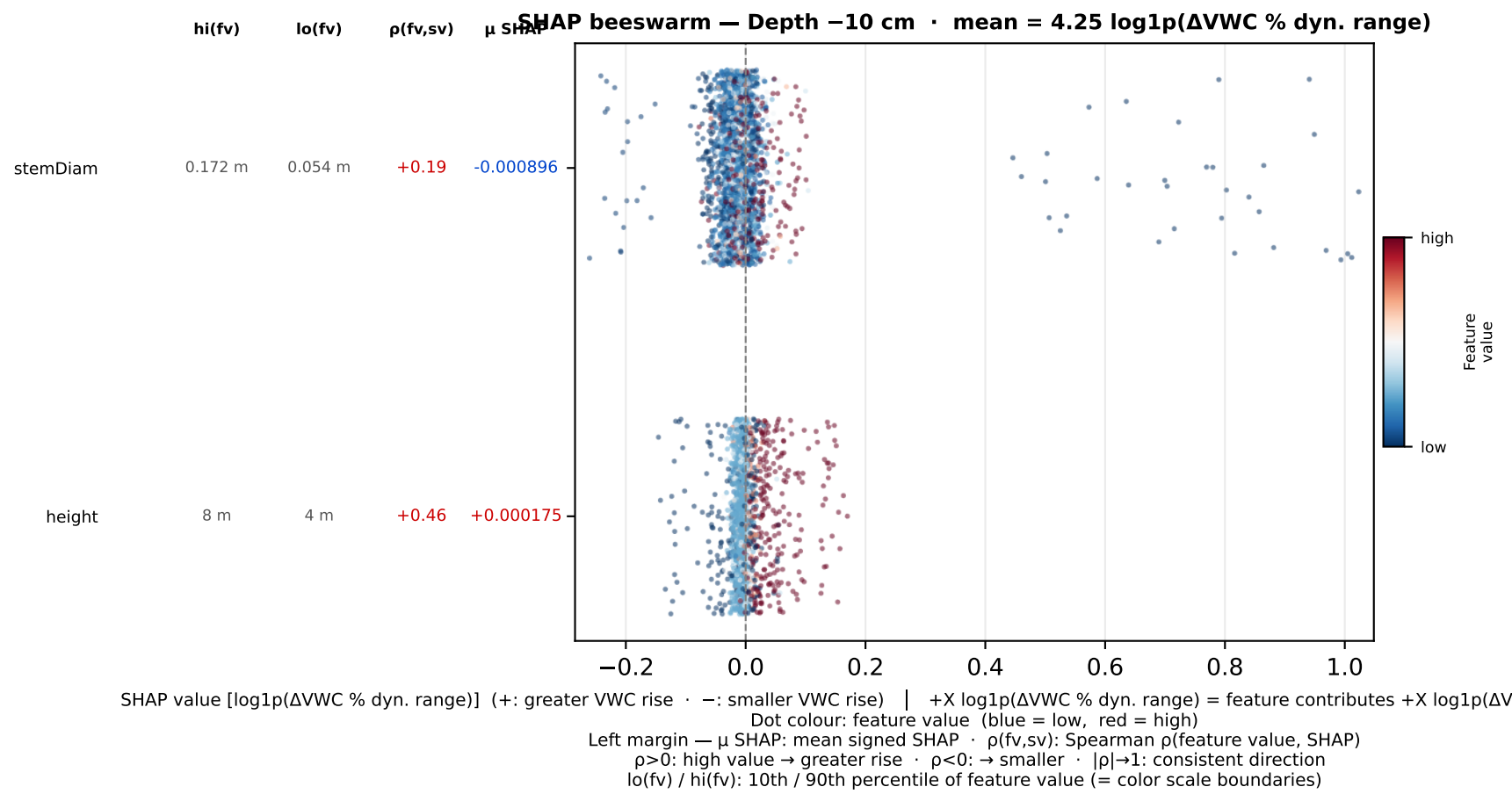
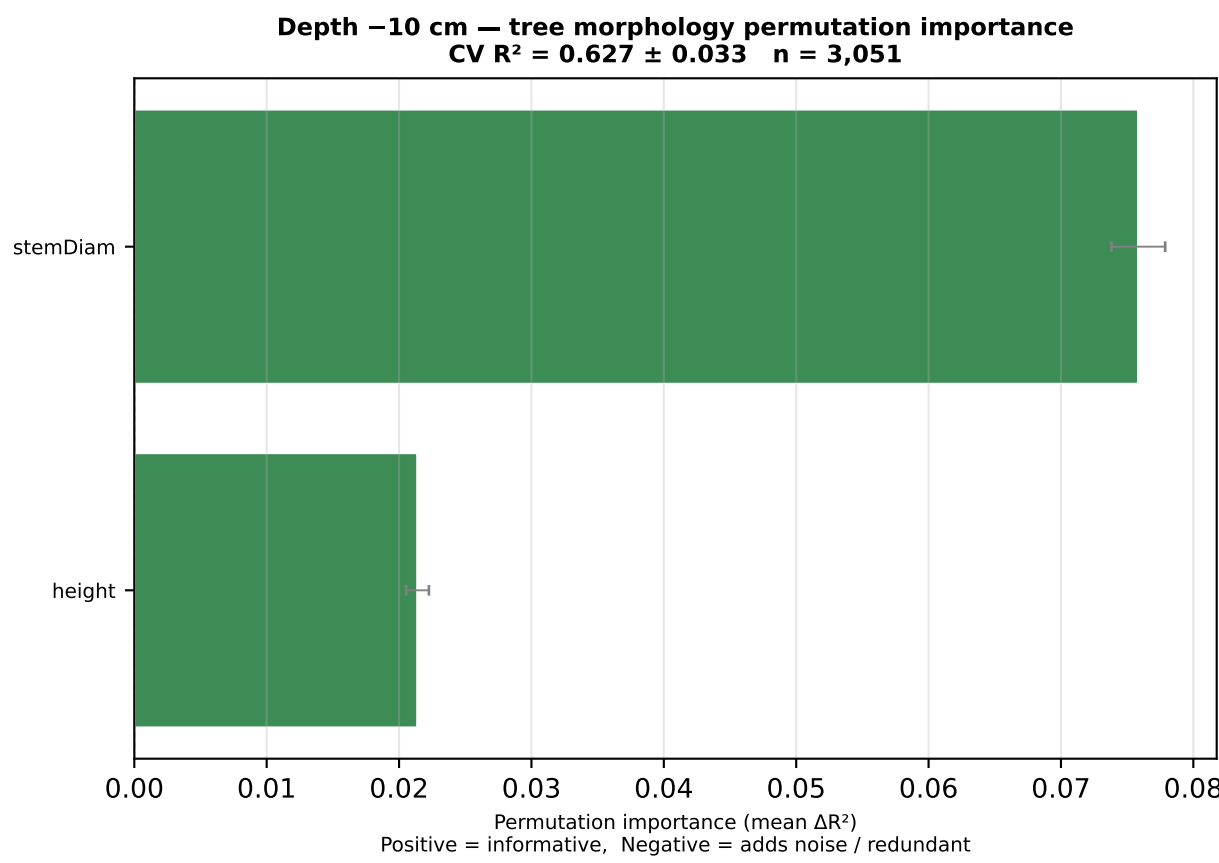
Features: TWI · TPI (2.5/5/7.5 m + crown-diam) · slope · amsl · flow accumulation · SVF · depth to groundwater

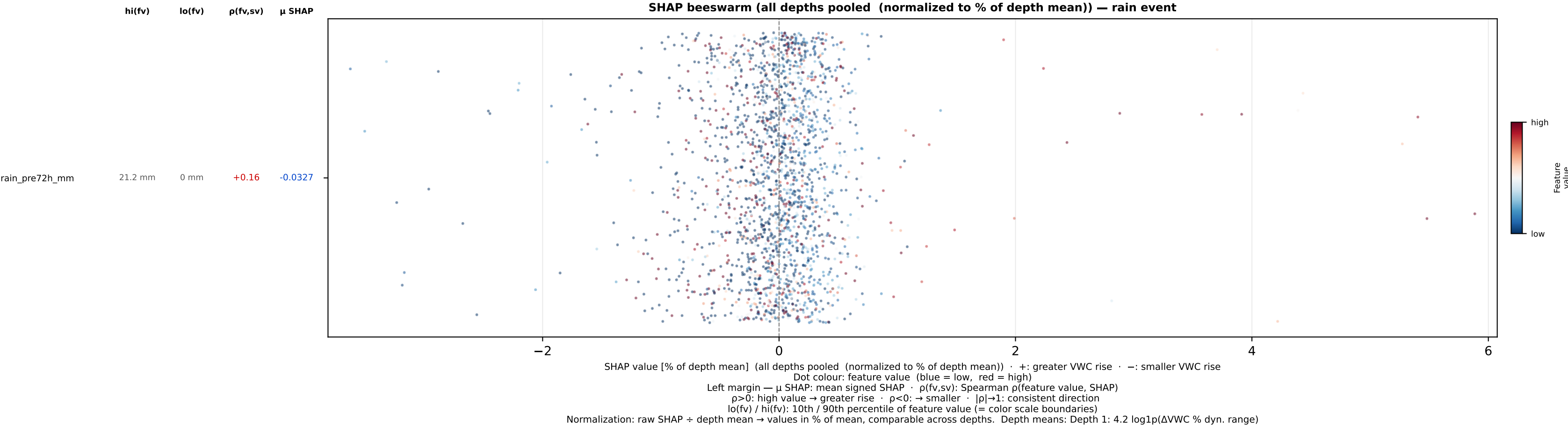
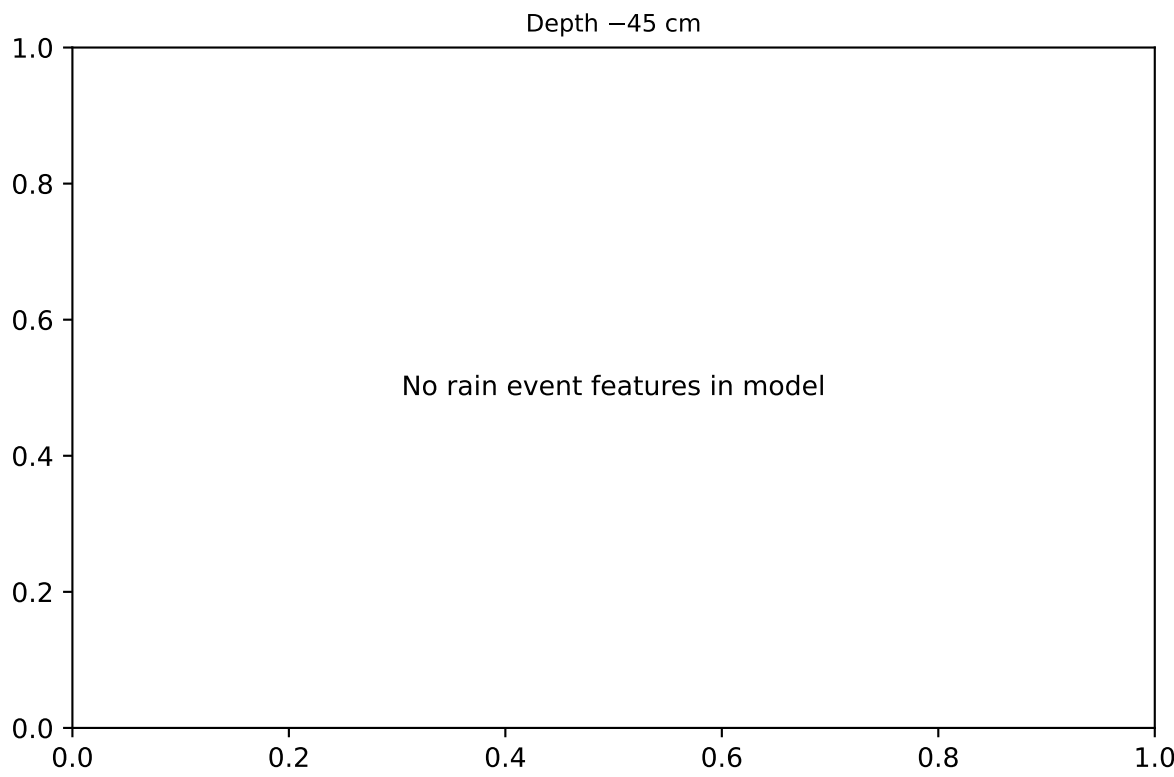
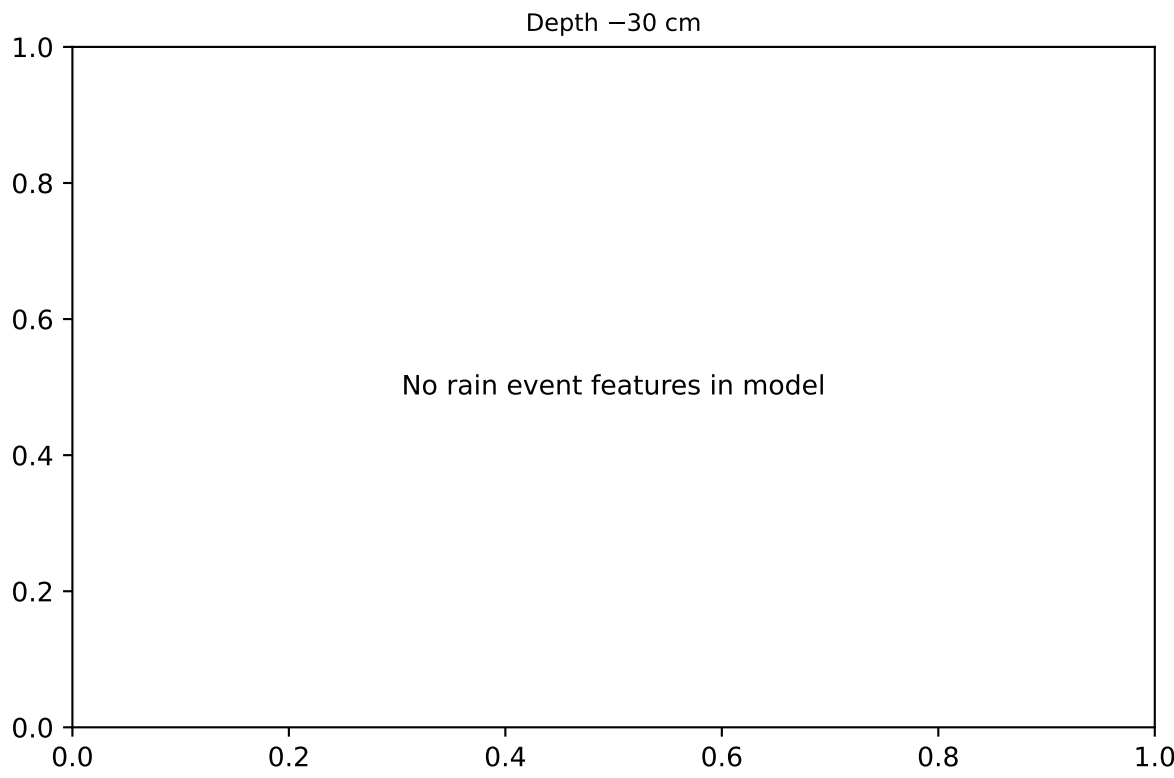
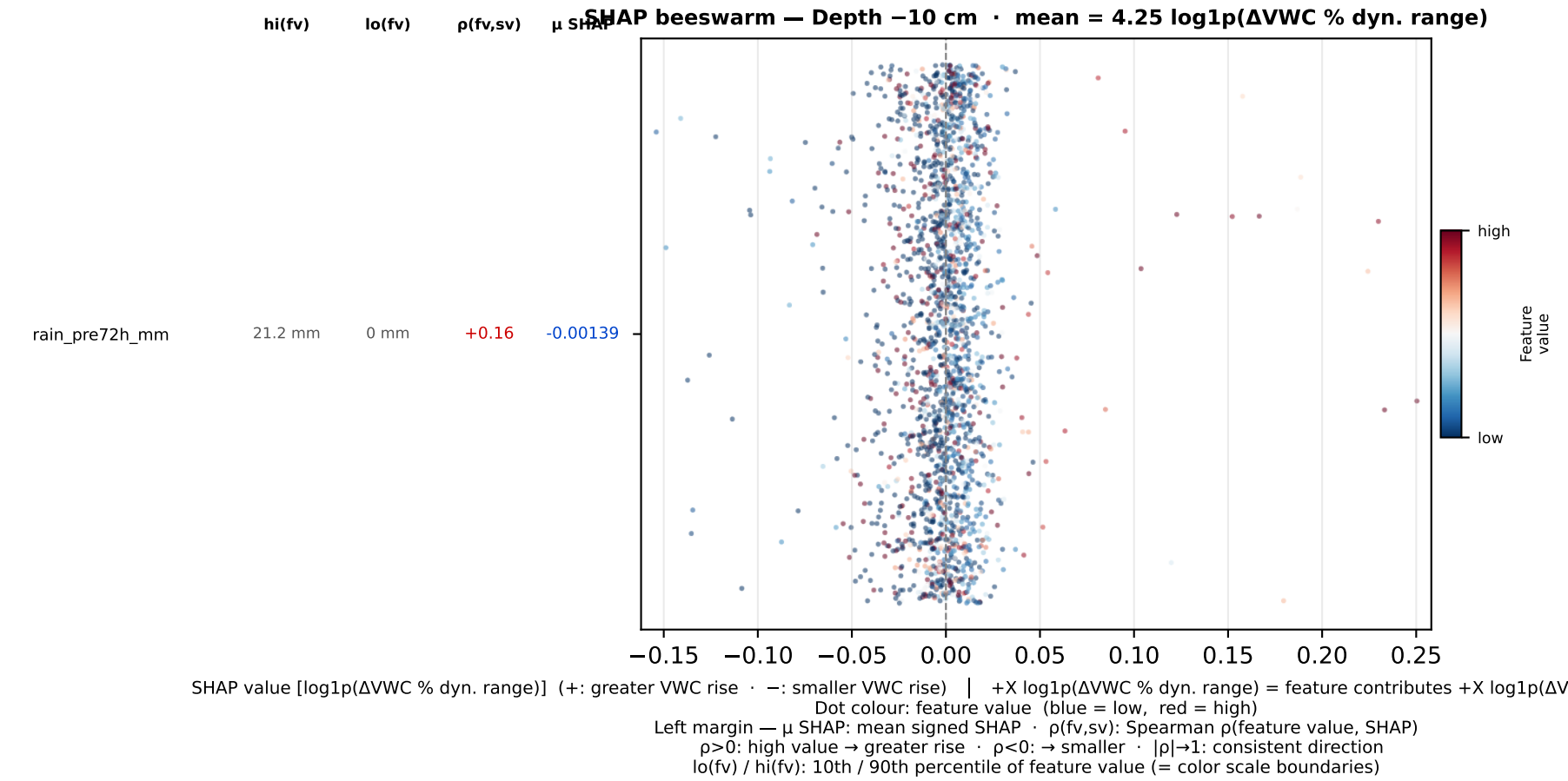
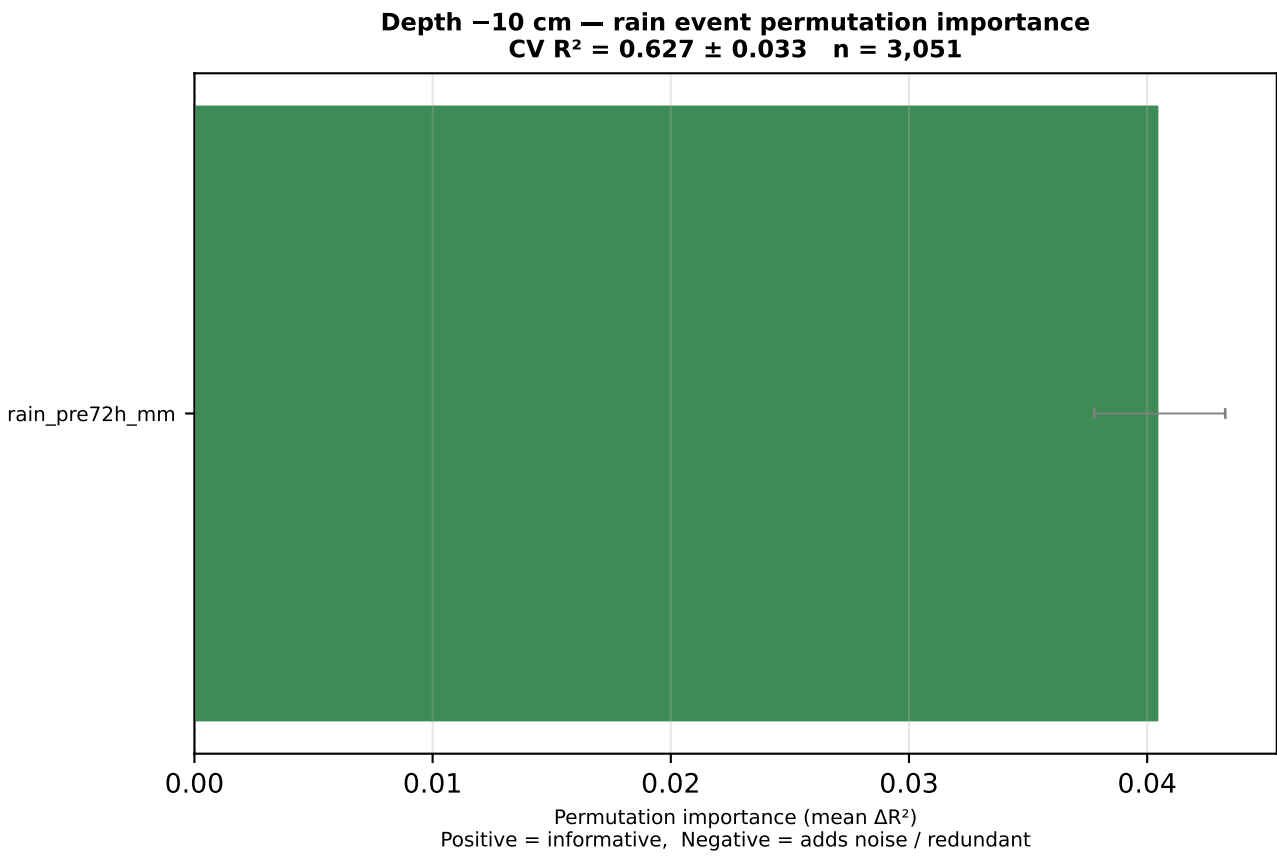


Tree Morphology Feature Analysis — Irrigation

Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (5,242 events)

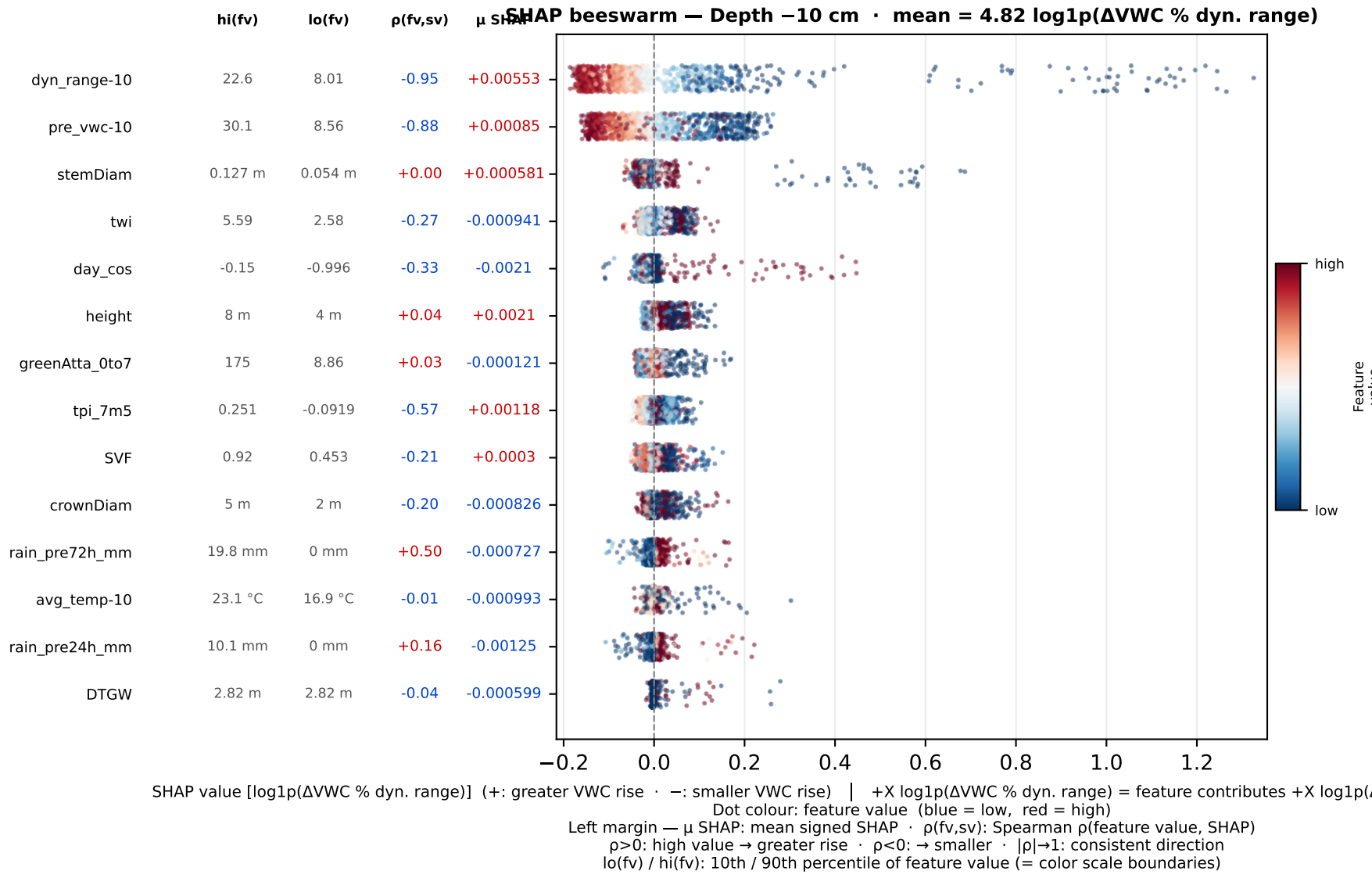
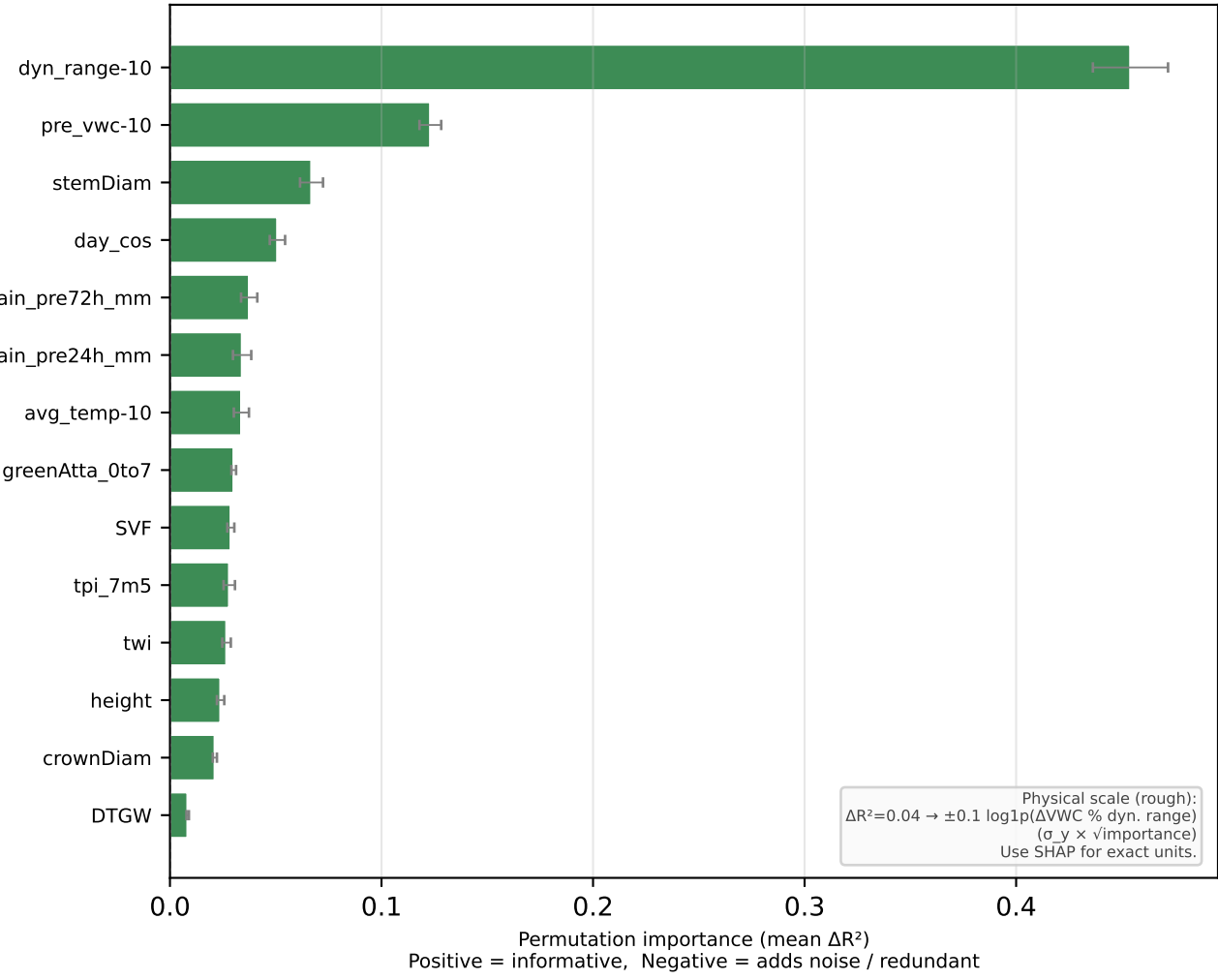
Features: crown diameter · height · stem diameter · vitality



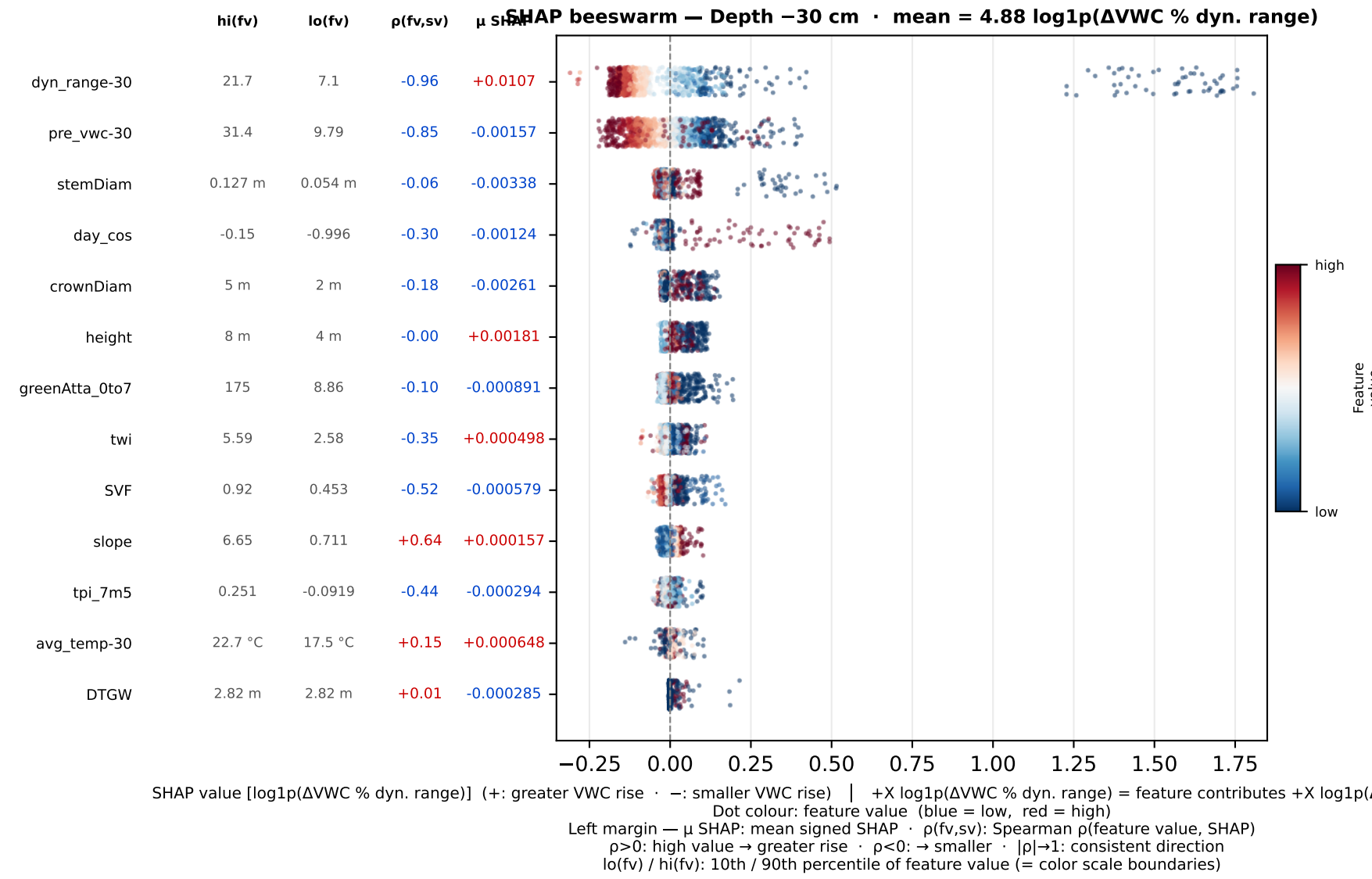
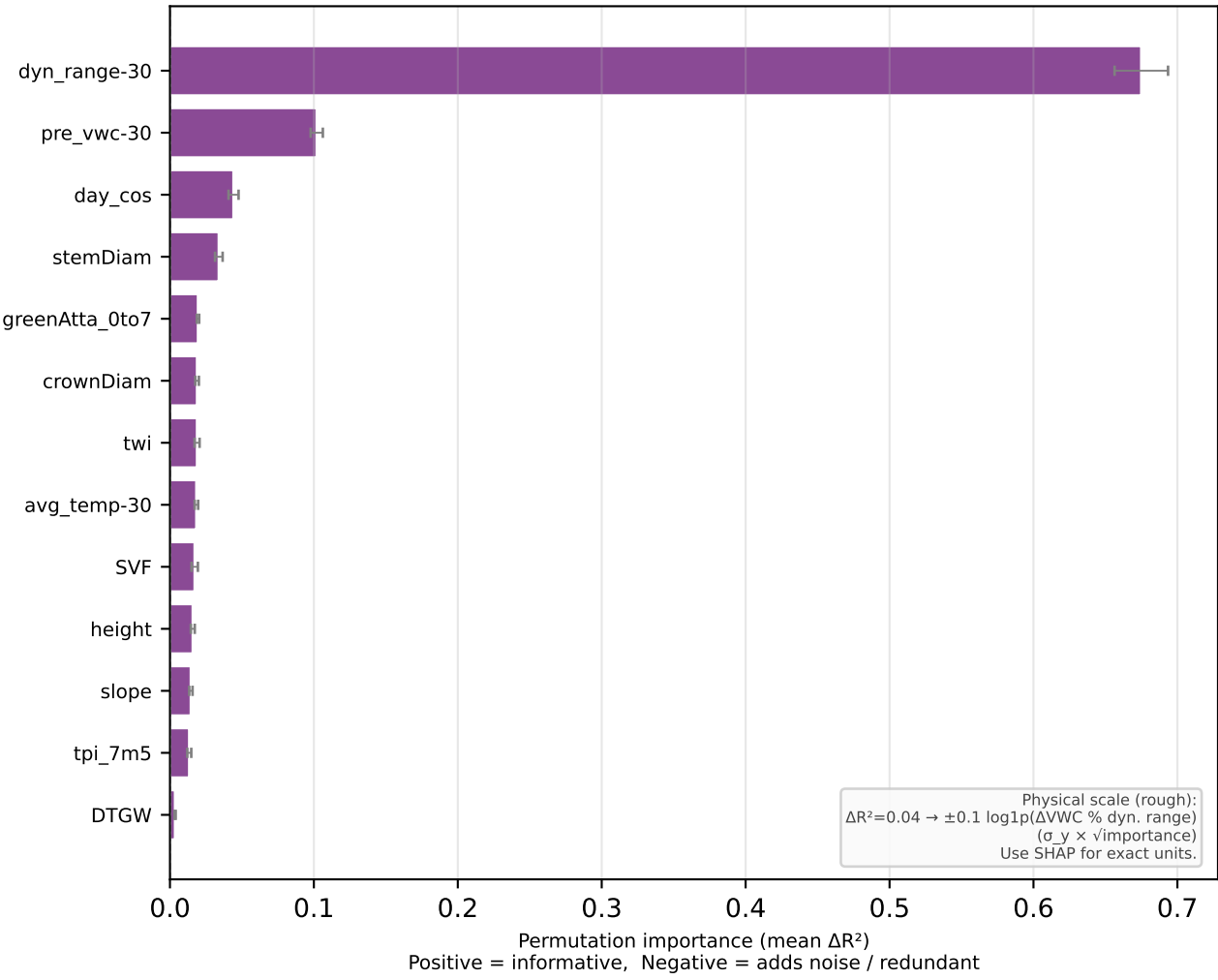


Random Forest Feature Importance — Irrigation strong
Target: ΔVWC as % of dynamic range at each depth (1,161 events)

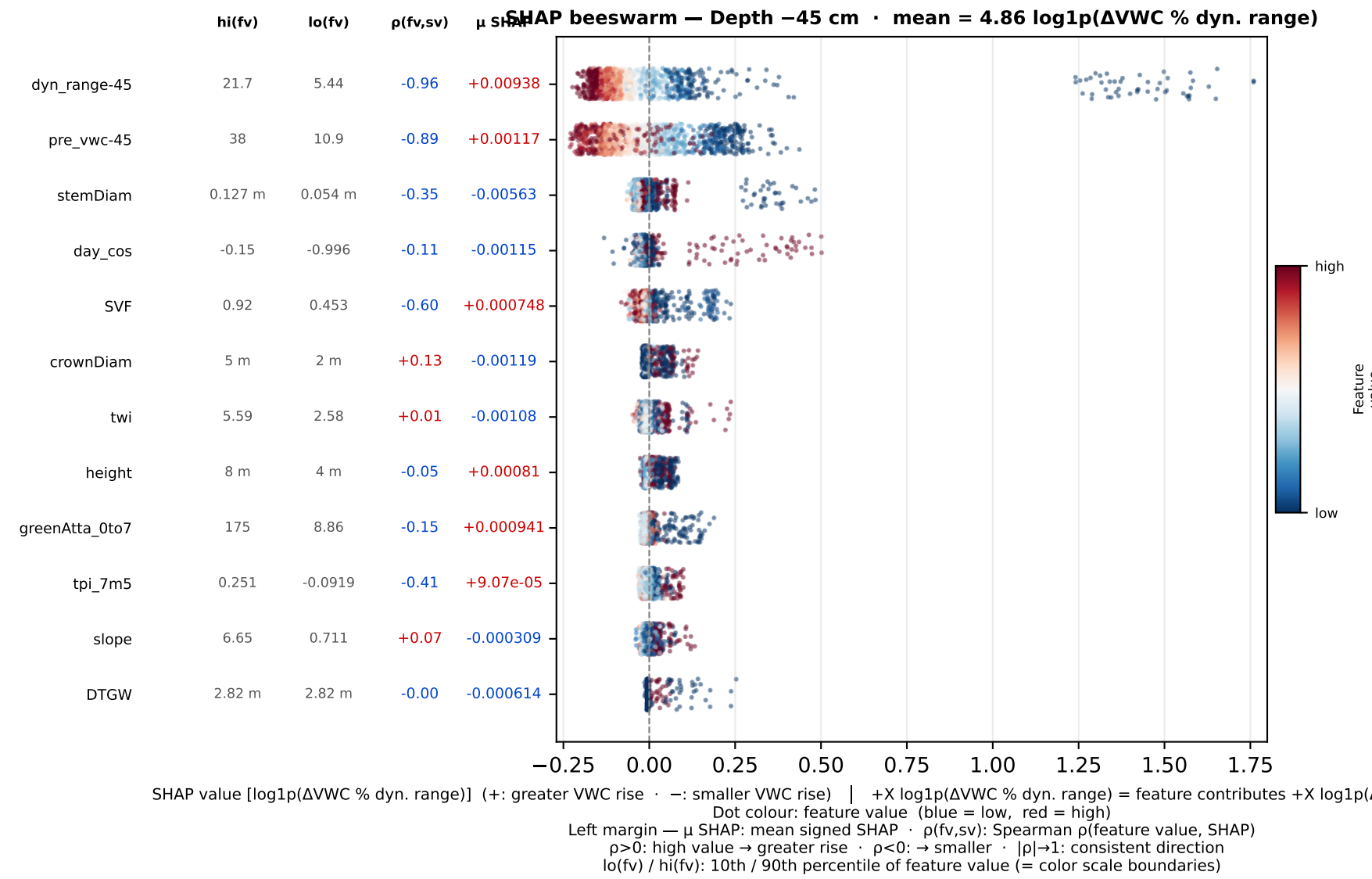
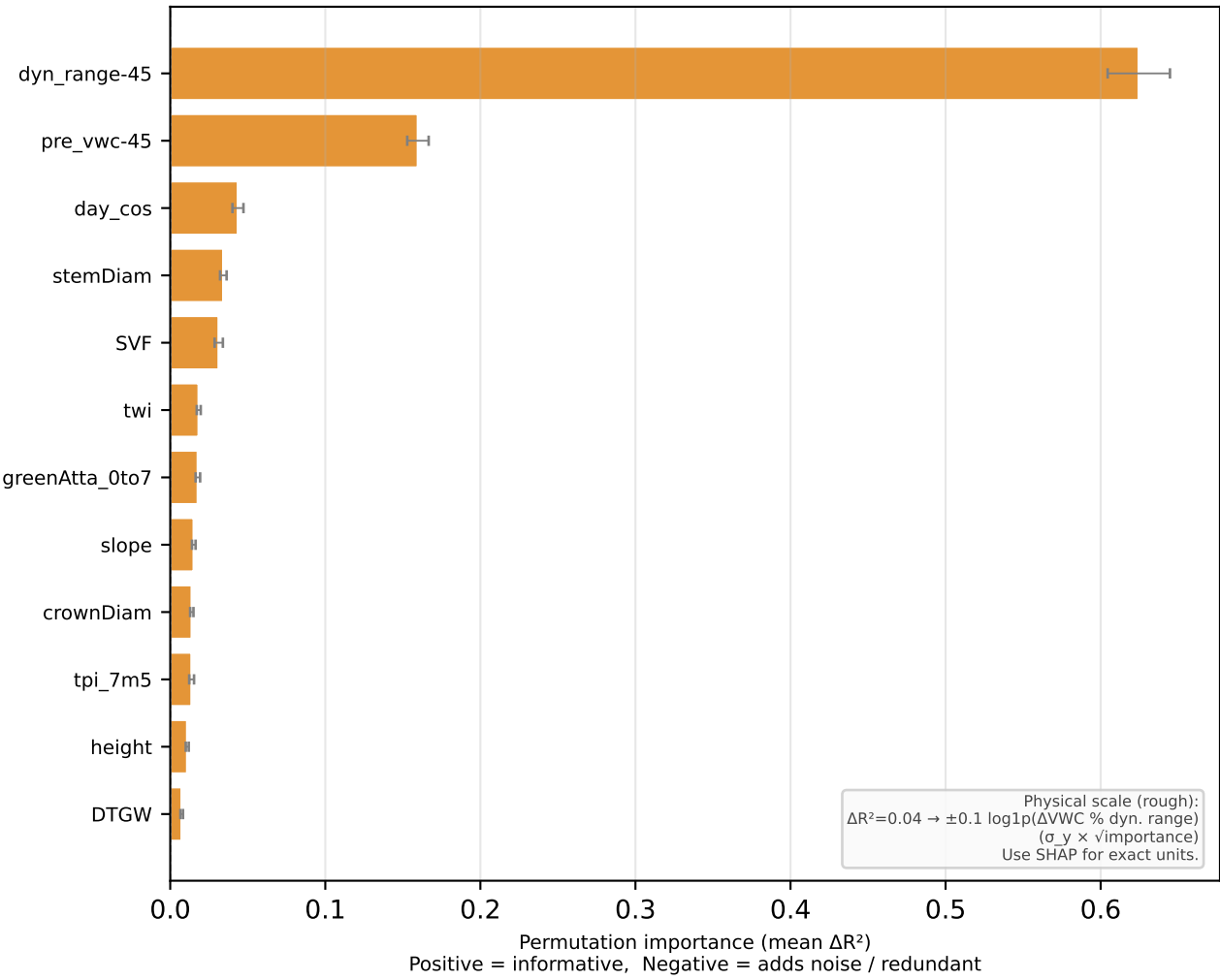
Depth -10 cm — permutation importance
CV R² = 0.583 ± 0.005 n = 1,161

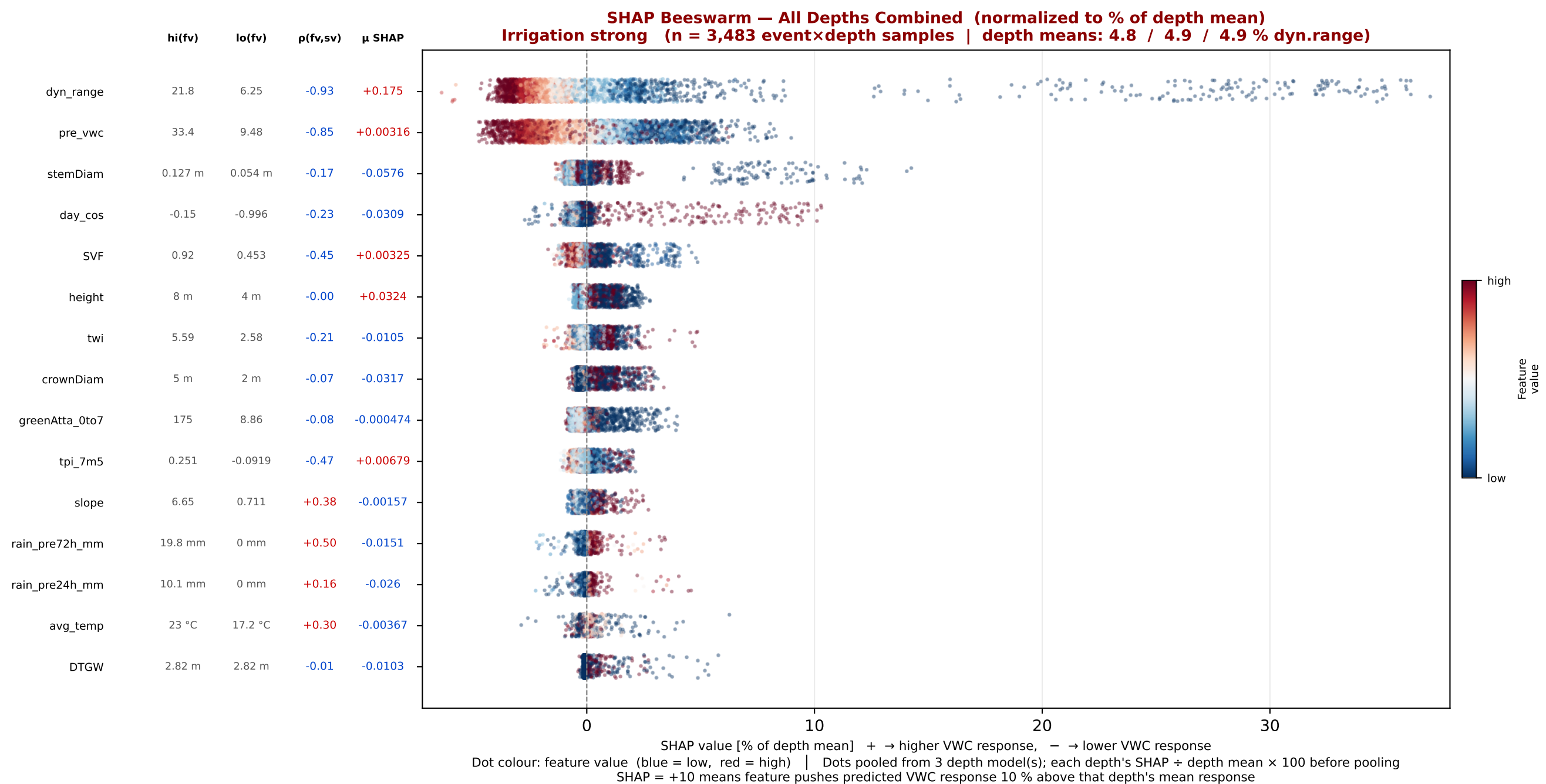


Depth -30 cm — permutation importance
CV R² = 0.718 ± 0.027 n = 1,161



Depth -45 cm — permutation importance
CV R² = 0.691 ± 0.025 n = 1,161





HOW TO READ THIS CHART (Combined depth view — SHAP normalized to % of depth mean)

SHAP value (x-axis) — unit: % of depth-specific mean VWC response

Each depth's SHAP values were divided by that depth's training mean before pooling.

This makes all depths directly comparable regardless of their absolute response magnitude.

- +10 % → this feature pushes the predicted VWC response 10 % above that depth's mean.
- -10 % → this feature pulls the predicted VWC response 10 % below that depth's mean.

Per-depth means used for normalization (3 depth(s)):

Depth 1: mean = 4.8 % dyn.range → SHAP +10 % of mean = +0.5 % dyn.range ≈ +0.087 % abs. VWC (at dyn.range = 18 %)

Depth 2: mean = 4.9 % dyn.range → SHAP +10 % of mean = +0.5 % dyn.range ≈ +0.088 % abs. VWC (at dyn.range = 18 %)

Depth 3: mean = 4.9 % dyn.range → SHAP +10 % of mean = +0.5 % dyn.range ≈ +0.088 % abs. VWC (at dyn.range = 18 %)

Example — top feature "dyn_range" (normalized SHAP IQR: -2.6 to +0.5 % of depth mean):

Low feature value → SHAP ≈ -2.6% High feature value → SHAP ≈ +0.5%

Difference: ~3.1 % of each depth's mean → in physical units per depth:

Depth 1 (mean 4.8 % dyn.range): Δ ≈ 0.1 % dyn.range ≈ 0.027 % abs. VWC

Depth 2 (mean 4.9 % dyn.range): Δ ≈ 0.2 % dyn.range ≈ 0.027 % abs. VWC

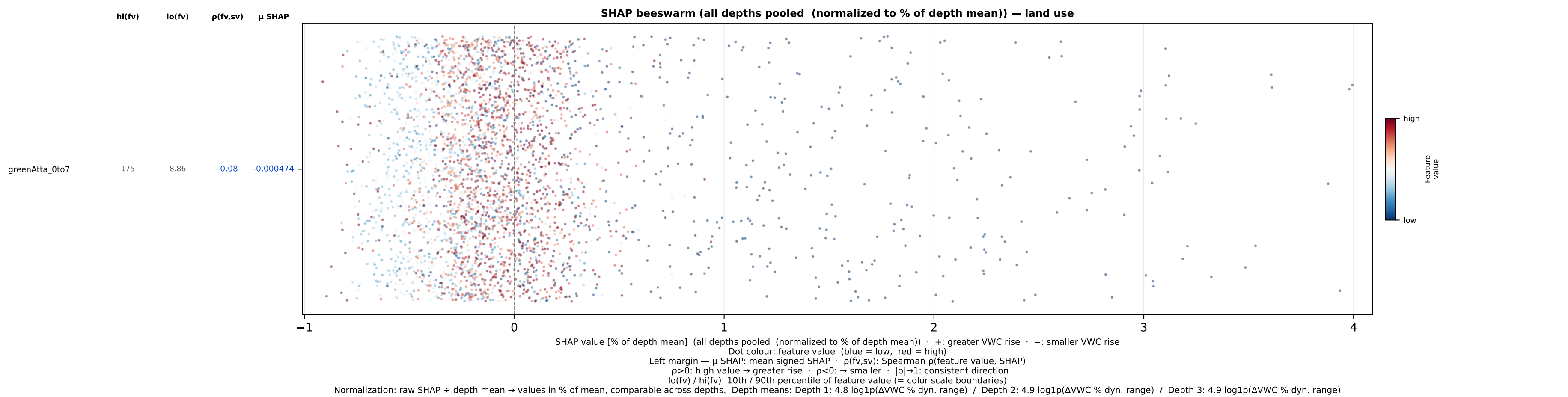
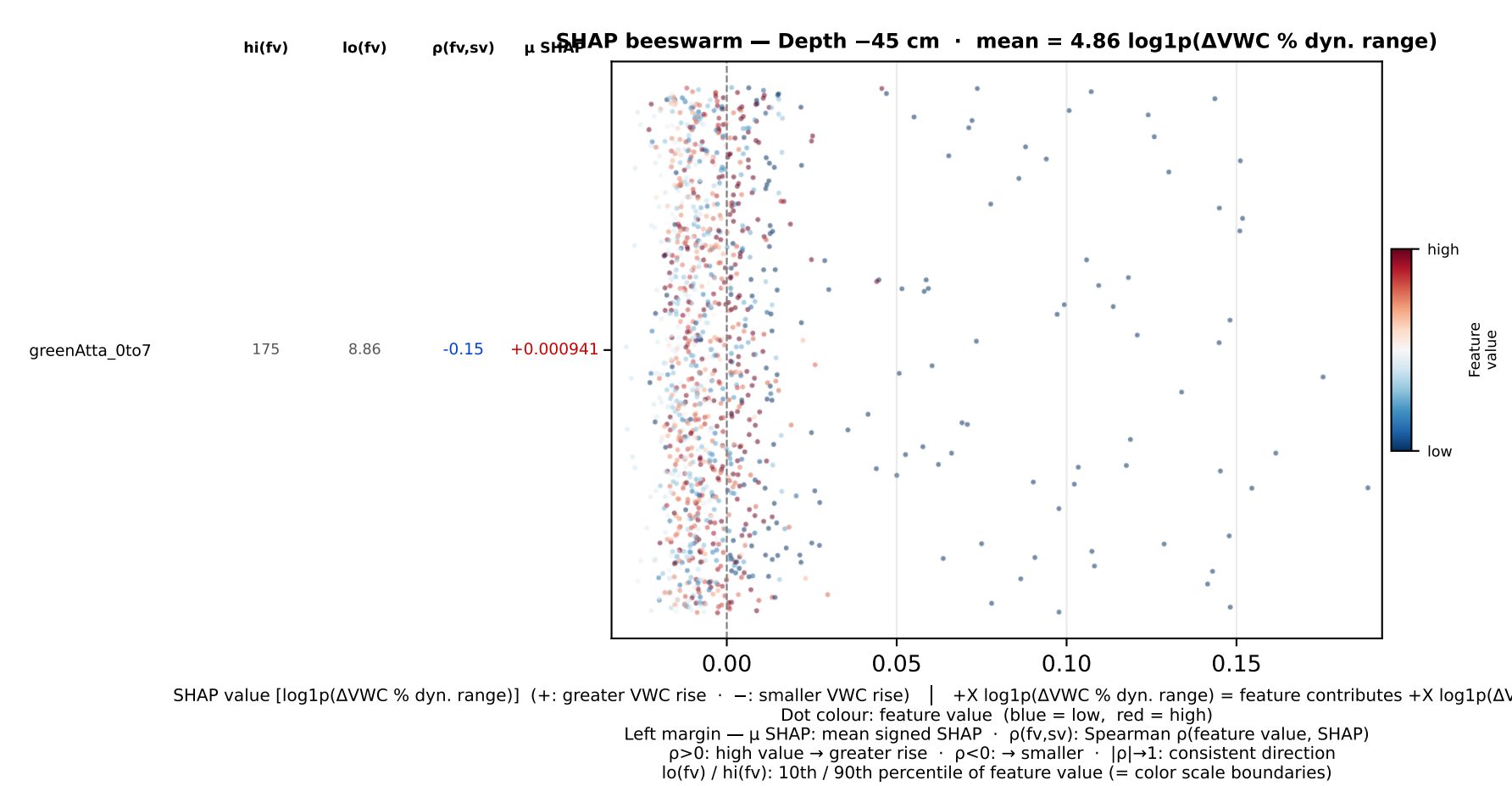
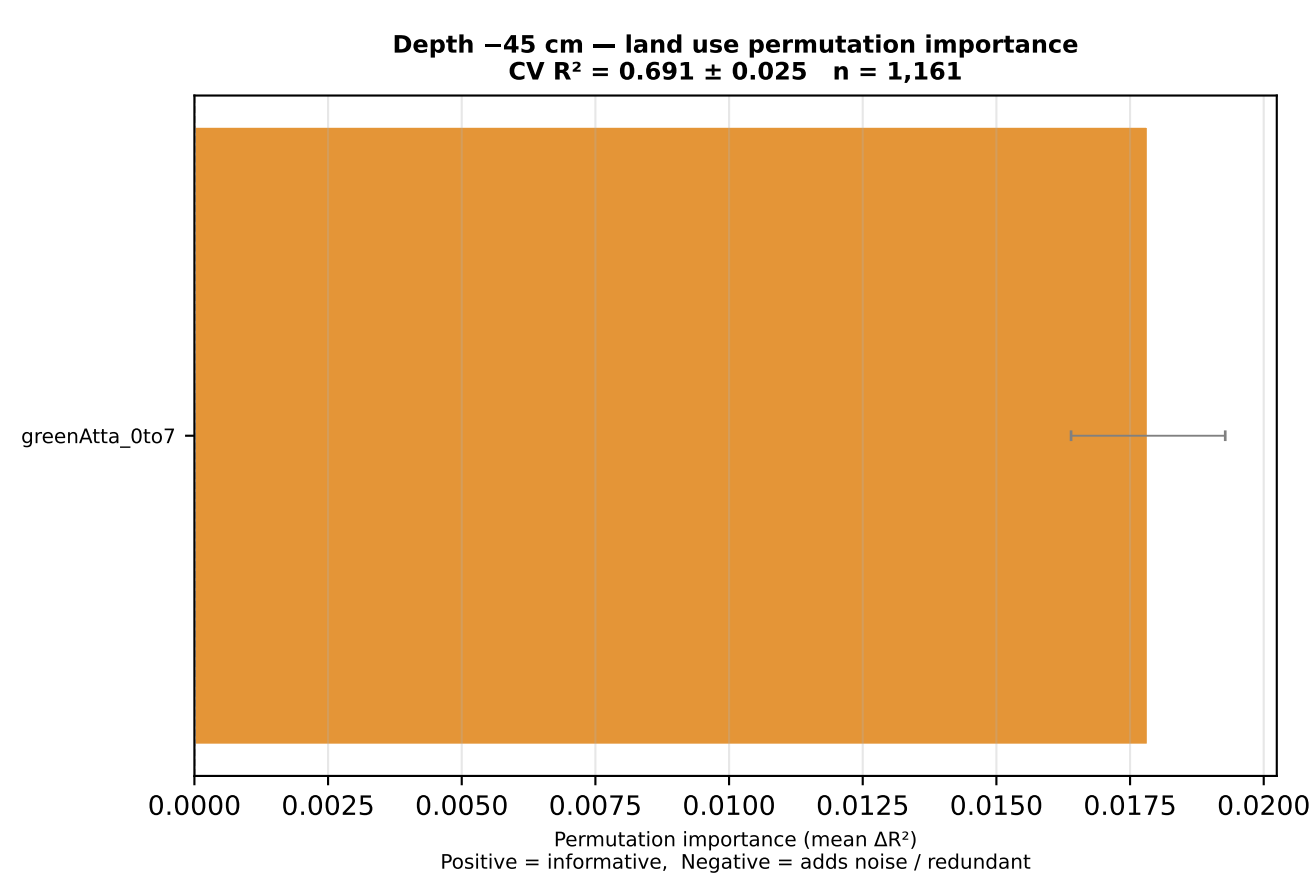
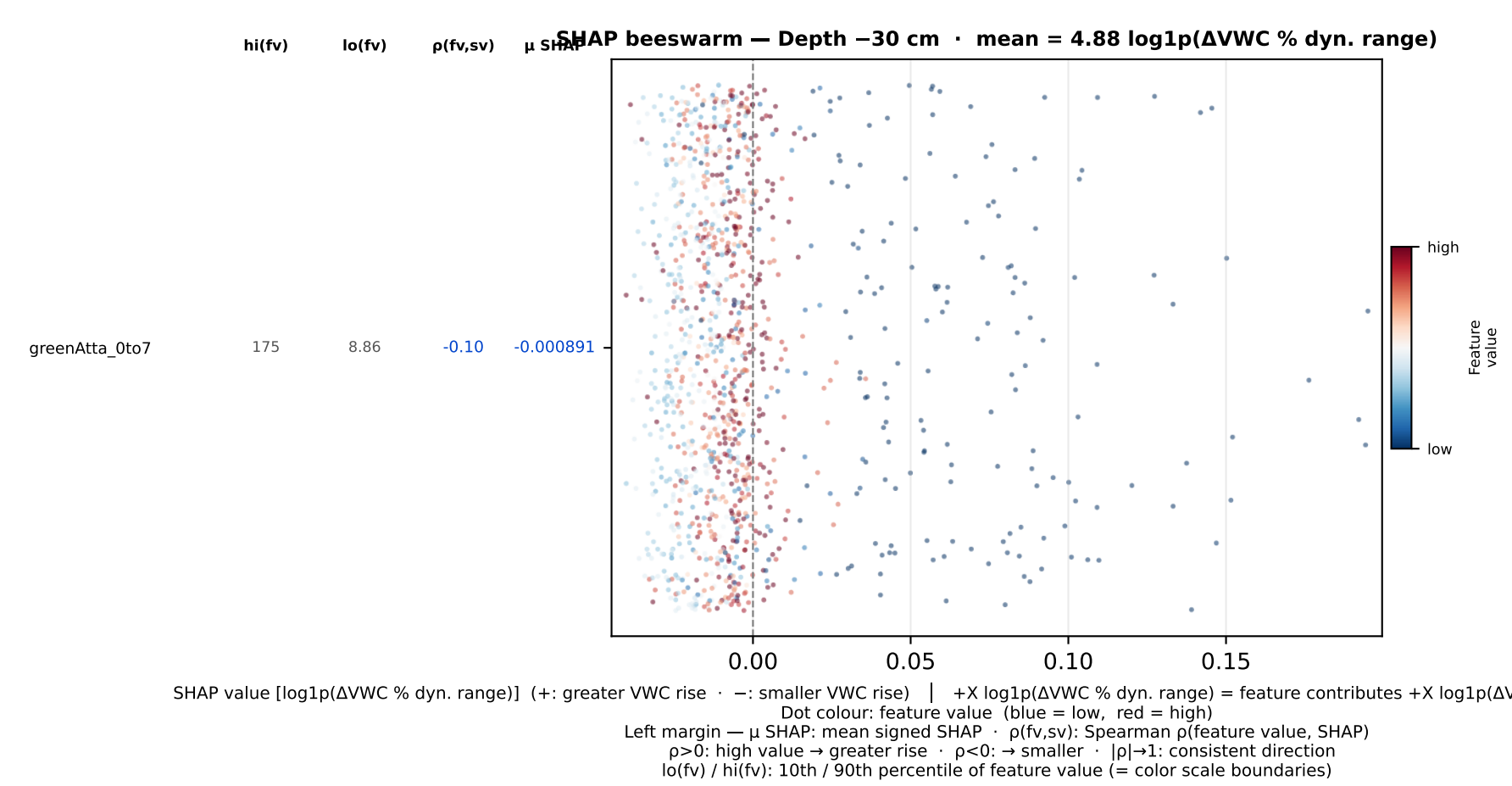
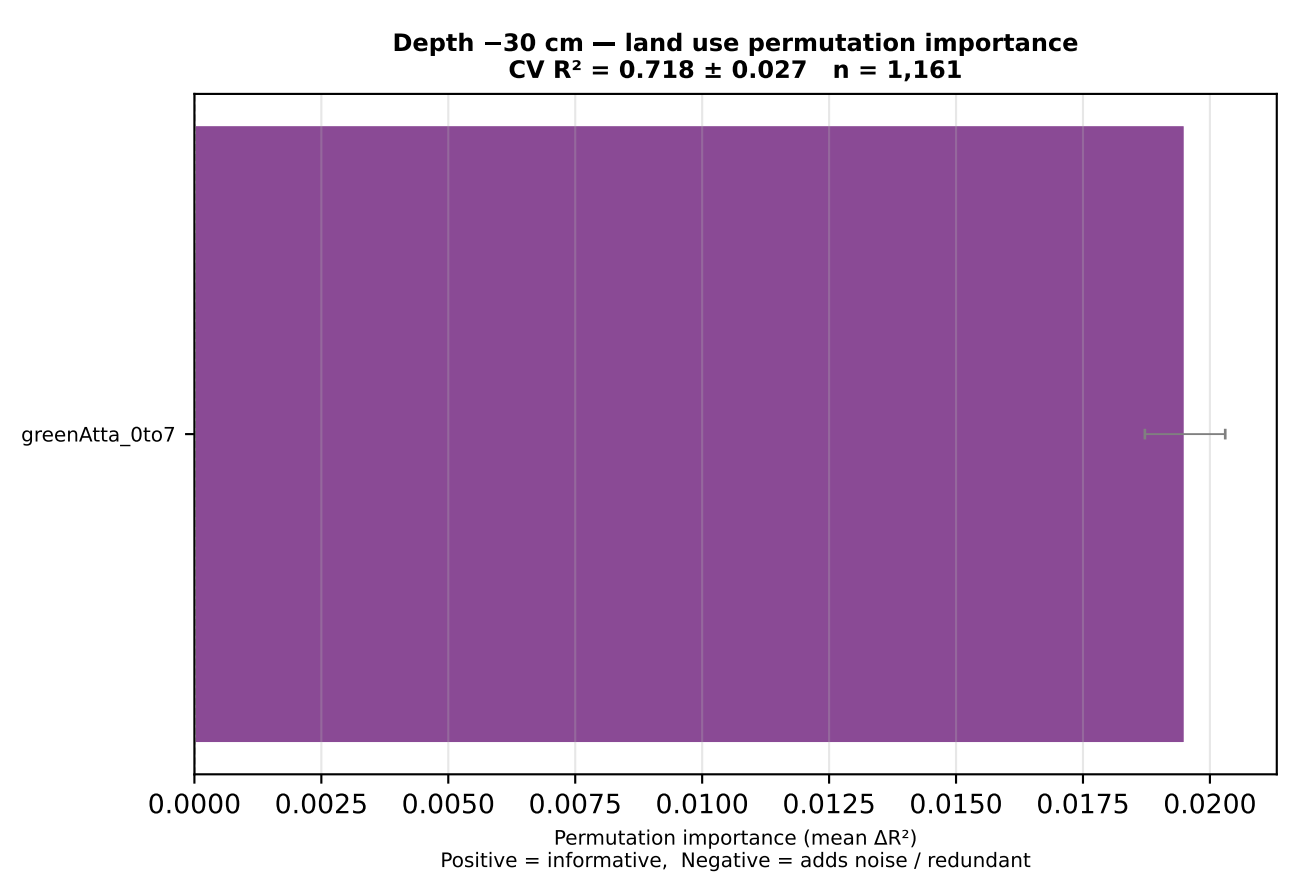
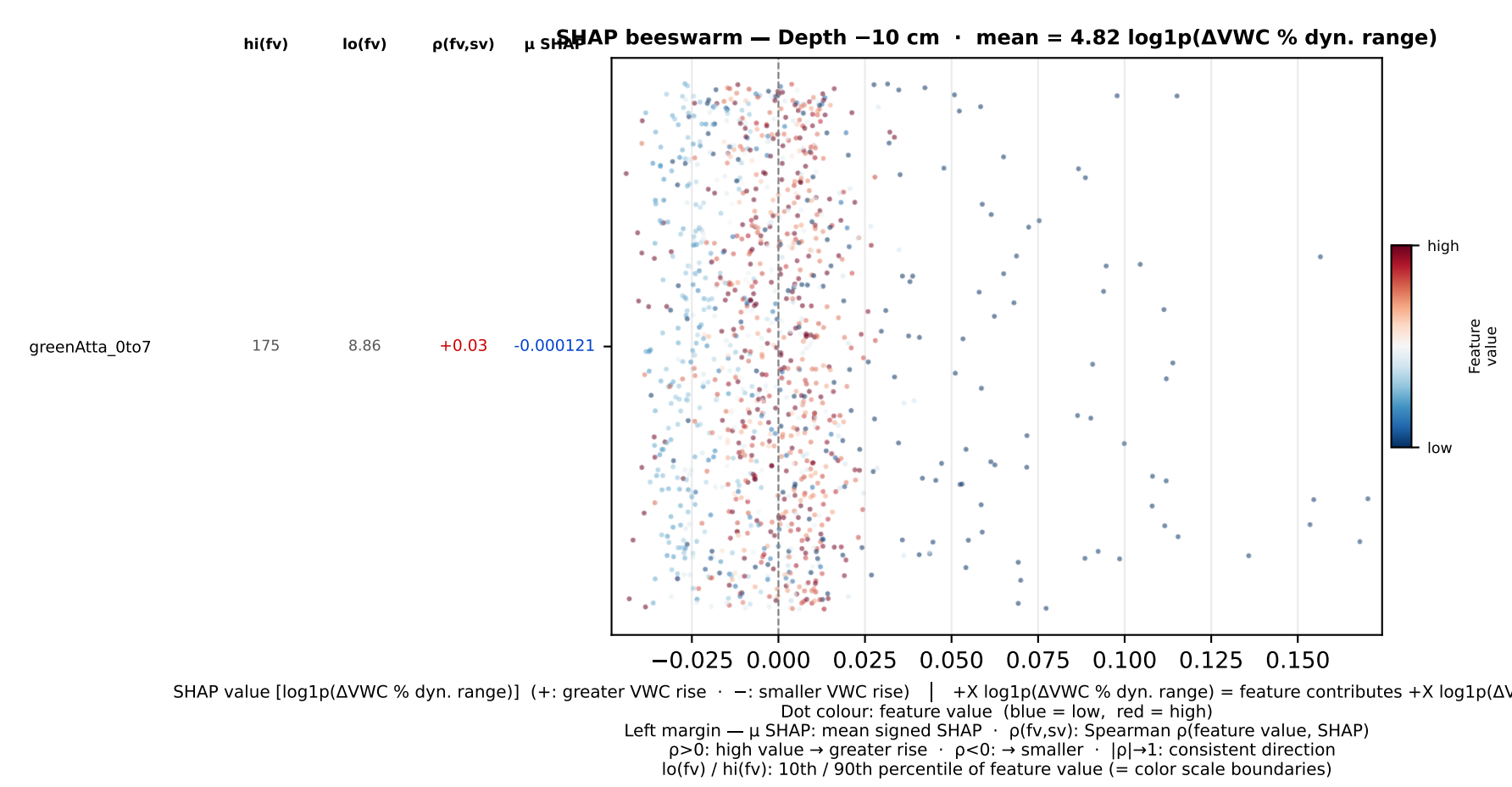
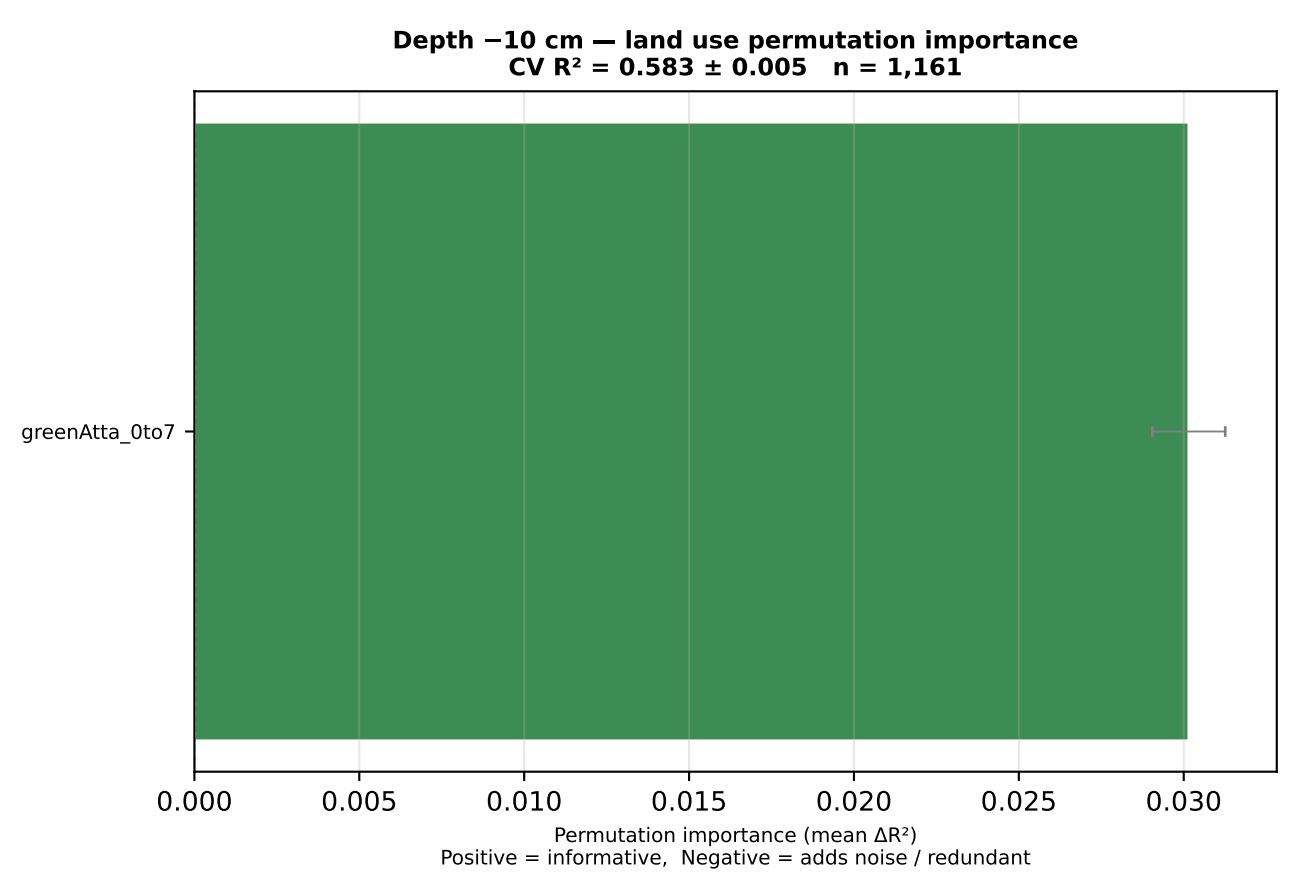
Depth 3 (mean 4.9 % dyn.range): Δ ≈ 0.2 % dyn.range ≈ 0.027 % abs. VWC

Total samples: 3,483 pooled across 3 depth(s) | μ SHAP and ρ in the left margin are computed on the normalized values.

Permutation importance (bar charts on preceding pages) — unit: mean ΔR² (NOT in % dyn.range!)

Physical scale estimate per depth: contribution ≈ σ_y × √importance

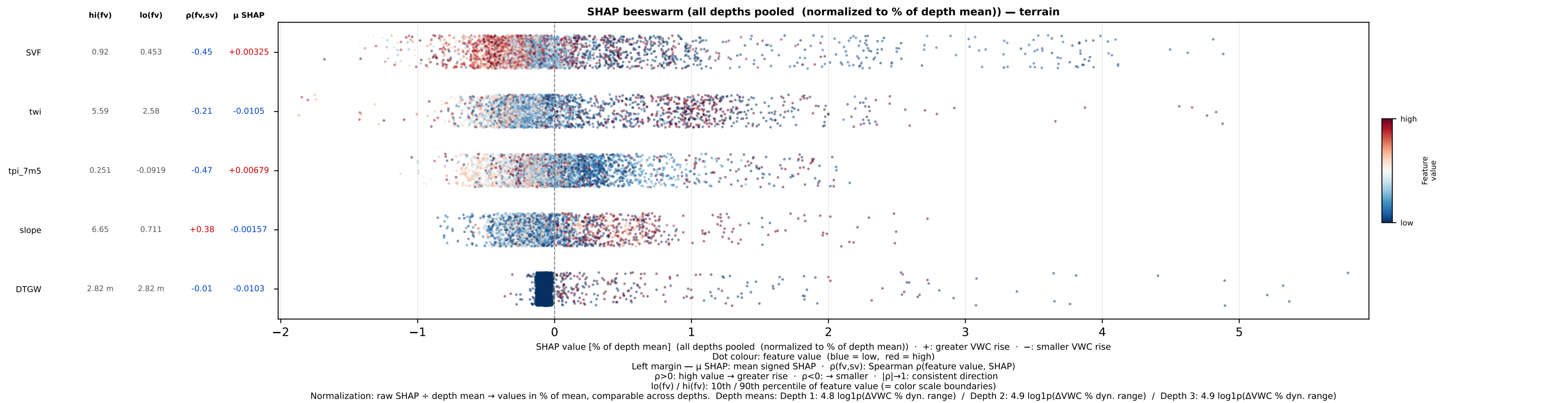
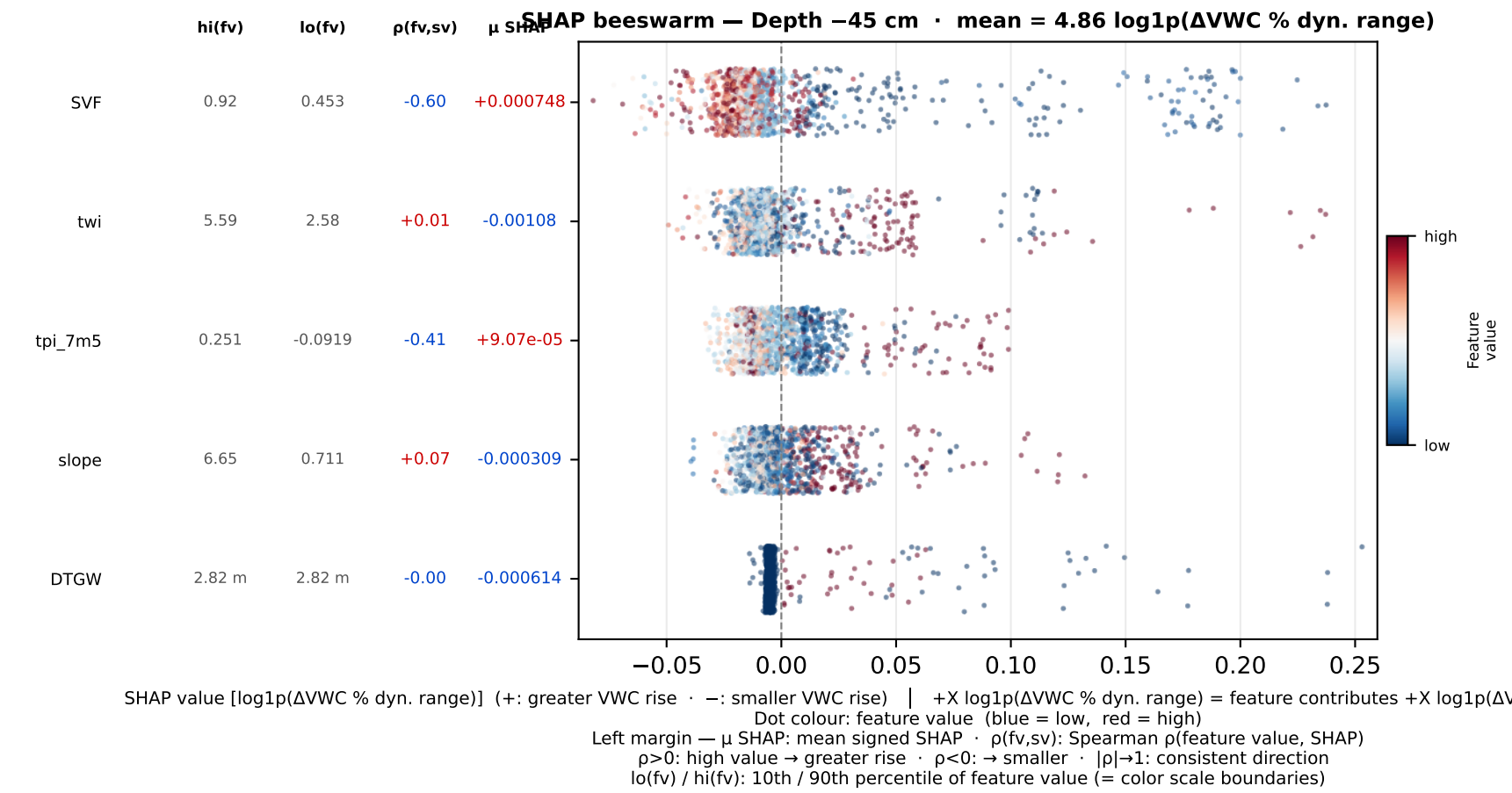
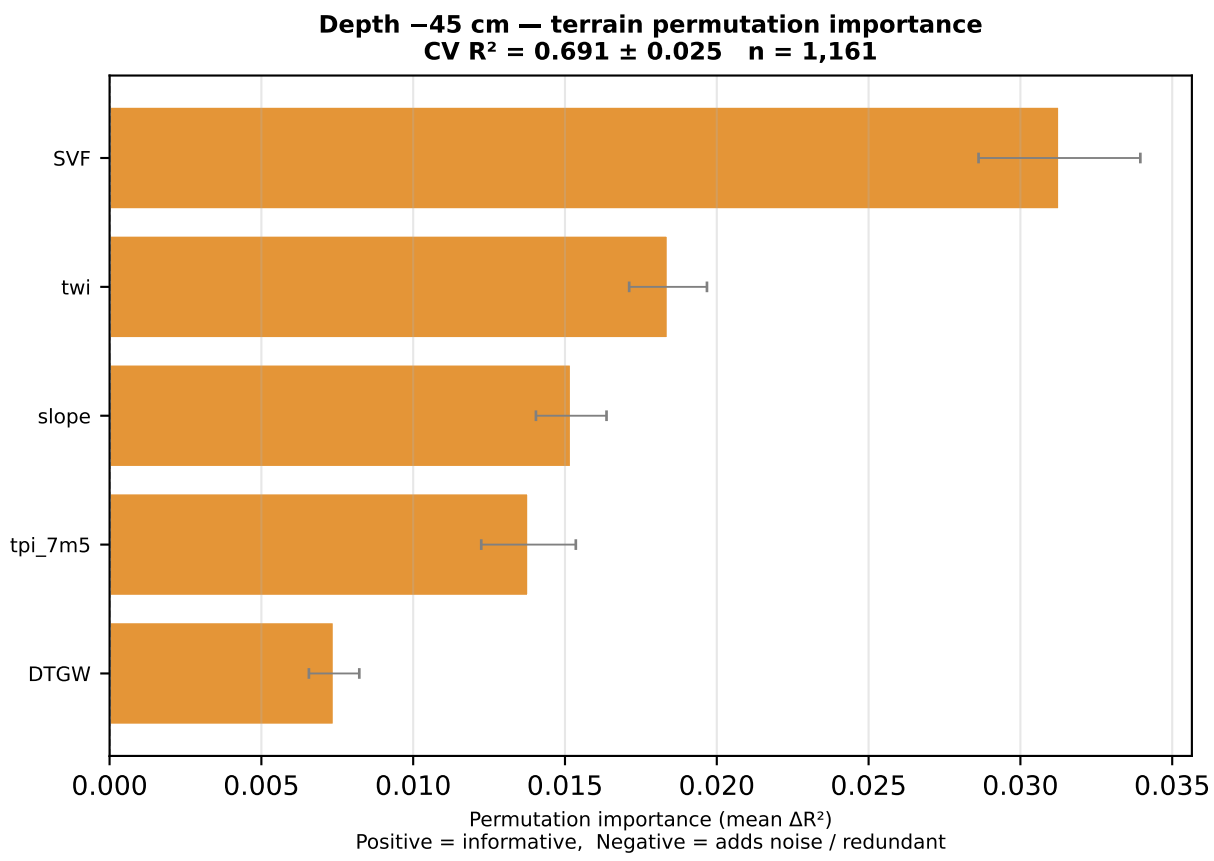
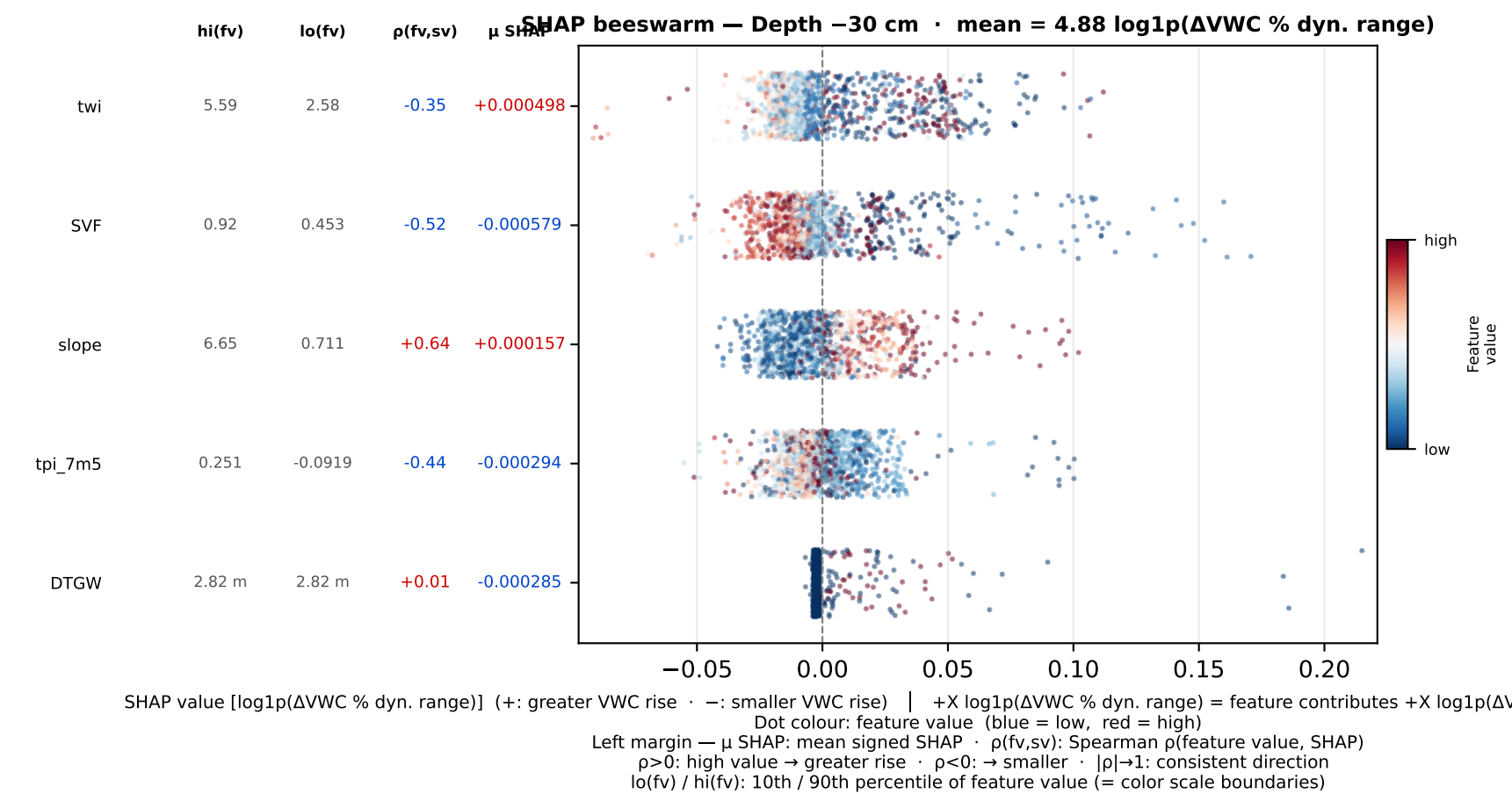
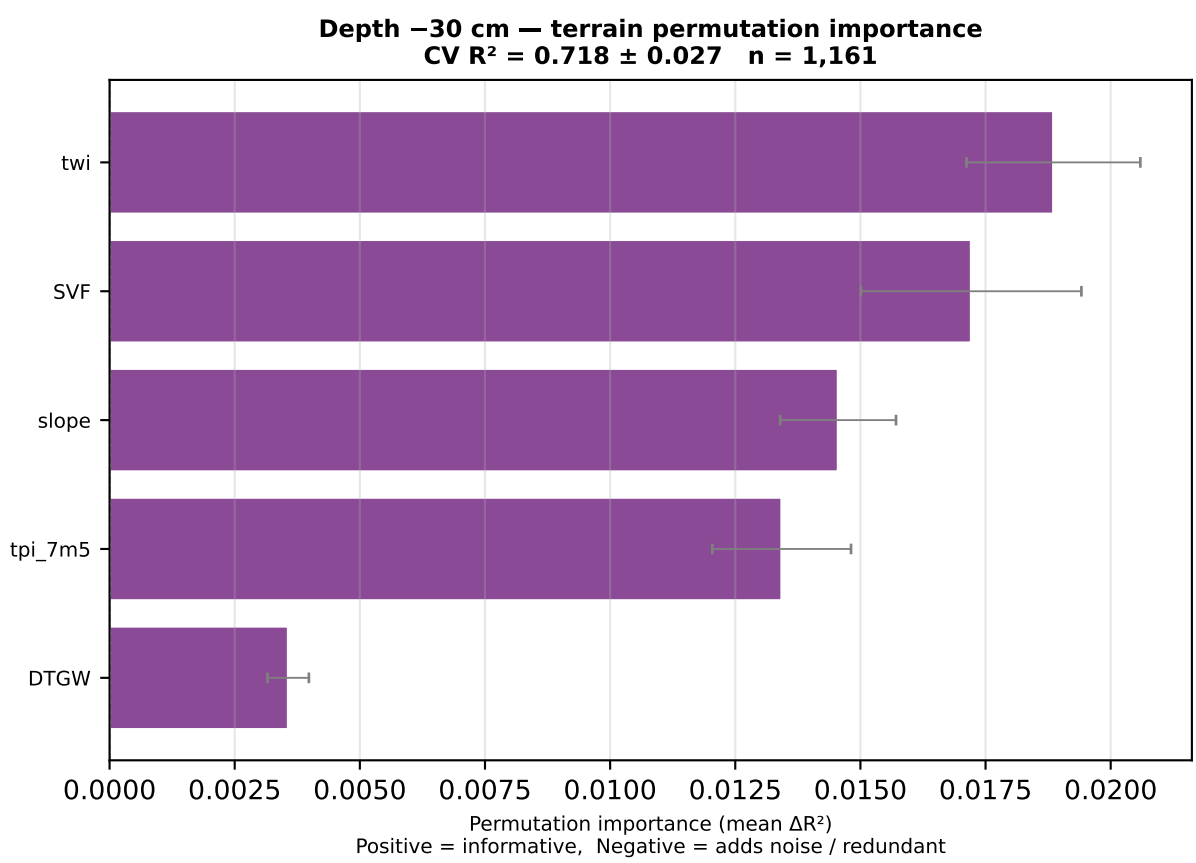
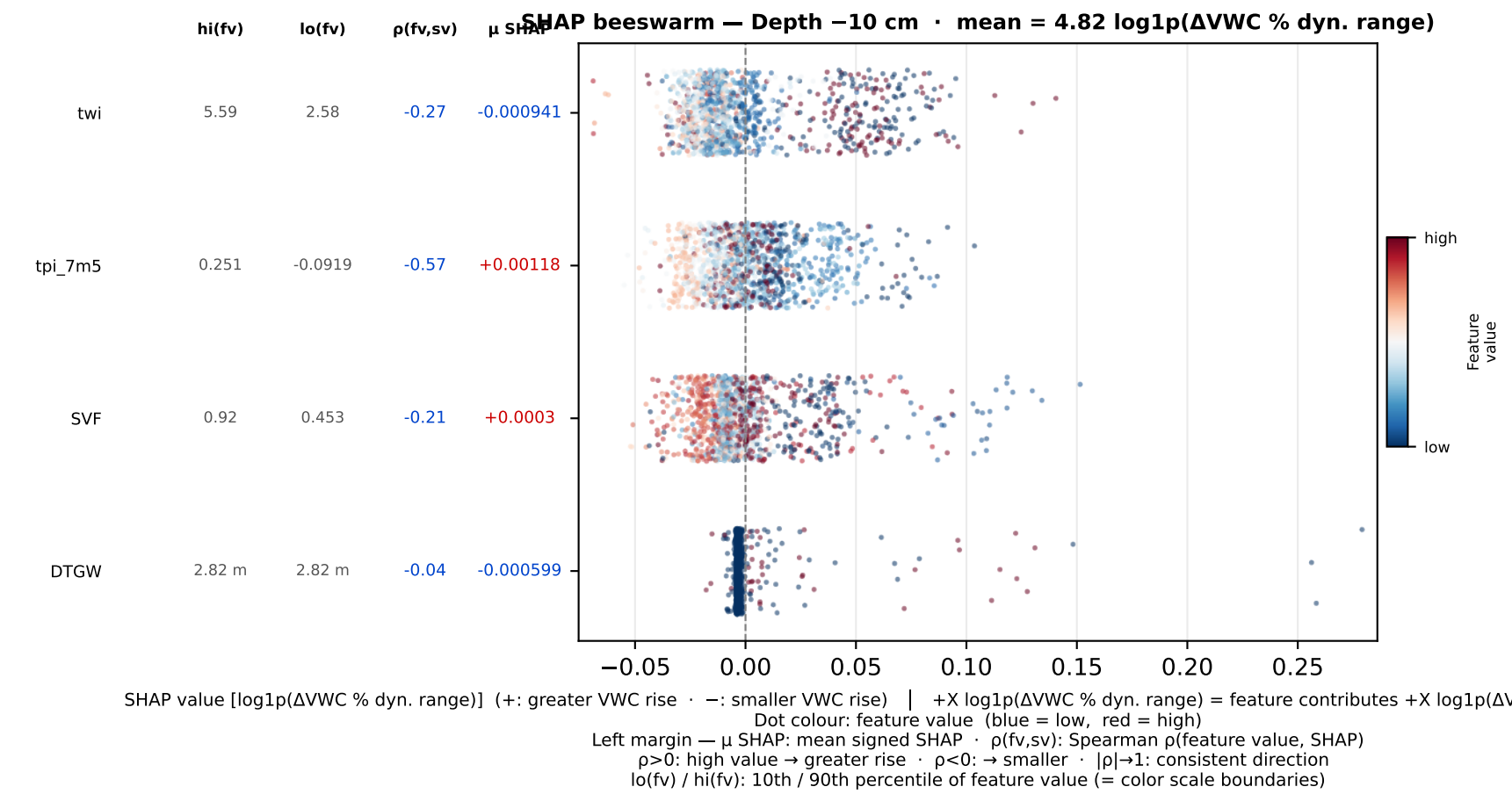
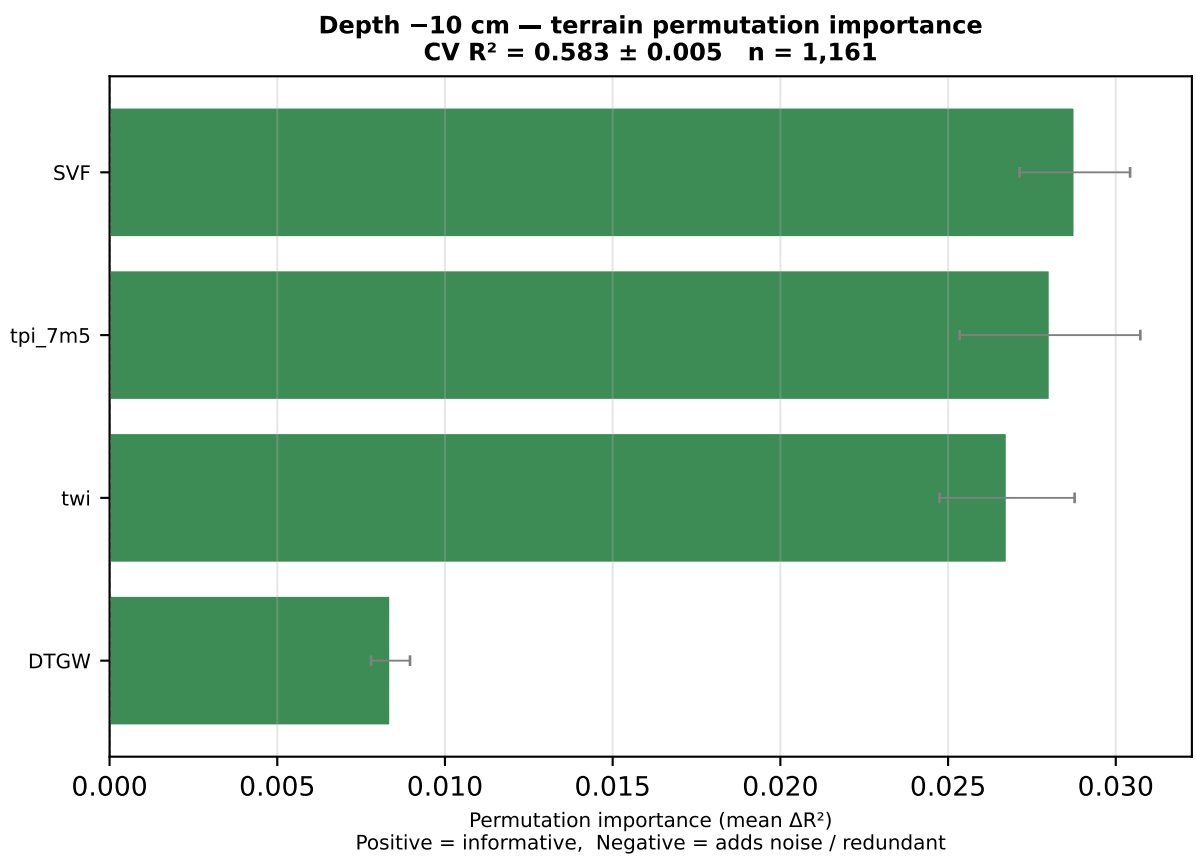
(use SHAP values, not permutation importance, for direct attribution in physical units)



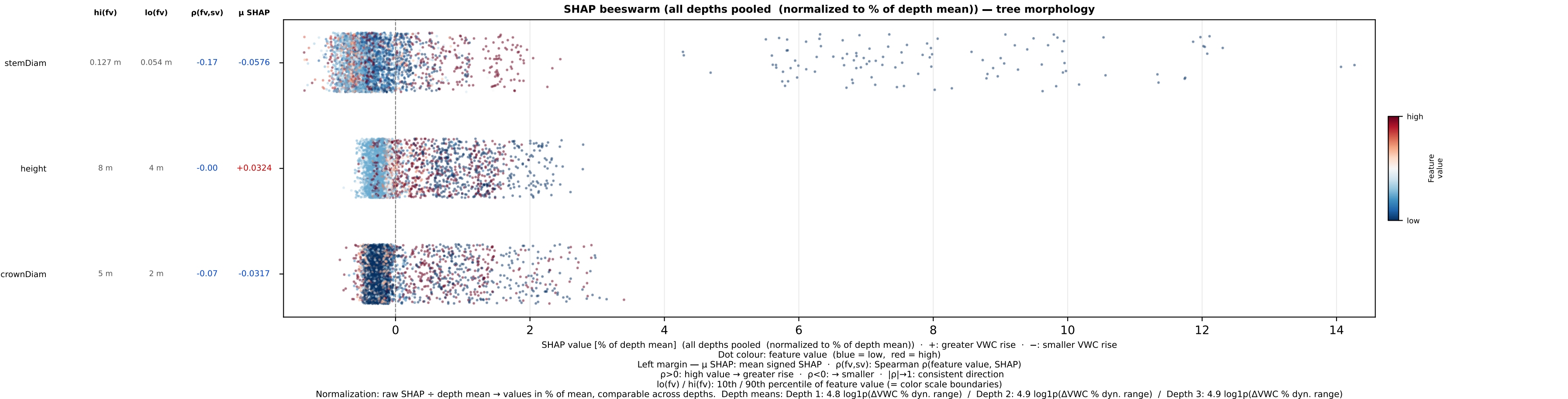
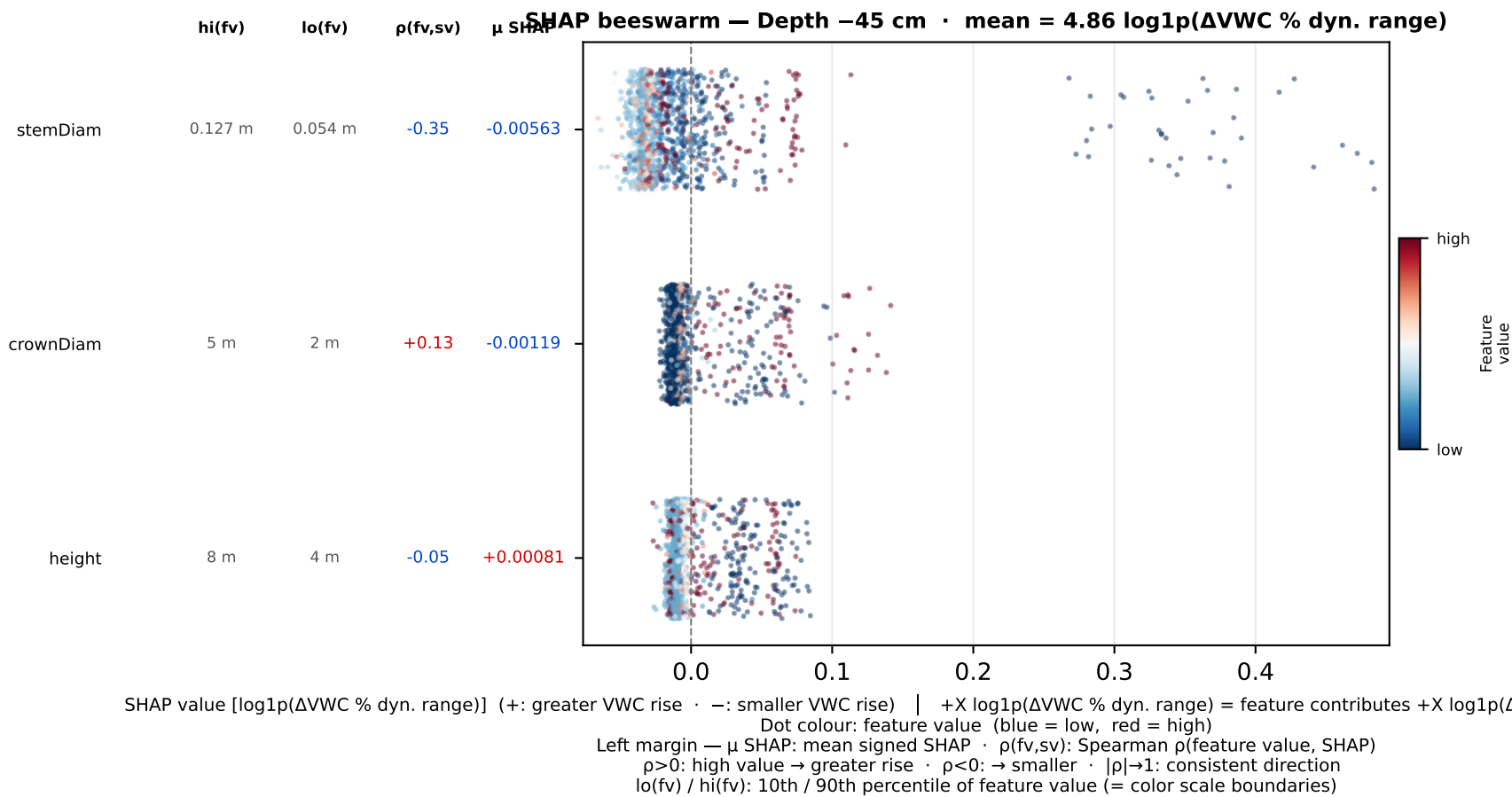
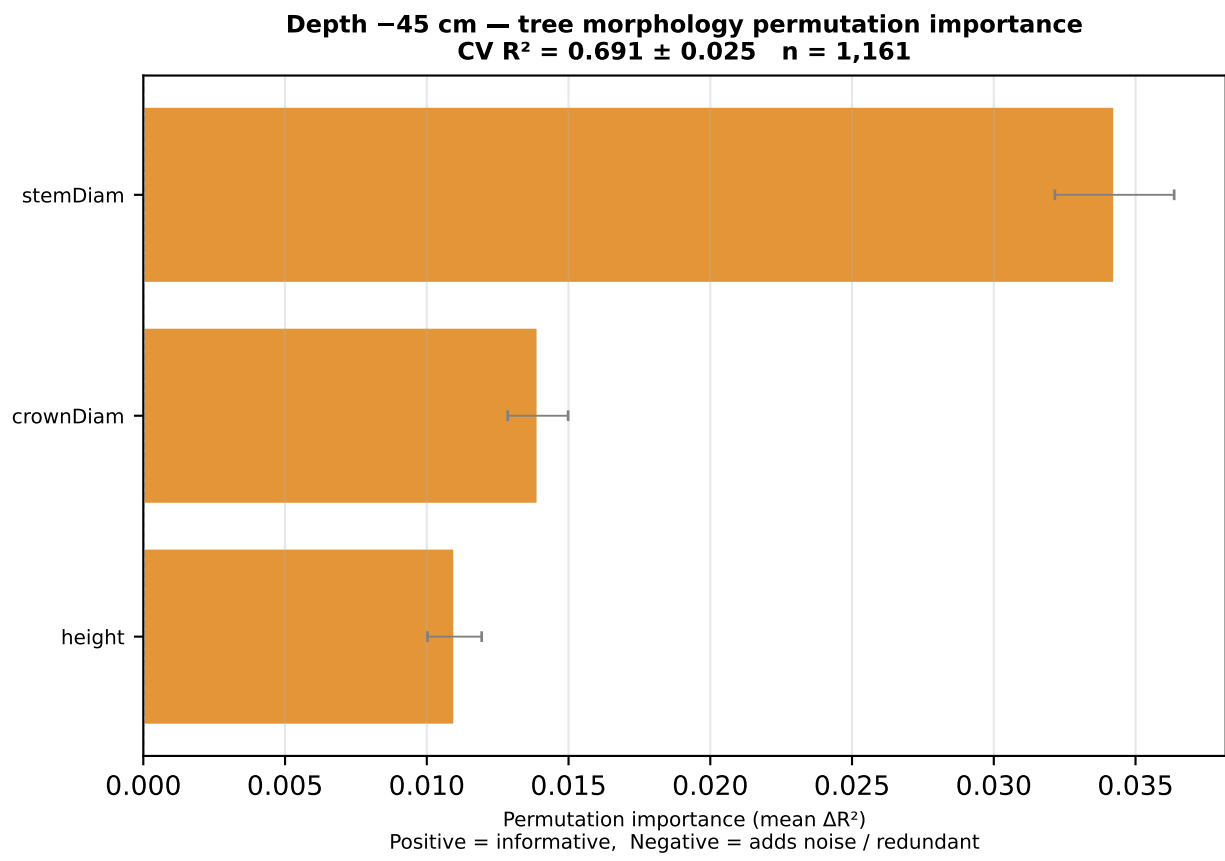
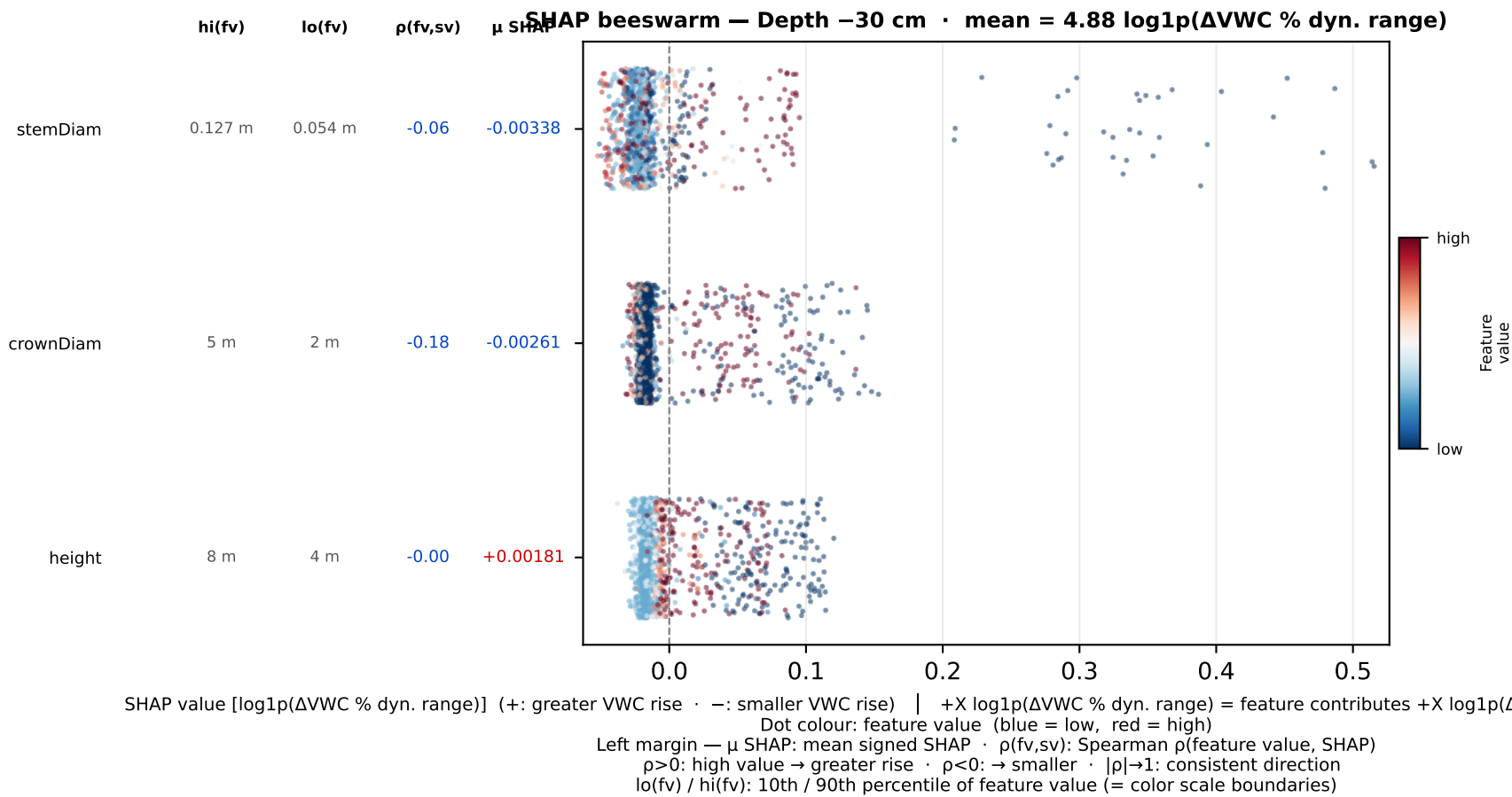
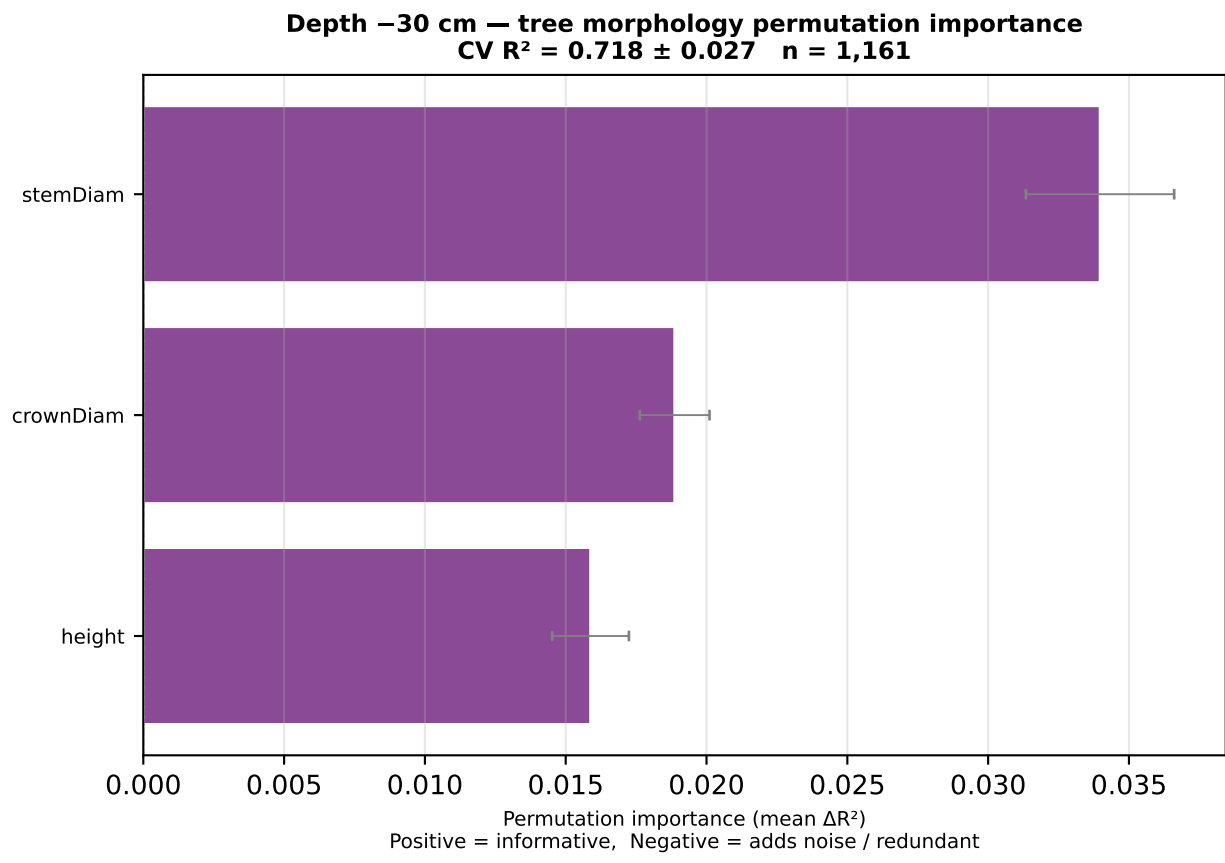
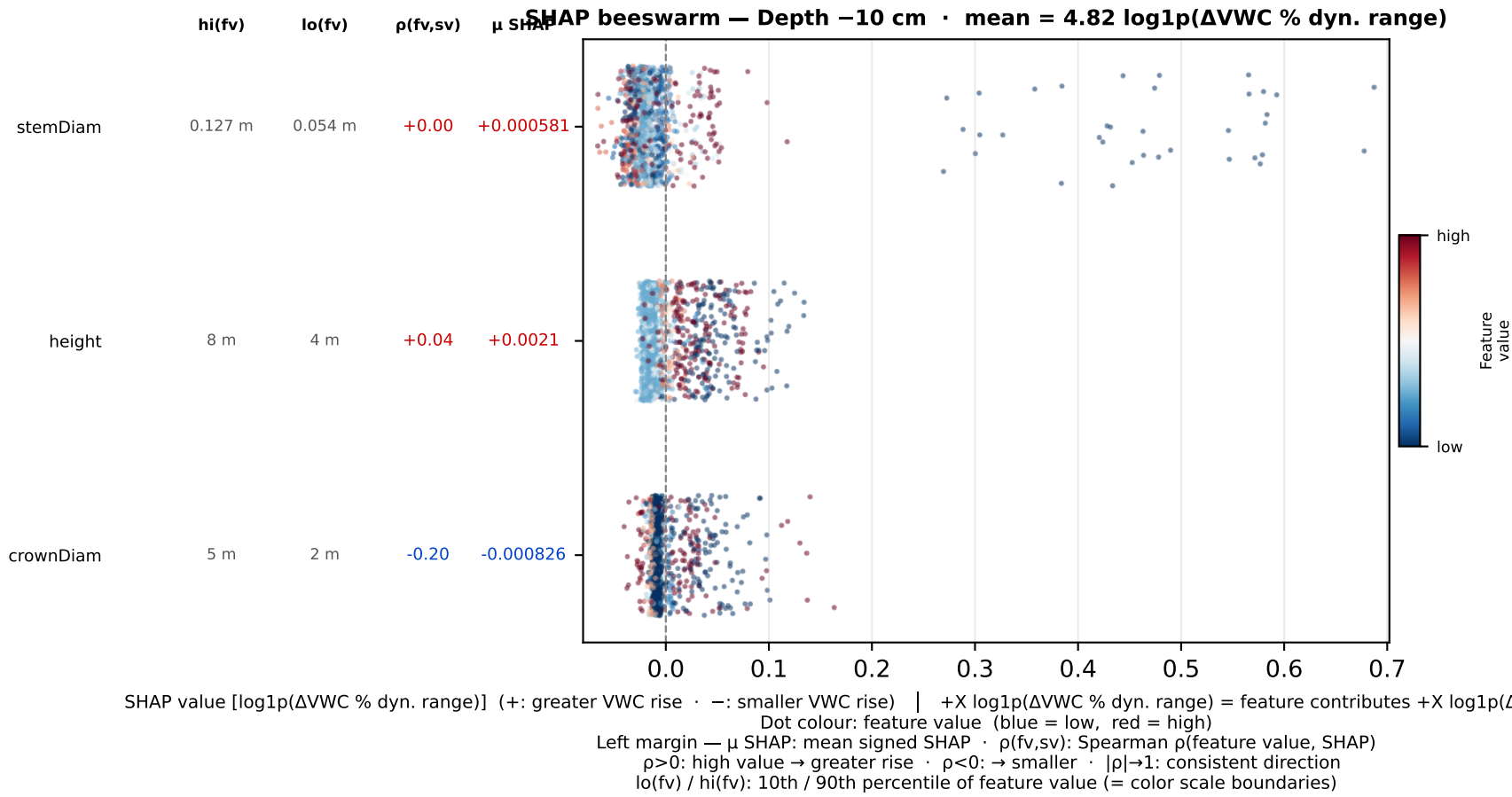
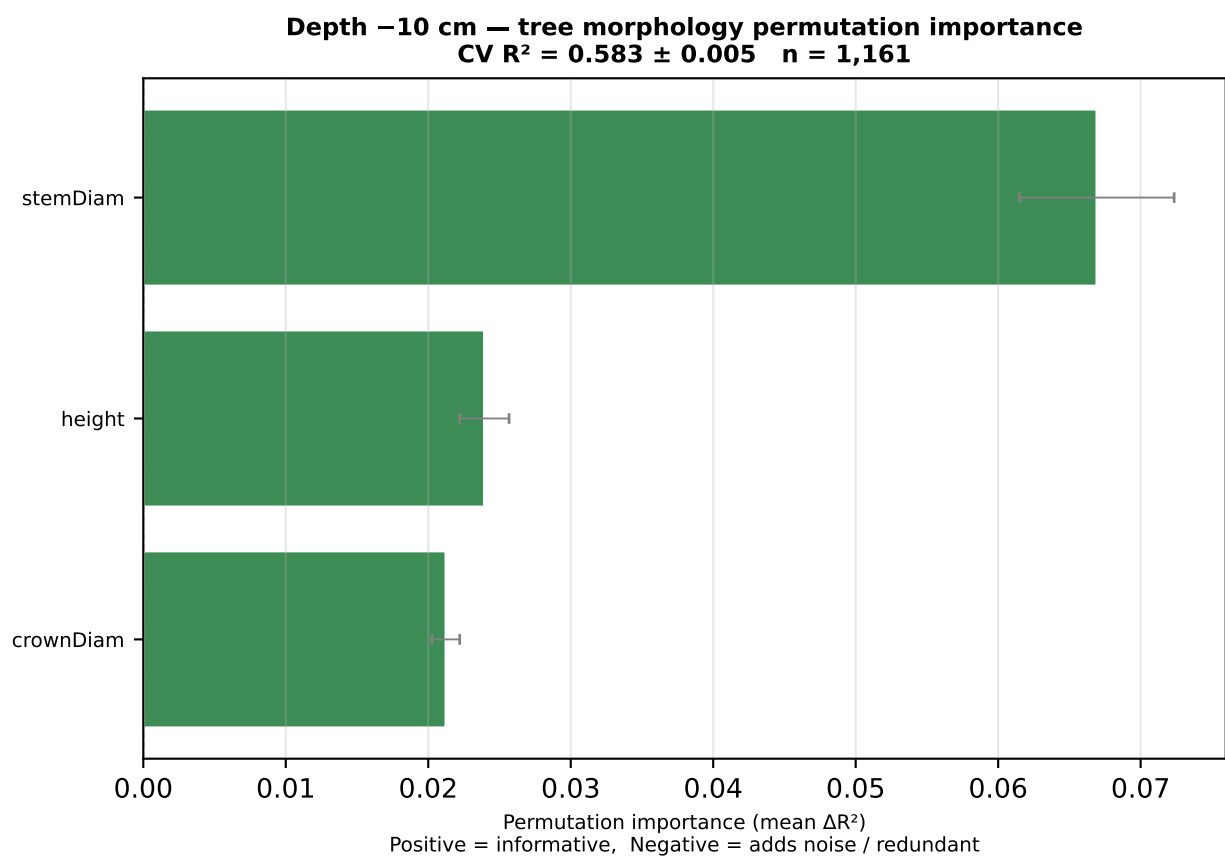
Terrain Feature Analysis — Irrigation strong

Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (1,161 events)

Features: TWI · TPI (2.5/5/7.5 m + crown-diam) · slope · amsl · flow accumulation · SVF · depth to groundwater



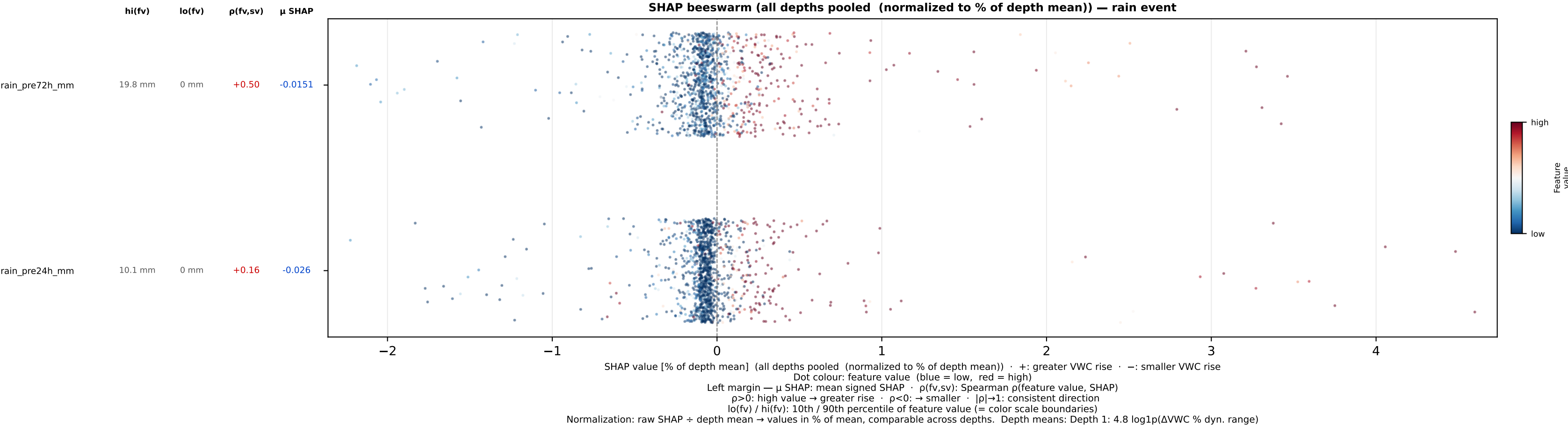
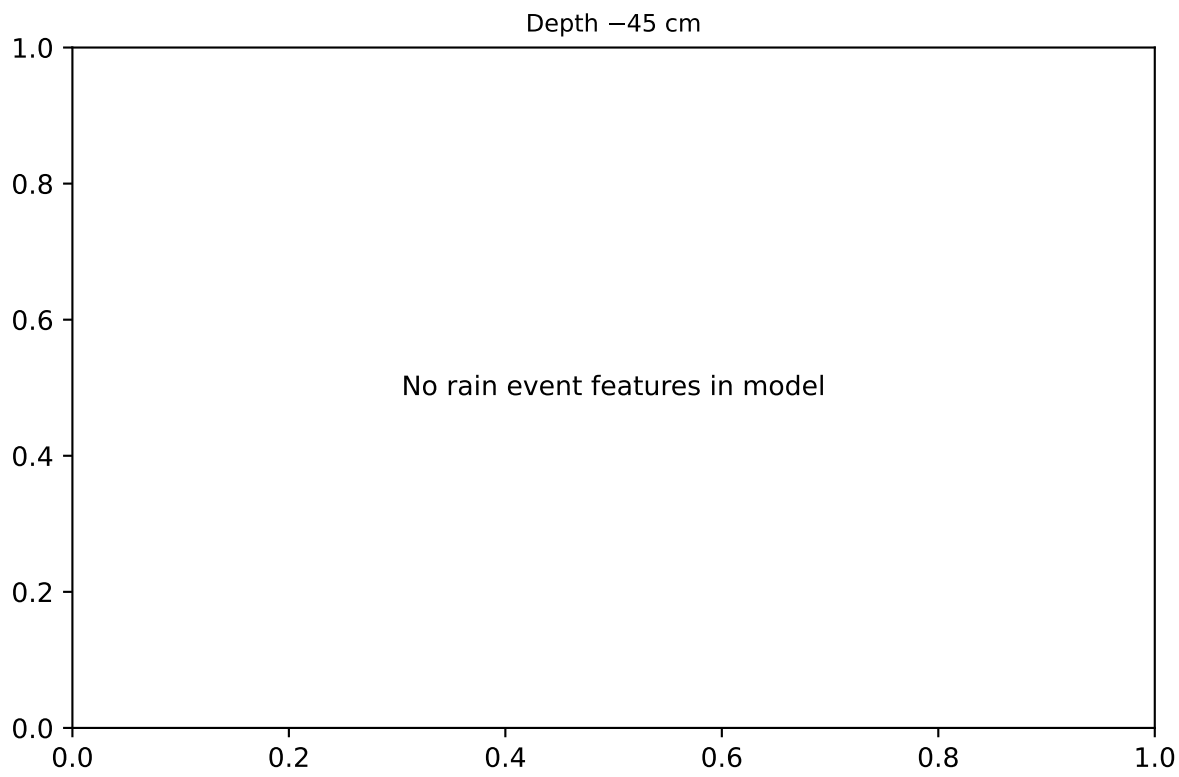
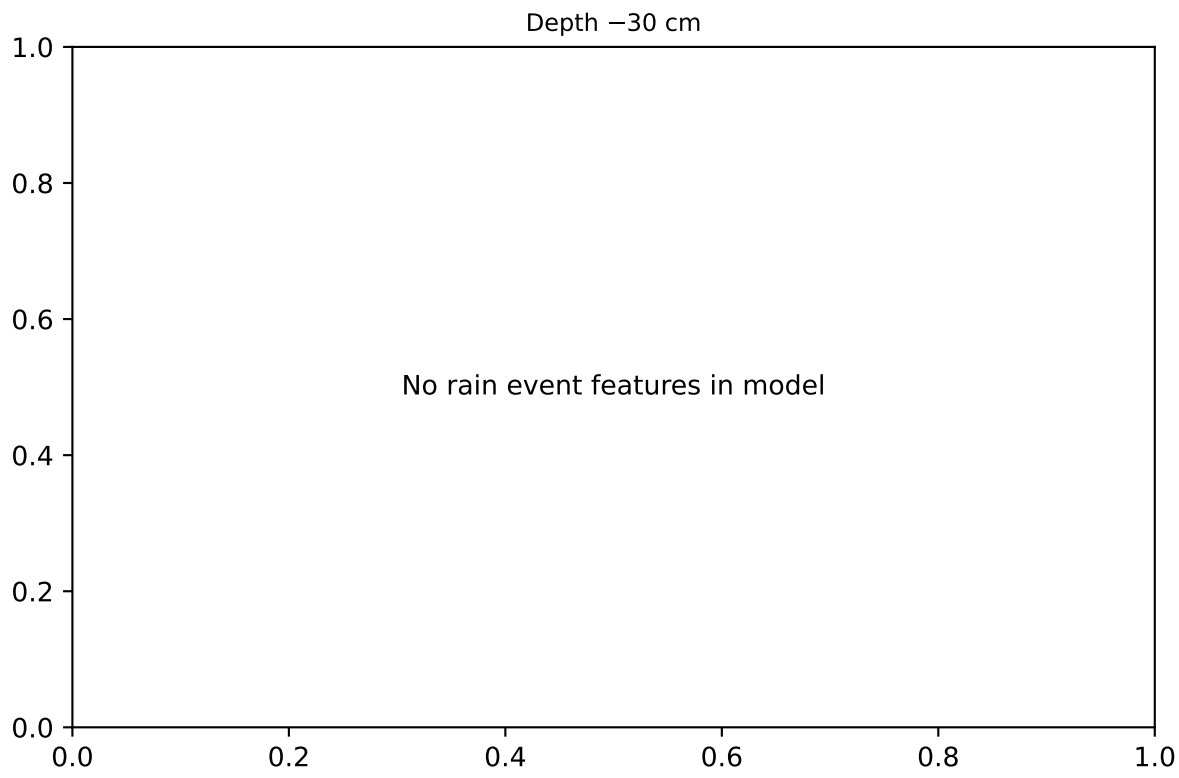
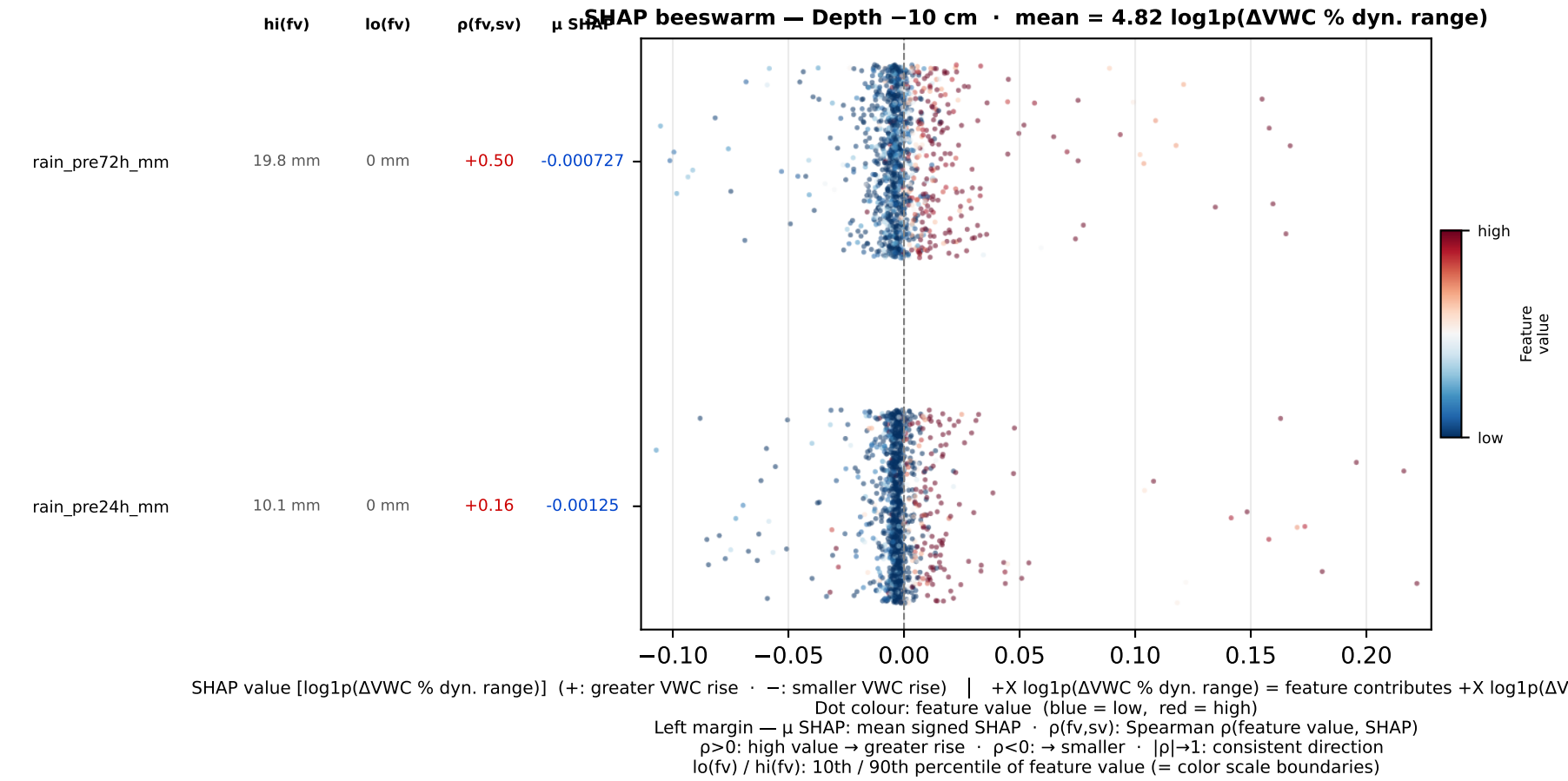
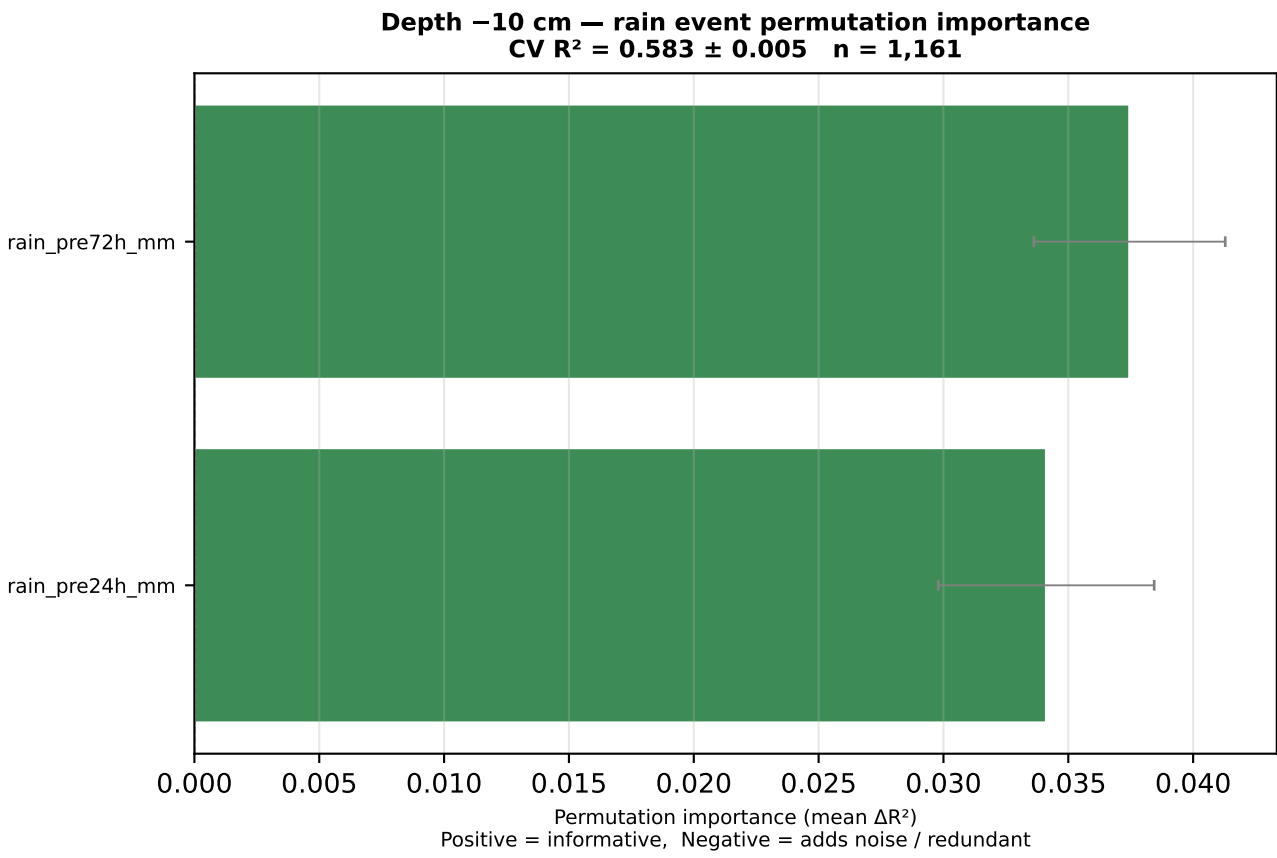
Tree Morphology Feature Analysis — Irrigation strong
Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (1,161 events)
Features: crown diameter · height · stem diameter · vitality



Rain Event Feature Analysis — Irrigation strong

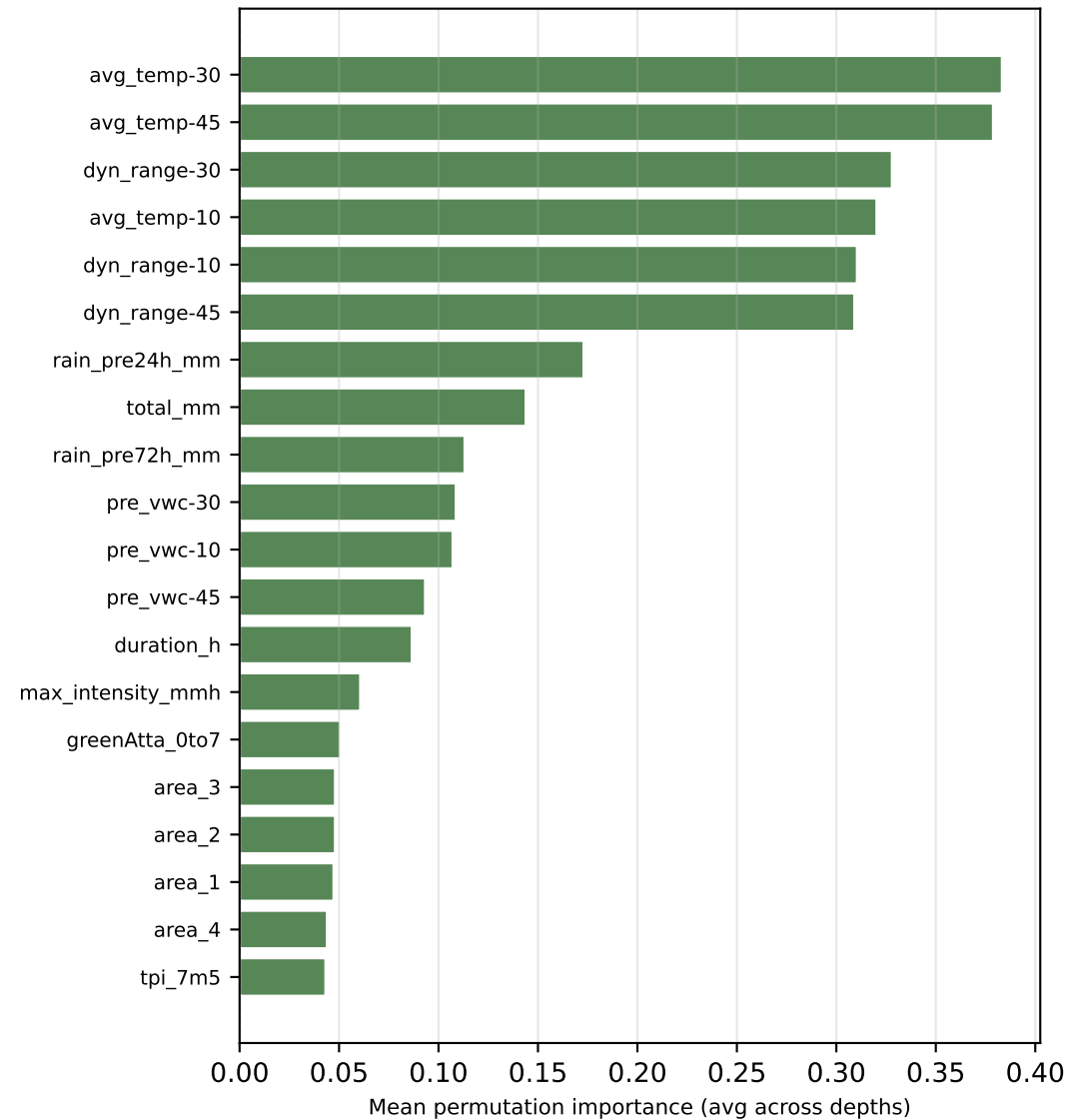
Target: VWC rise [log1p(ΔVWC % dyn. range)] per depth (1,161 events)

Features: total rain · pre/post event rain · peak intensity · StAcc · duration · Huff quartile · rain distribution percentages · day-of-year

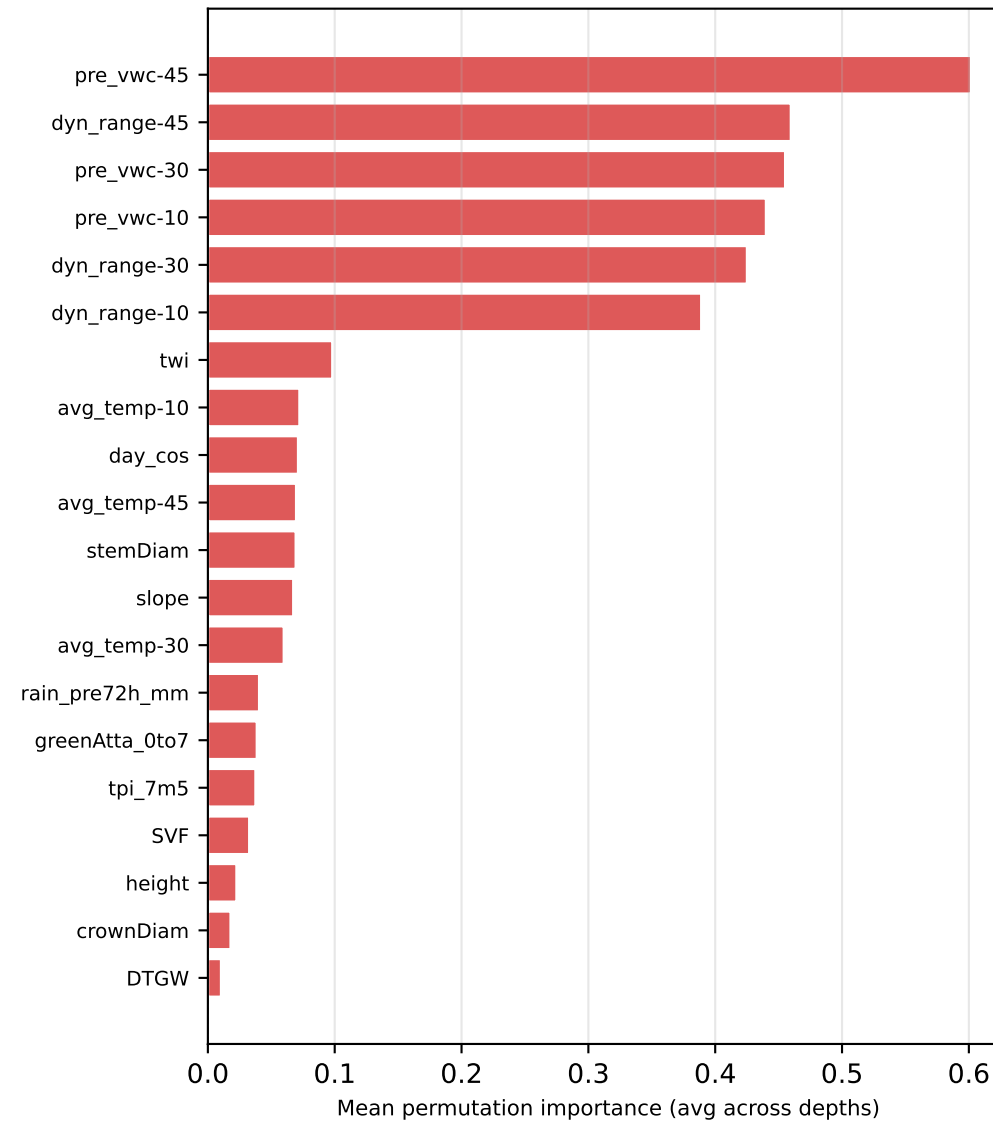


Feature Importance Summary — mean permutation importance across depths

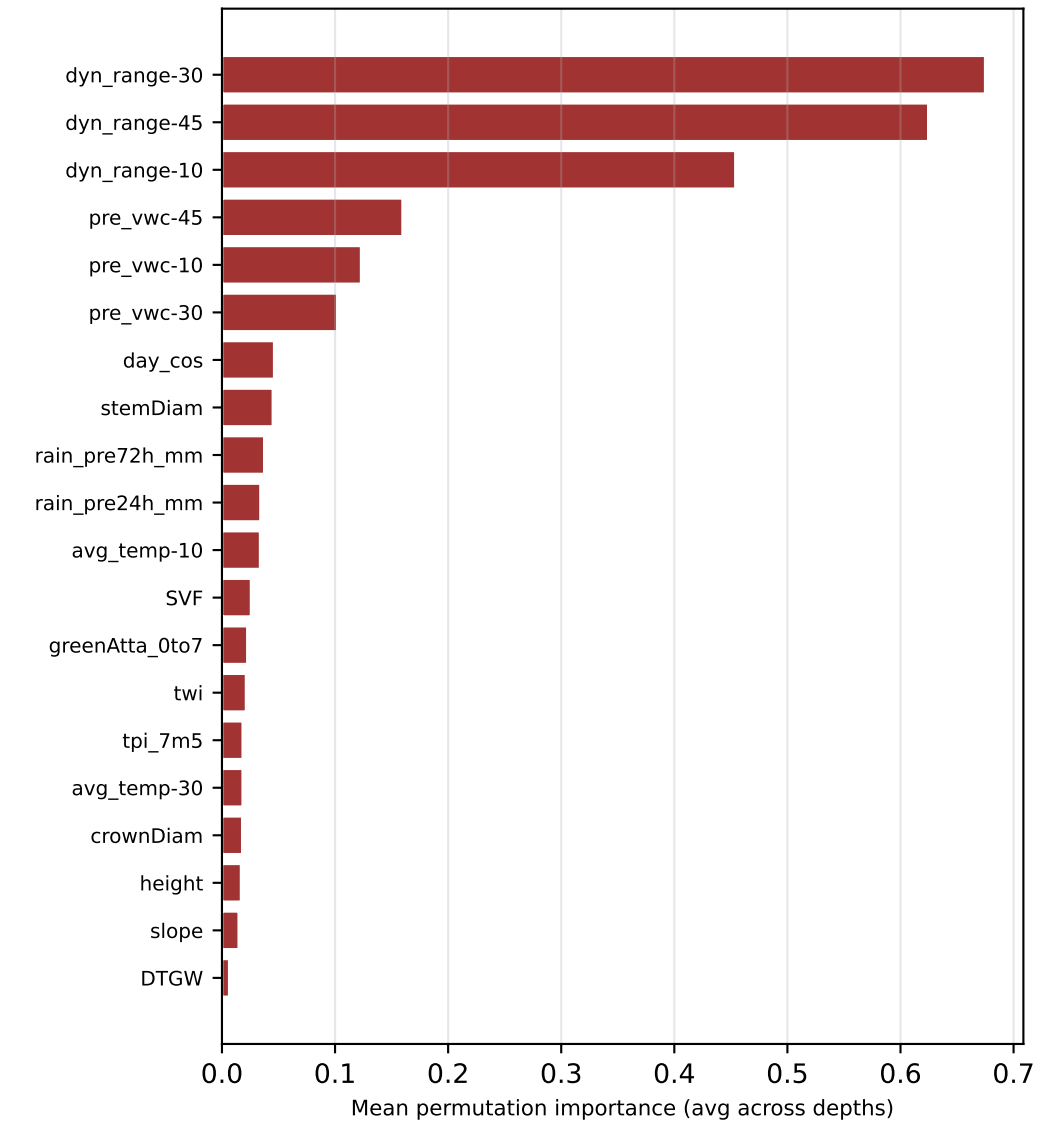
BSC XX11 + Huff Q1/Q2 (front-loaded rain)
avg $R^2 = 0.526$



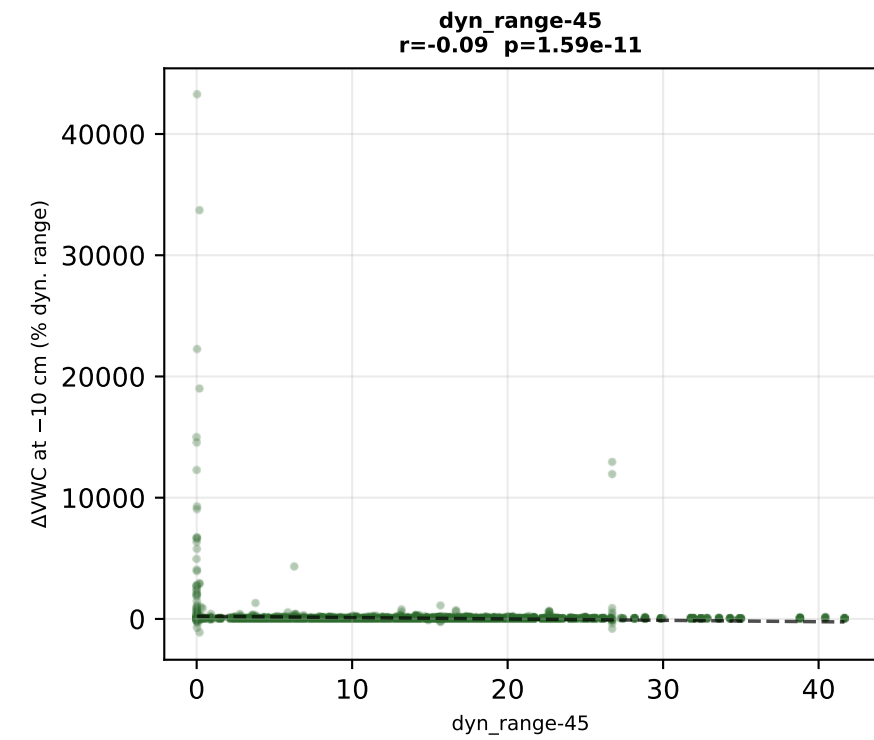
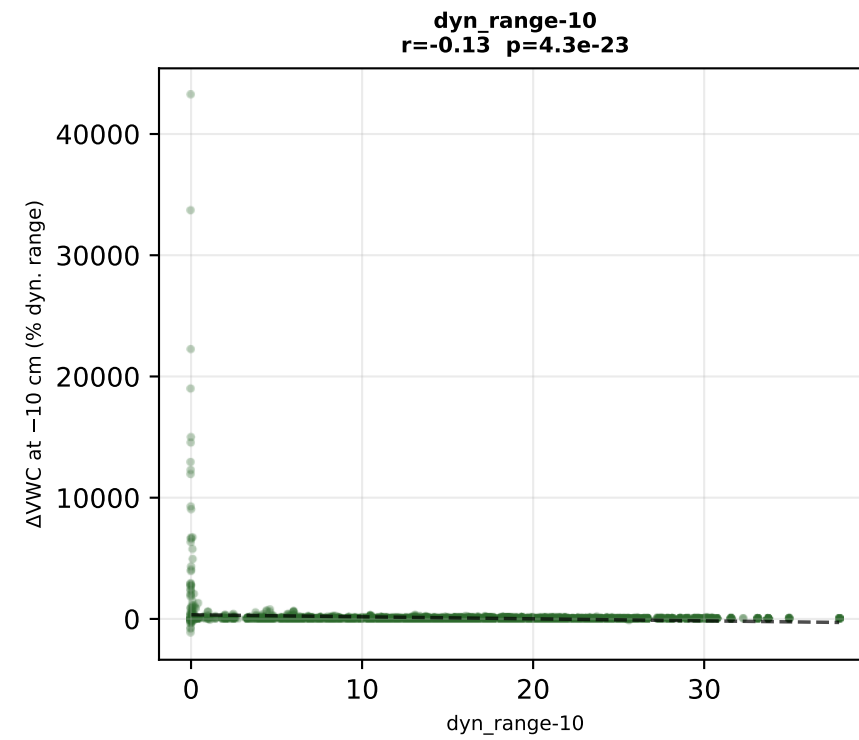
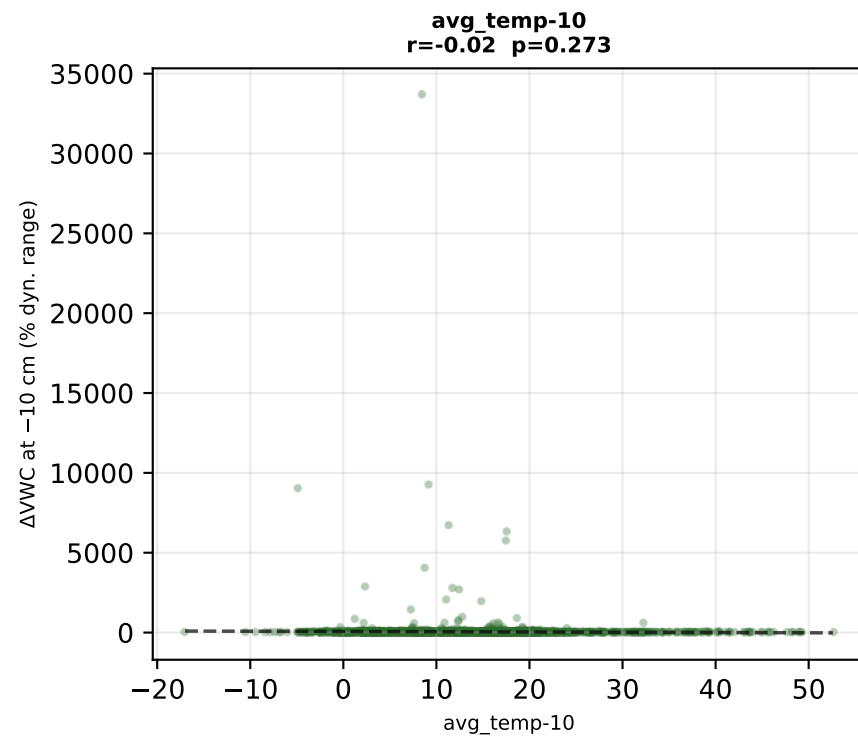
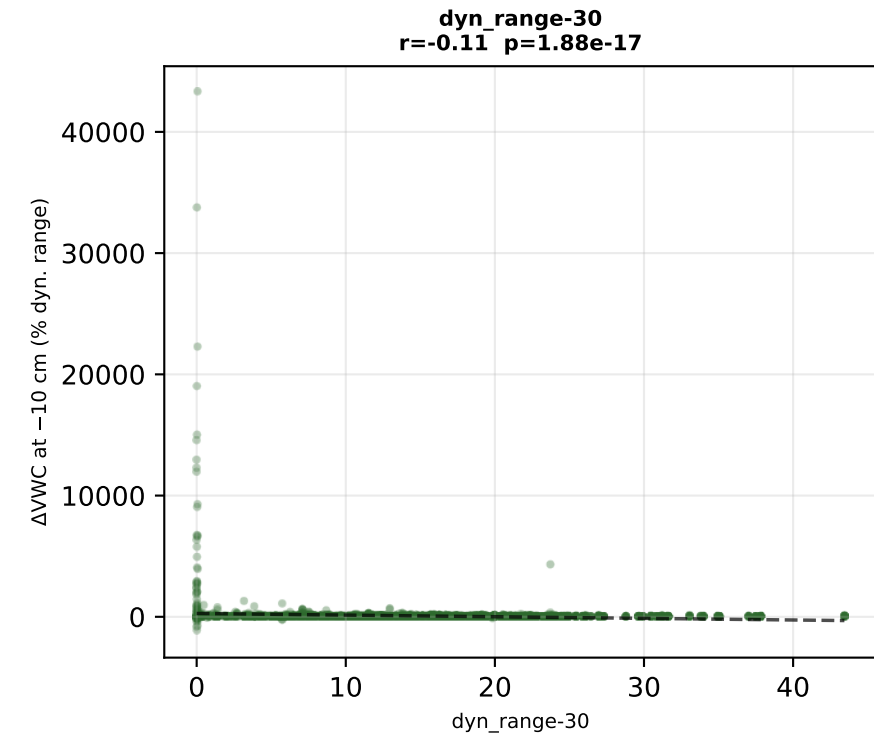
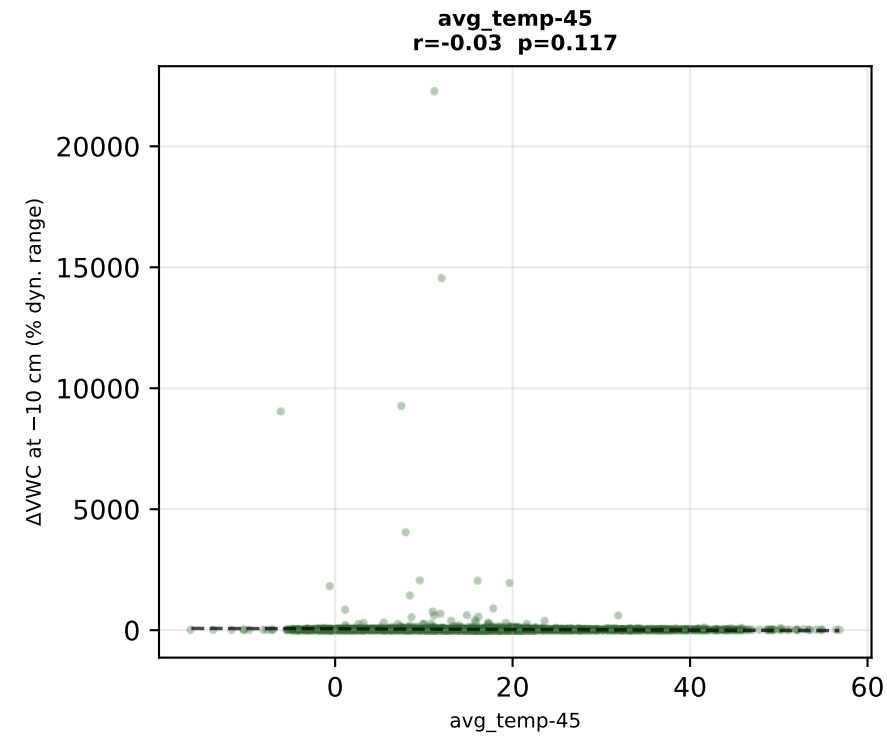
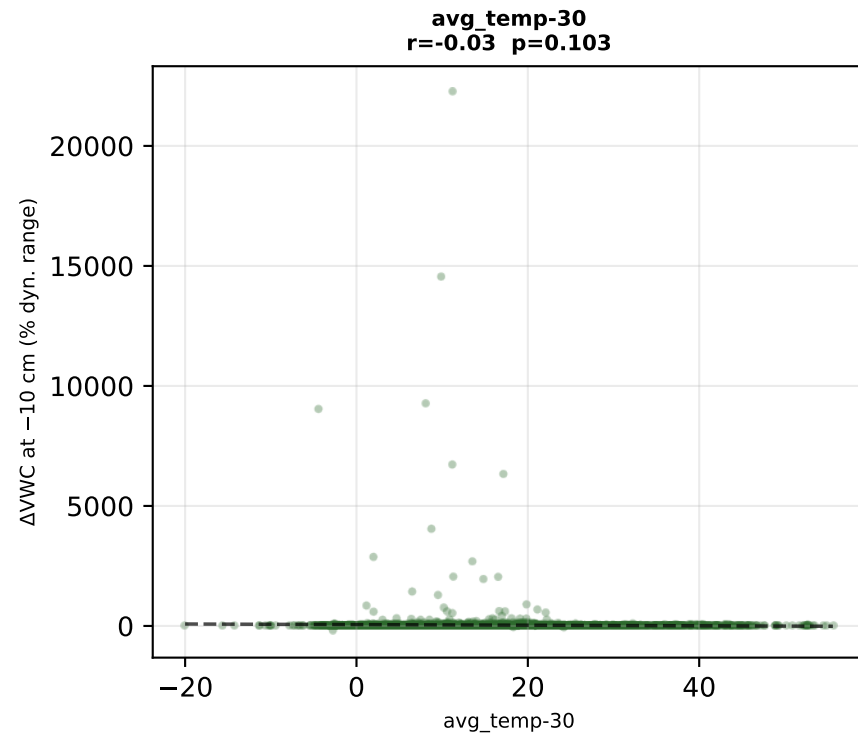
Irrigation
avg $R^2 = 0.665$



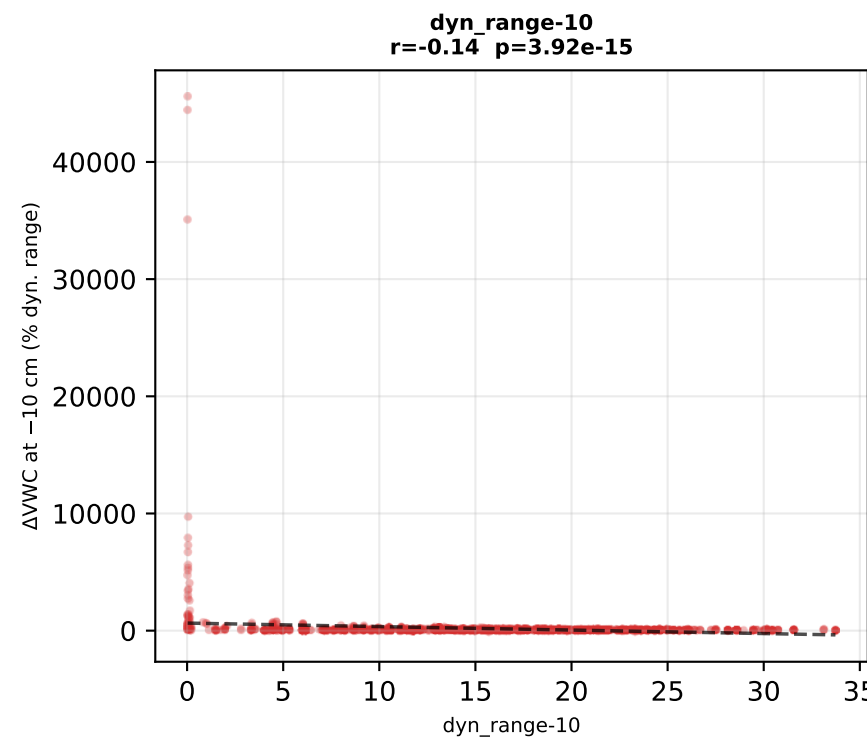
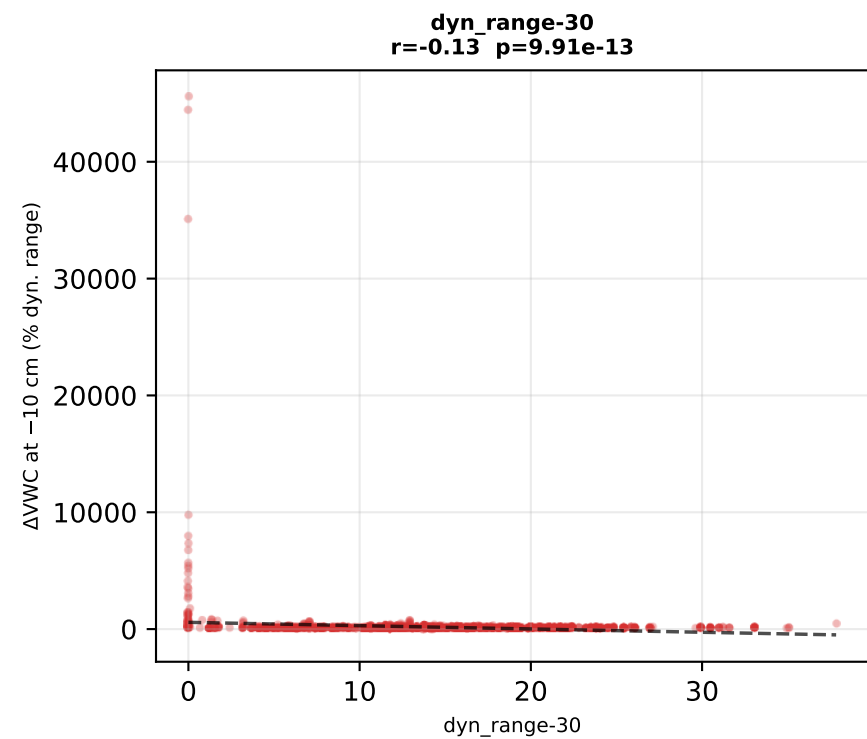
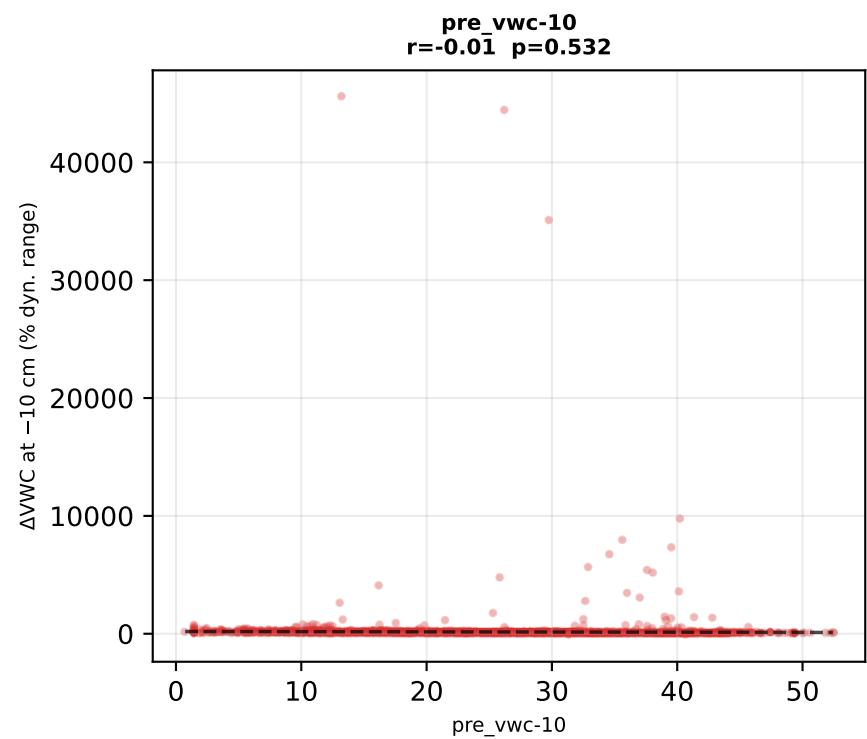
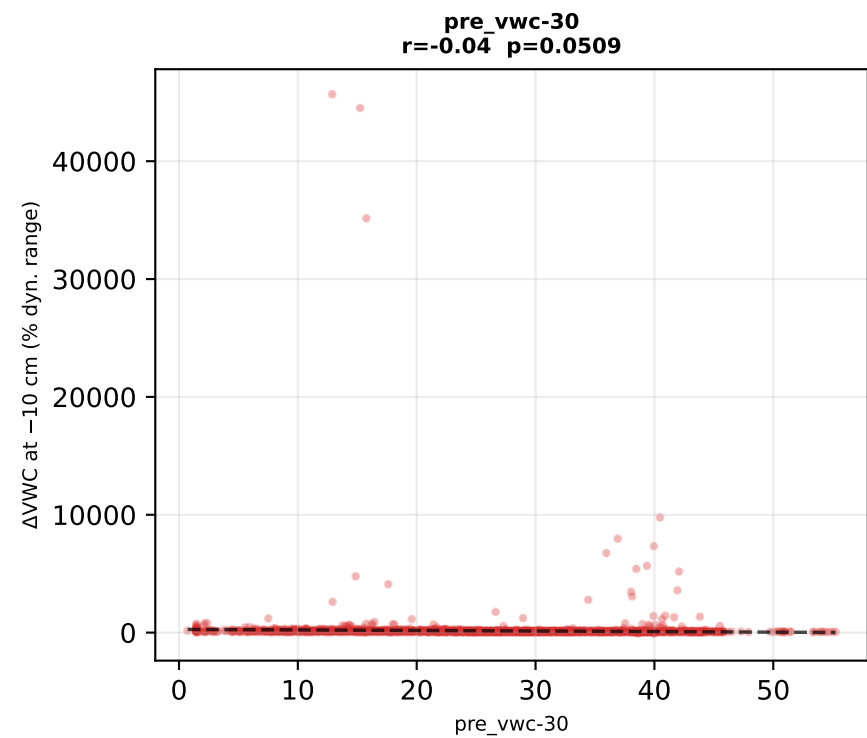
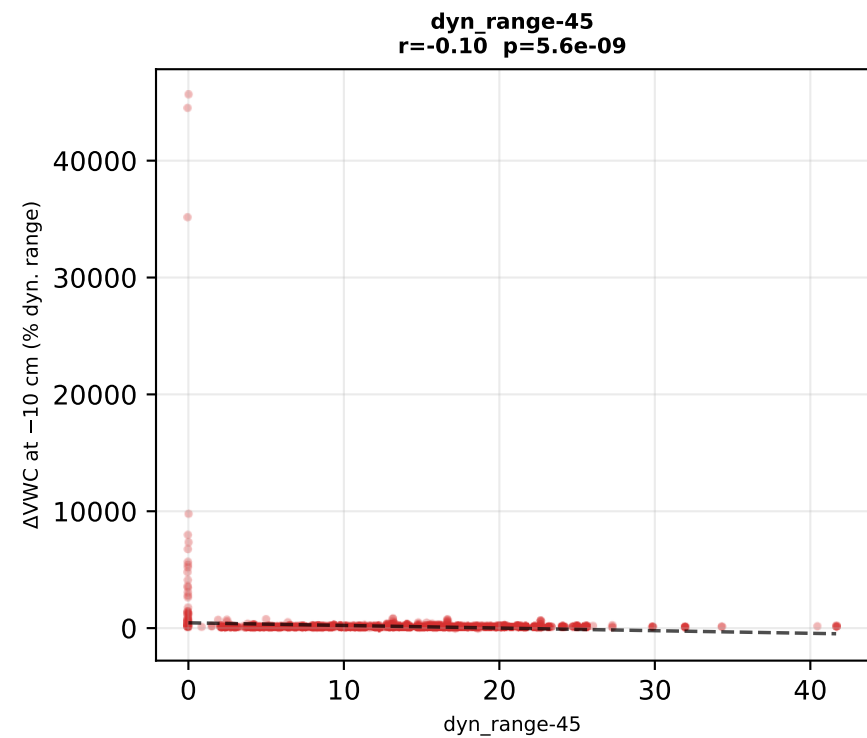
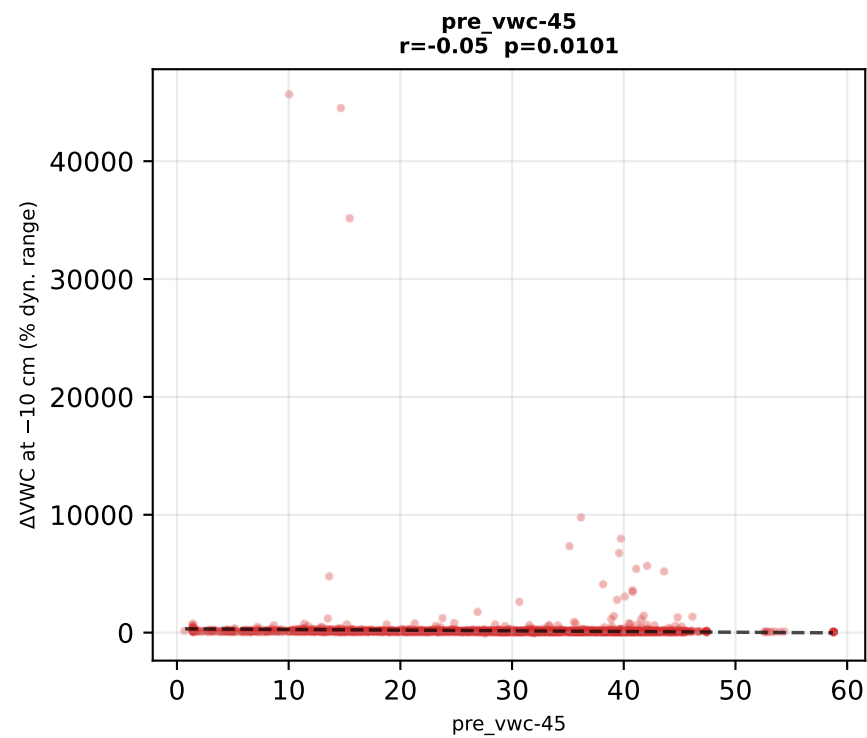
Irrigation strong
avg $R^2 = 0.661$



Top feature scatter plots — BSC XX11 + Huff Q1/Q2 (front-loaded rain)
Target: Δ VWC at -10 cm (% dyn. range) (n = 10,911)



Top feature scatter plots — Irrigation
Target: Δ VWC at -10 cm (% dyn. range) (n = 5,242)



Top feature scatter plots — Irrigation strong
Target: ΔVWC at -10 cm (% dyn. range) (n = 1,161)

